

Data Converter

Lecture 2

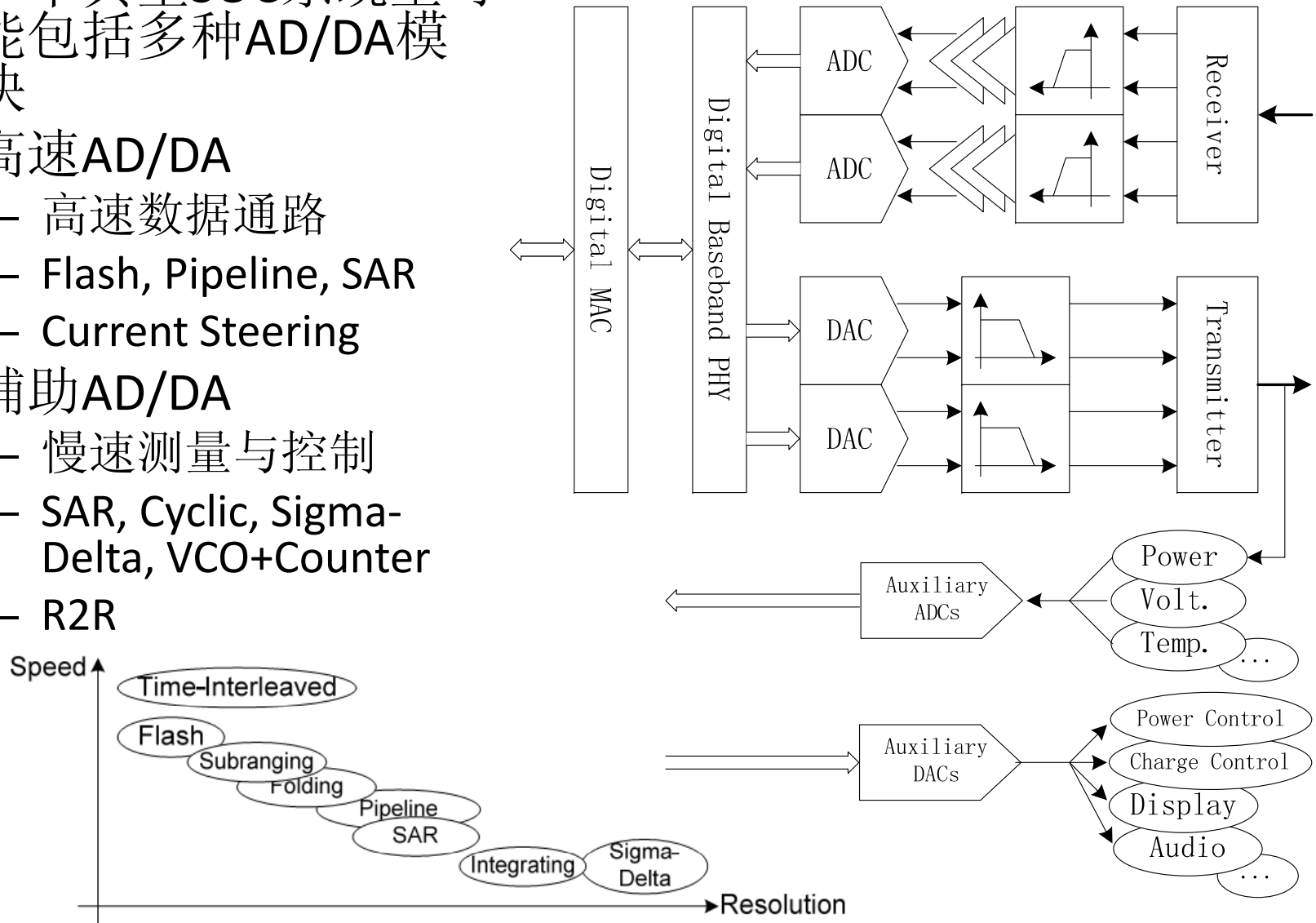
ADC/DAC Structures

Li Fule

Summer 2010

典型应用情况

- 一个典型SOC系统里可能包括多种AD/DA模块
- 高速AD/DA
 - 高速数据通路
 - Flash, Pipeline, SAR
 - Current Steering
- 辅助AD/DA
 - 慢速测量与控制
 - SAR, Cyclic, Sigma-Delta, VCO+Counter
 - R2R



Lecture 2

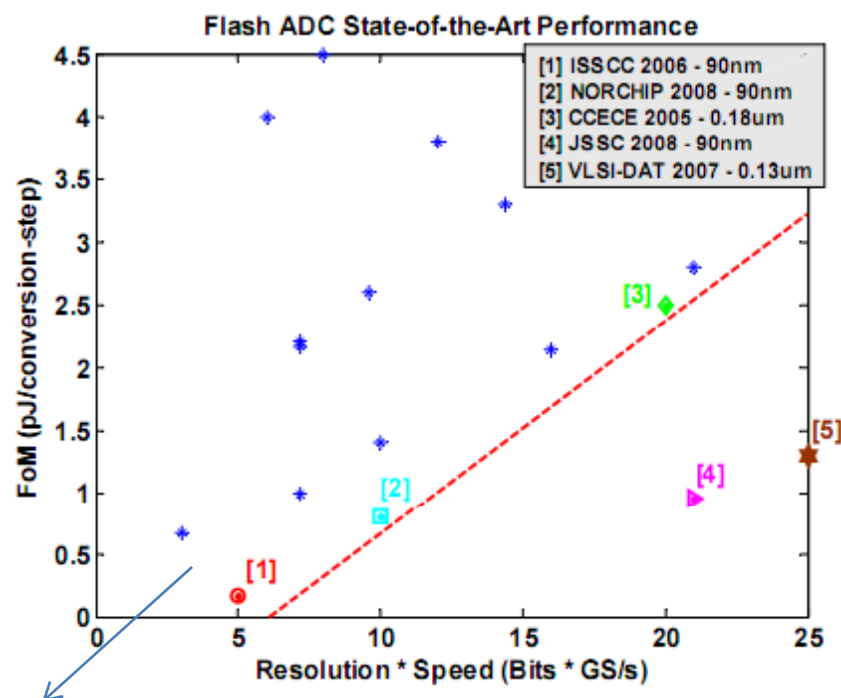
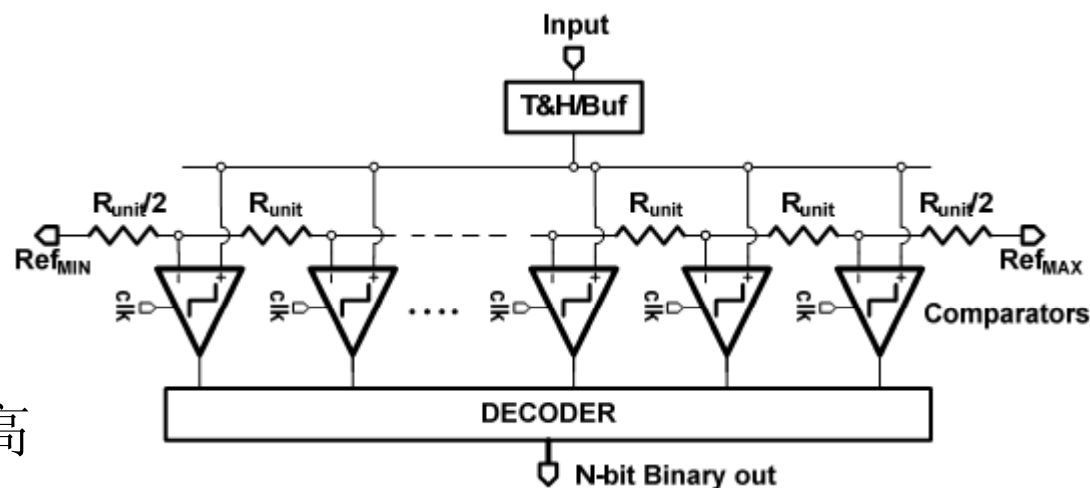
ADC/DAC Structures

- **Flash ADC**
- Pipeline ADC
- SAR ADC
- Sigma-Delta Modulator
- Time-interleaved
- R2R DAC
- Current-Steering DAC

Flash ADC

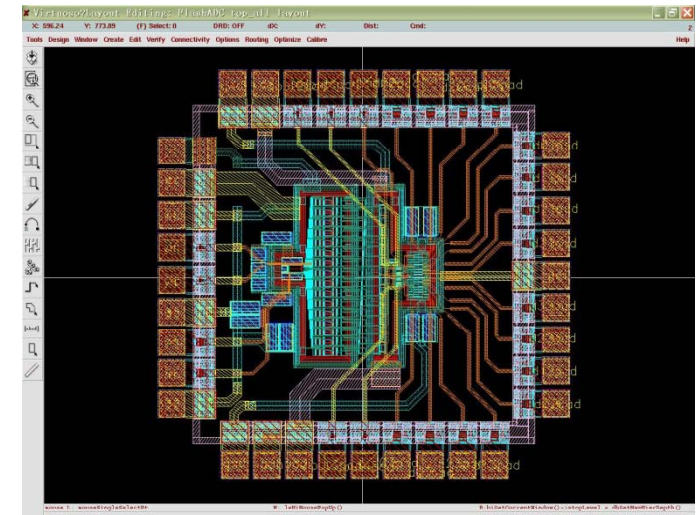
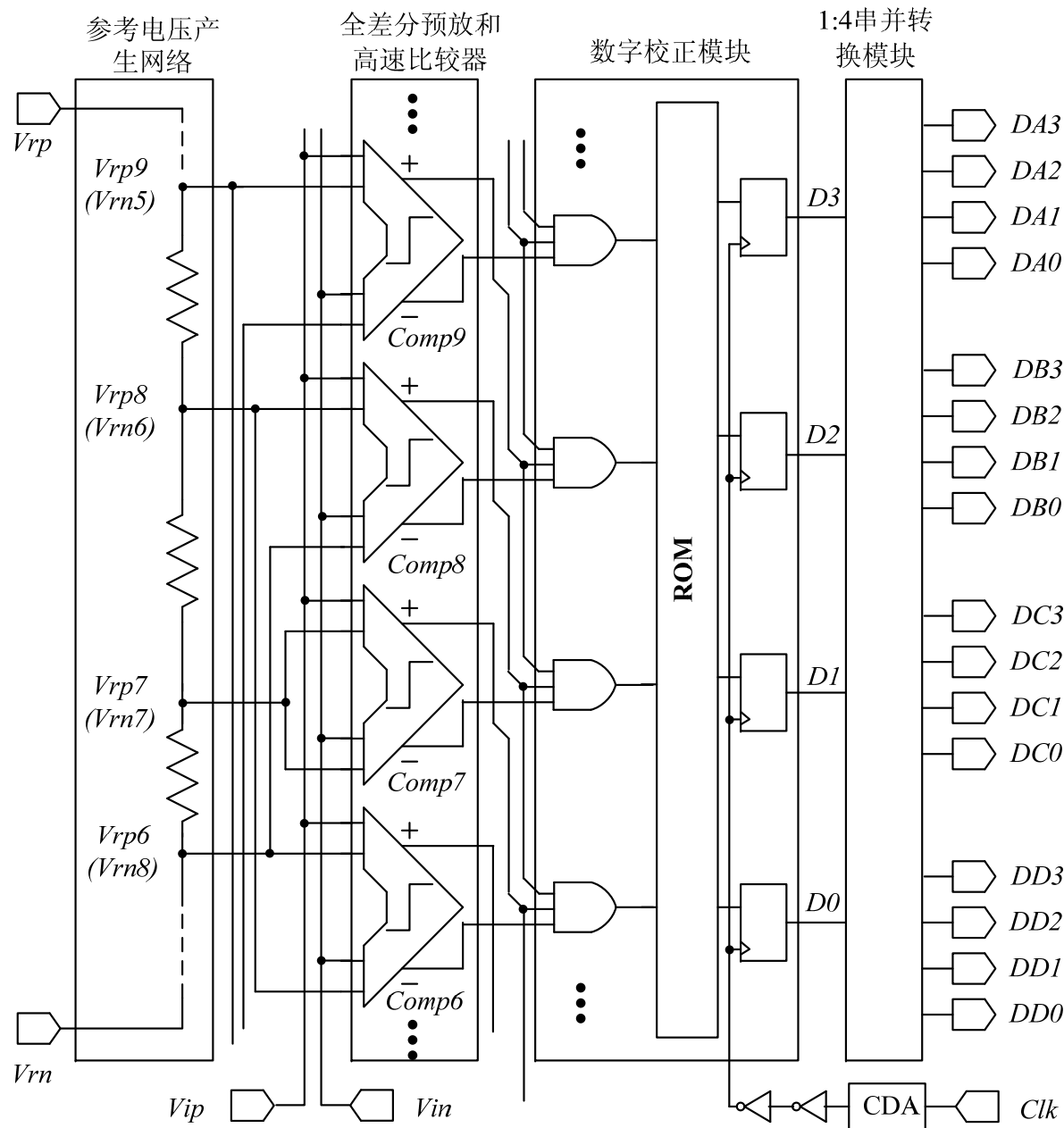
可实现性能
3~8 bit
 $xM \sim xGS/s$

- 全并行操作，转换率最高
- 单步转换，延时最小
 - 对与反馈控制很重要
- 适合于低分辨率转换器
 - 面积、功耗、输入电容
- 雷达系统——射频采样？
- 电路非理想因素限制精度
- T/H Buf ?
 - No T/H, Scalable with technology, but clock skew?
- 主要电路技术
 - 失调抵消、插值平均



EPKAN ALPMAN, A 7-Bit 2.5GS/sec Time-Interleaved C-2C SAR ADC
For 60GHz Multi-Band OFDM-Based Receivers. PHD Thesis

Flash ADC



最简单，最高速
的A/D转换结构

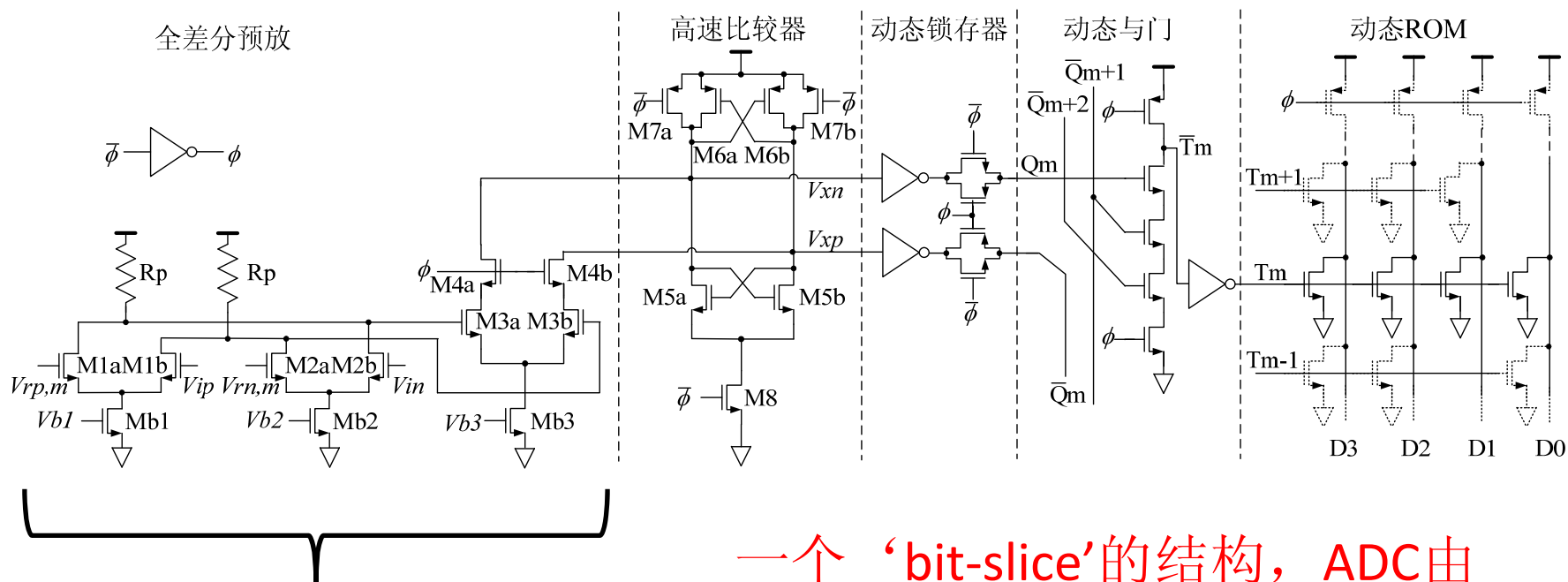
4-bit, 2Gsps ADC

Smic 0.18um CMOS

测试到700Msps，性能良好

(清华2008学生设计)

Flash ADC

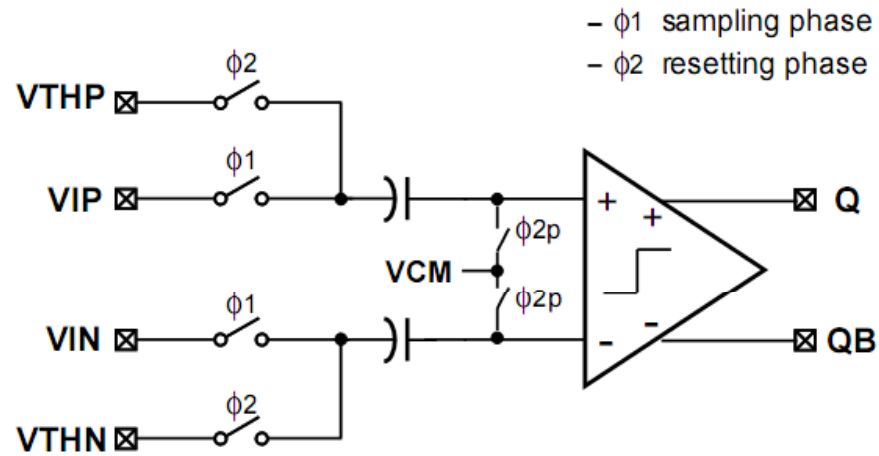
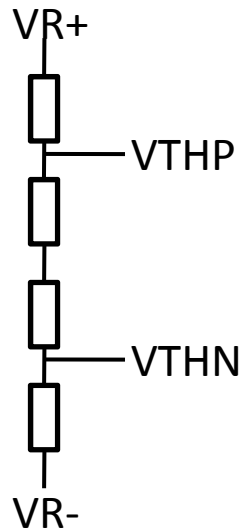


实现差分相减

一个 ‘bit-slice’ 的结构，ADC由 $2^n - 1$ 个这样的结构并联而成

差分比较：先差分相减，然后判断极性

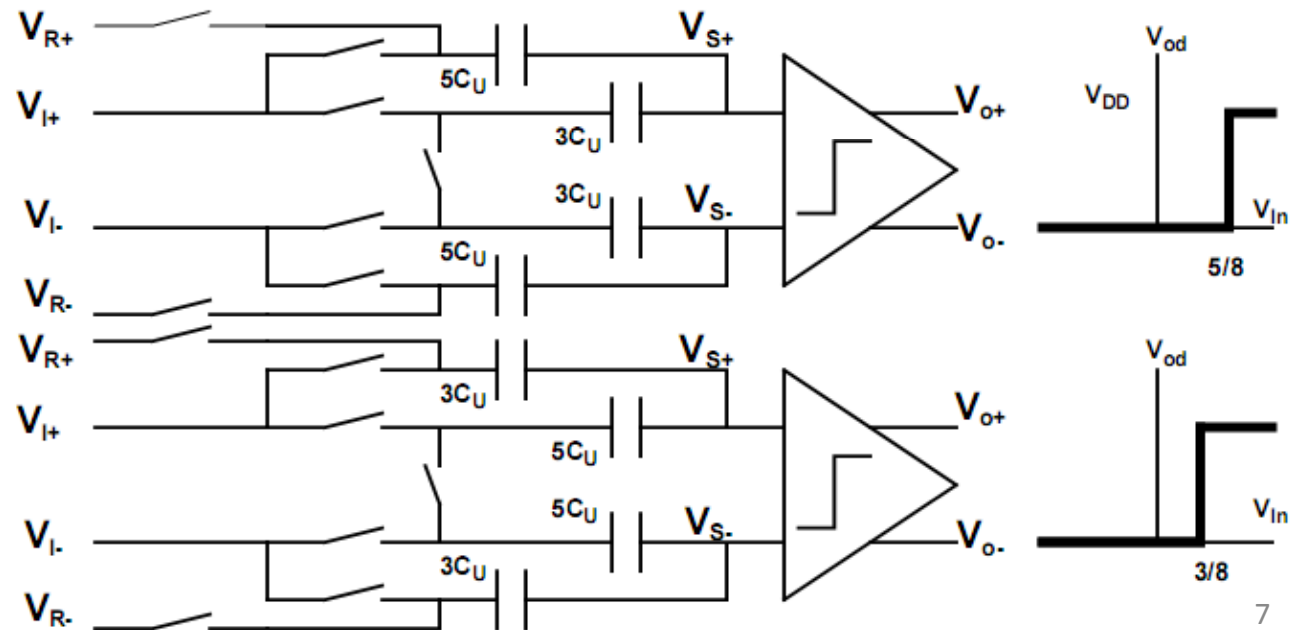
电容耦合方式



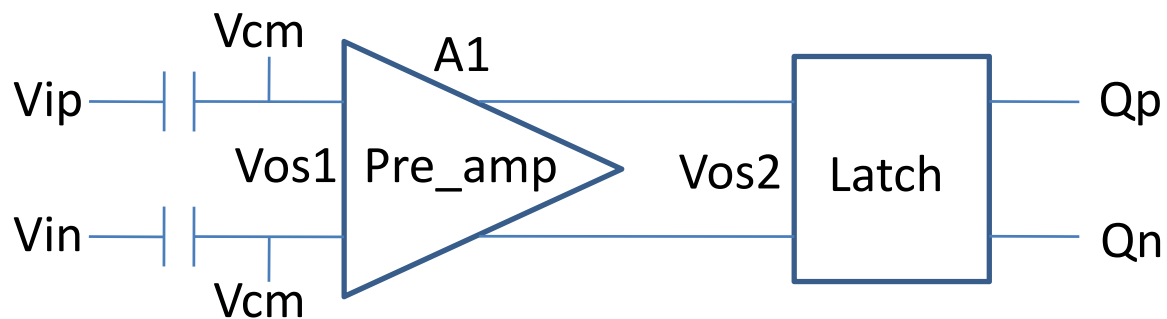
- $\phi 1$ sampling phase
- $\phi 2$ resetting phase

阈值由电阻
分压给出

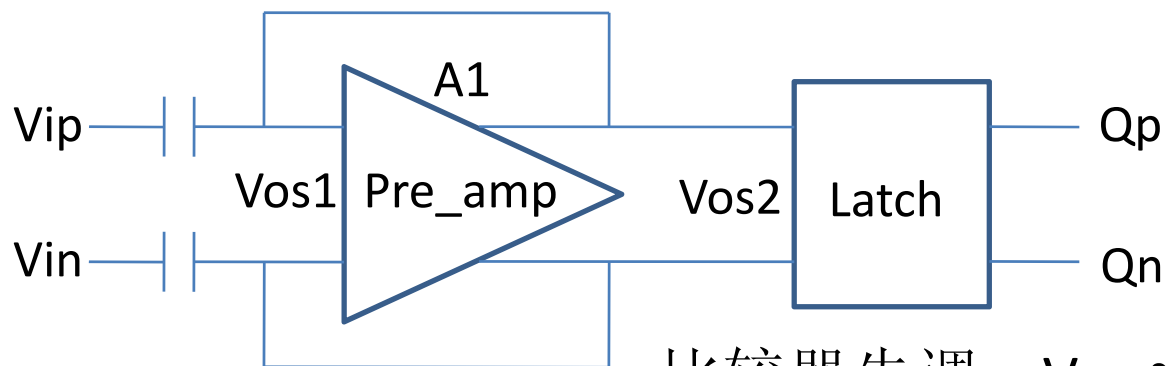
阈值由电容
比例决定



比较器失调

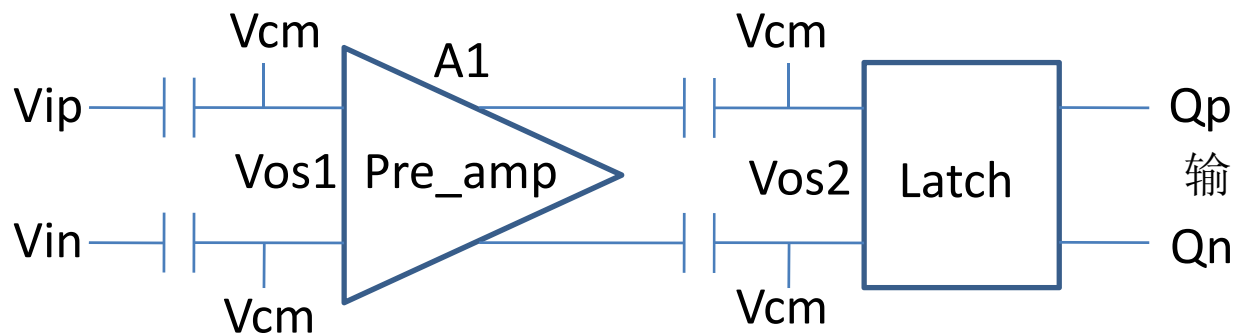


比较器失调: $V_{os} = V_{os1} + V_{os2}/A1$



输入失调电压存储

比较器失调: $V_{os} \approx (V_{os1} + V_{os2})/A1$



输出失调电压存储

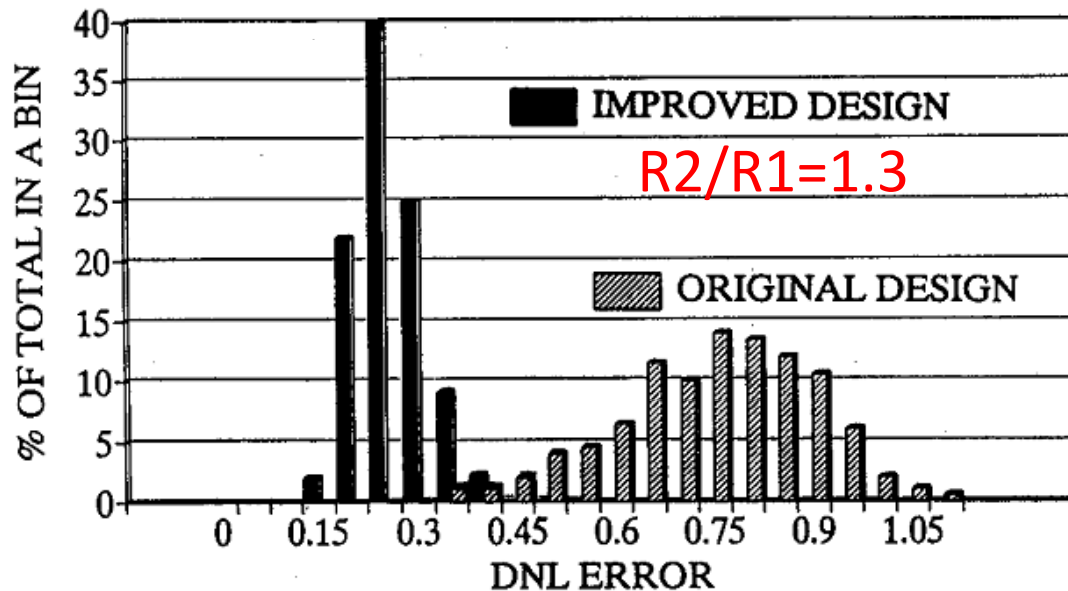
比较器失调: $V_{os} = V_{os2}/A1$

Averaging

$$ECF = 100 * (1 - DNL_{CORRECTED} / DNL_{UNCORRECTED})$$

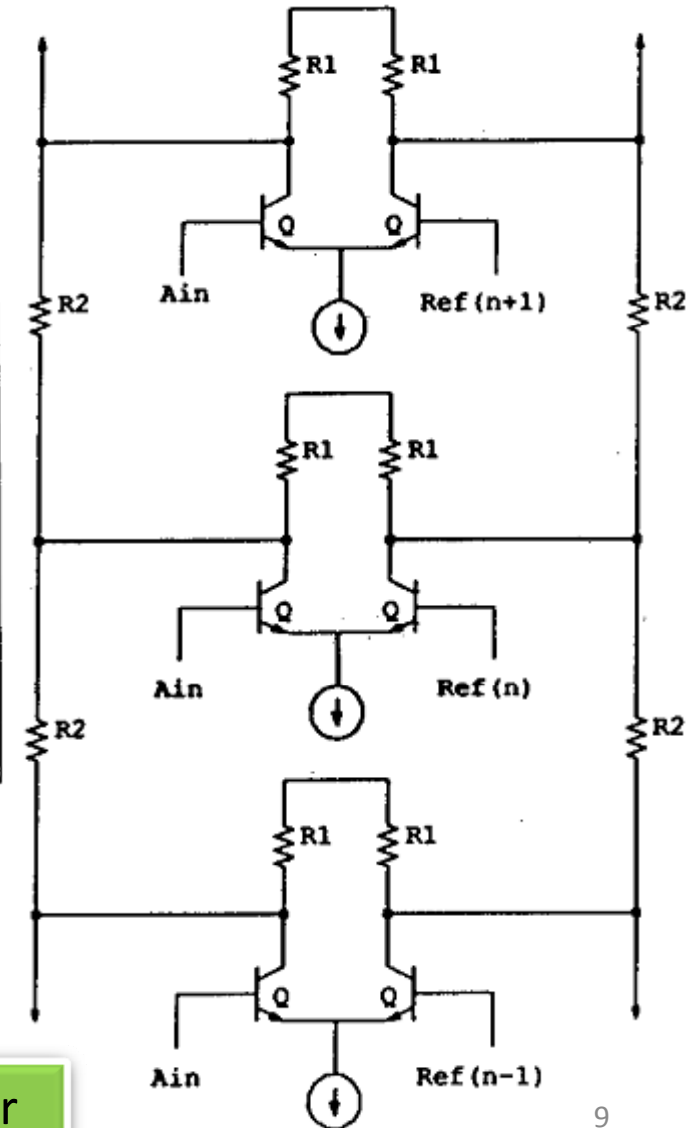
$$ECF = (1 - (R_x * (1 - R_1 / (R_1 + R_x)) / (2 * (R_1 + R_x / 2))) * 100$$

$$R_x = (R_2 + (R_2^2 + 4 * R_2 * R_1)^{1/2}) / 2$$

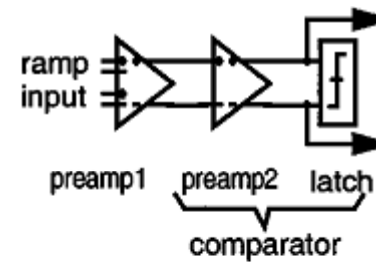
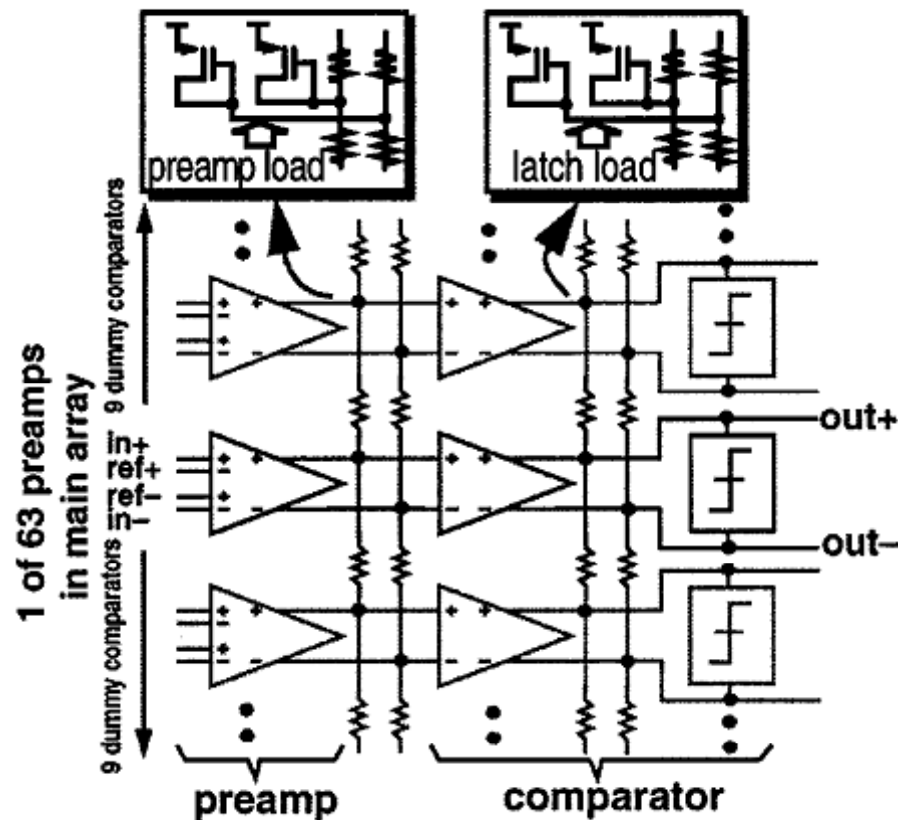


Ref: Kevin Kattmann, et, al. A Technique for Reducing Differential Non-Linearity Errors in Flash A/D Converters. ISSCC 91

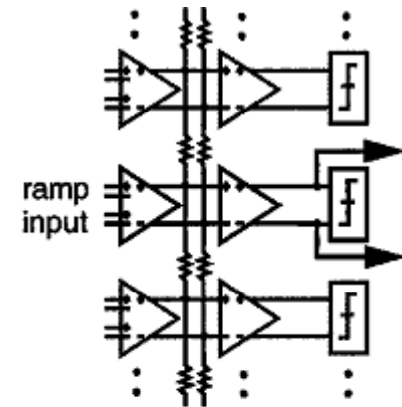
在量程两端需要dummy amplifier



Averaging

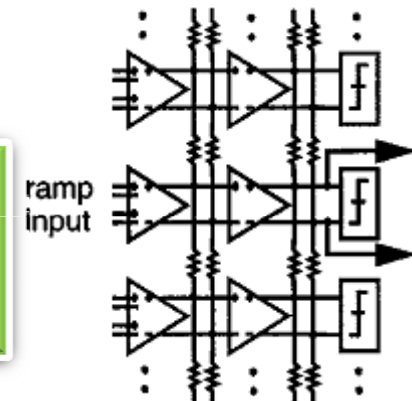


$\sigma = 11 \text{ mV}$
(no averaging)



$\sigma = 9 \text{ mV}$
(preamp averaging)

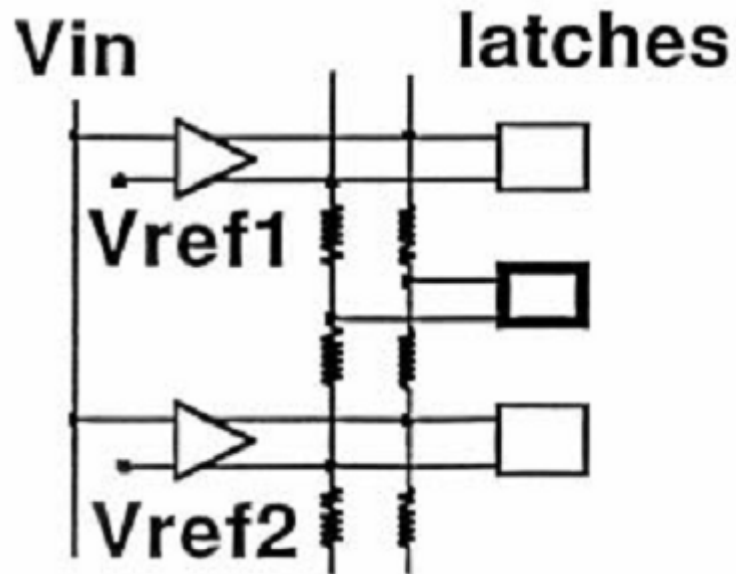
在量程两端须加
足够的dummy
comparators



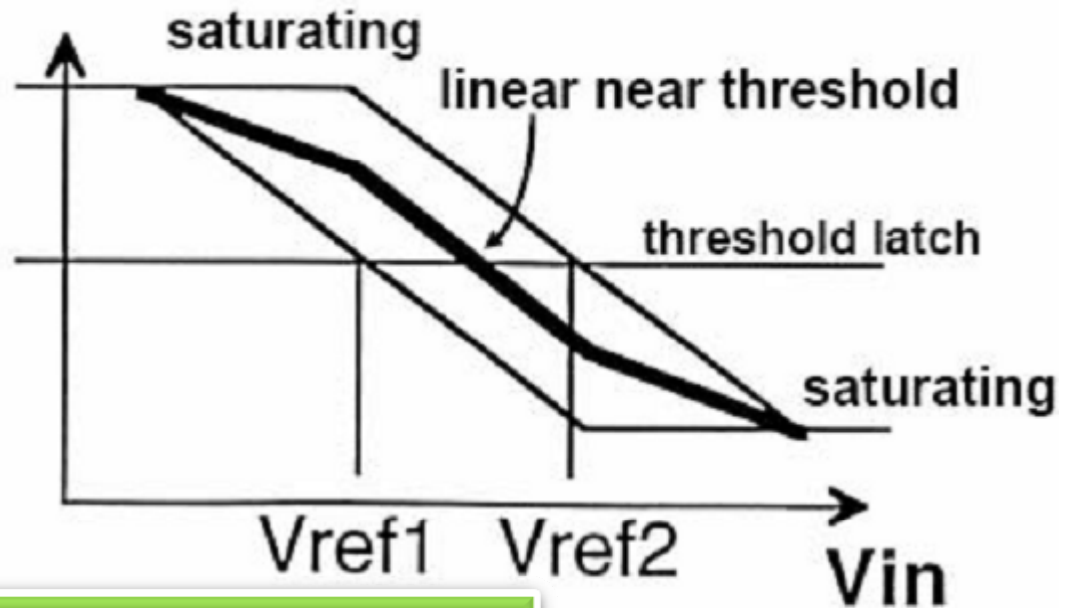
$\sigma = 3.7 \text{ mV}$
(preamp & comparator
averaging)

Ref: Michael Choi, et, al. A 6-b 1.3-Gsample/s A/D Converter in 0.35-um CMOS. JSSC Vol.36, No.12, Dec. 2001

Interpolating



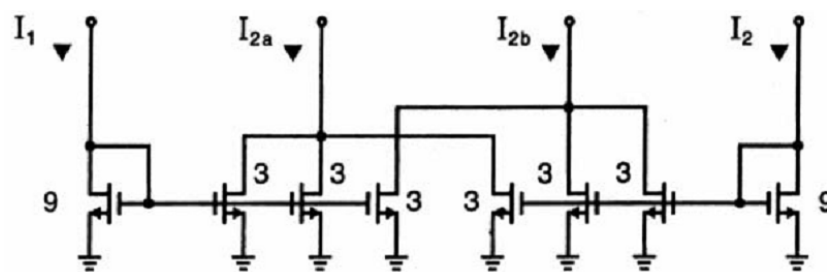
Input amplifiers
which saturate



降低功耗、输入电容
对offset有平均作用
中间的latch有R, Cp延迟

Van de Grift, JSSC Dec. 87, 944-953; Steyaert CICC 1993

电流内插

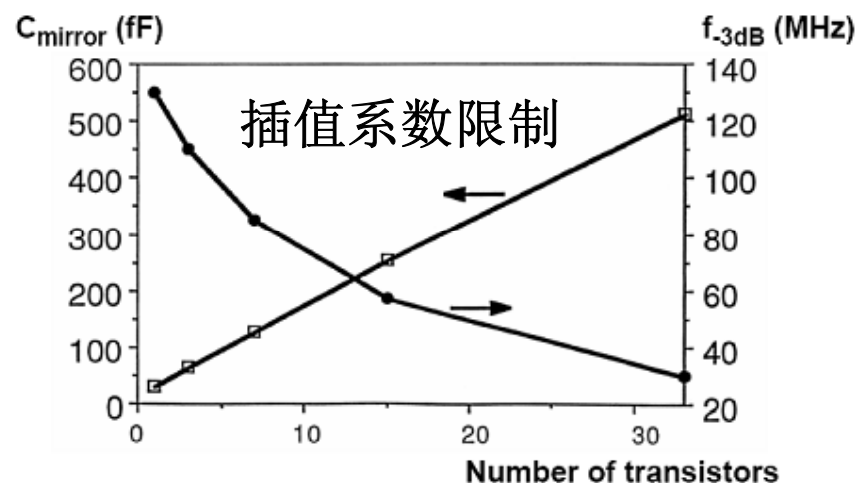


$$I_{2a} = \frac{2}{3} I_1 + \frac{1}{3} I_2$$

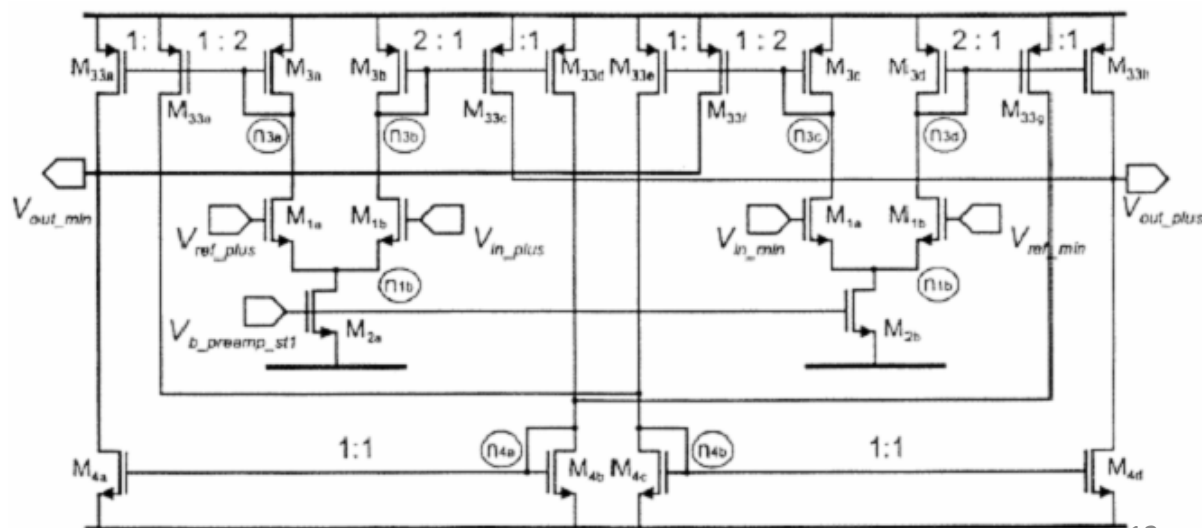
$$I_{2b} = \frac{1}{3} I_1 + \frac{2}{3} I_2$$

Interpolating by 3
between output currents I_1 & I_2
Requires $1/3$ input amps.: $C_{in}/3$

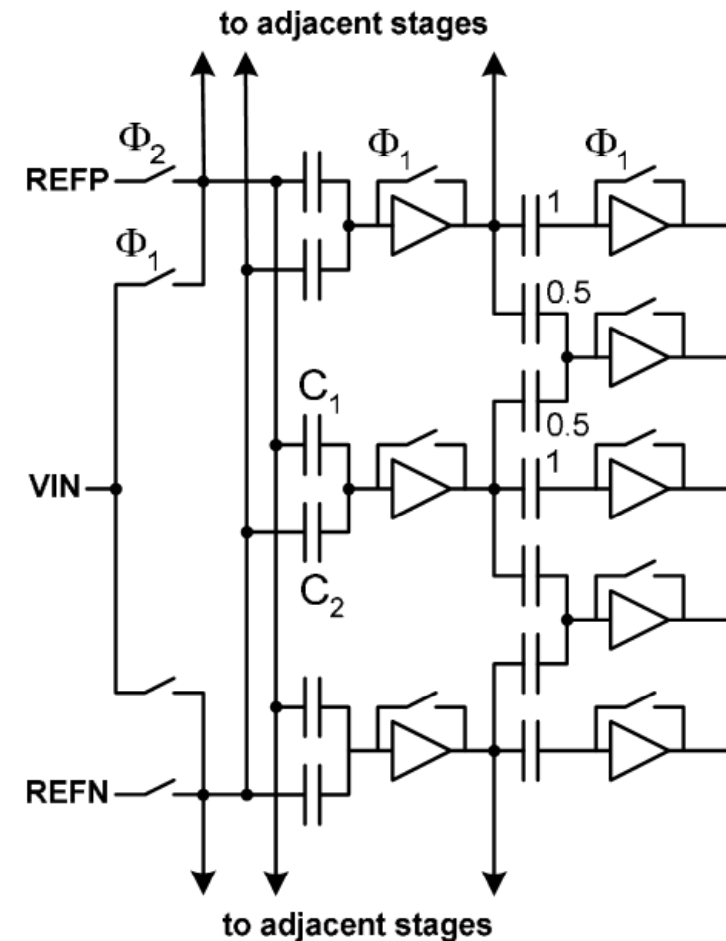
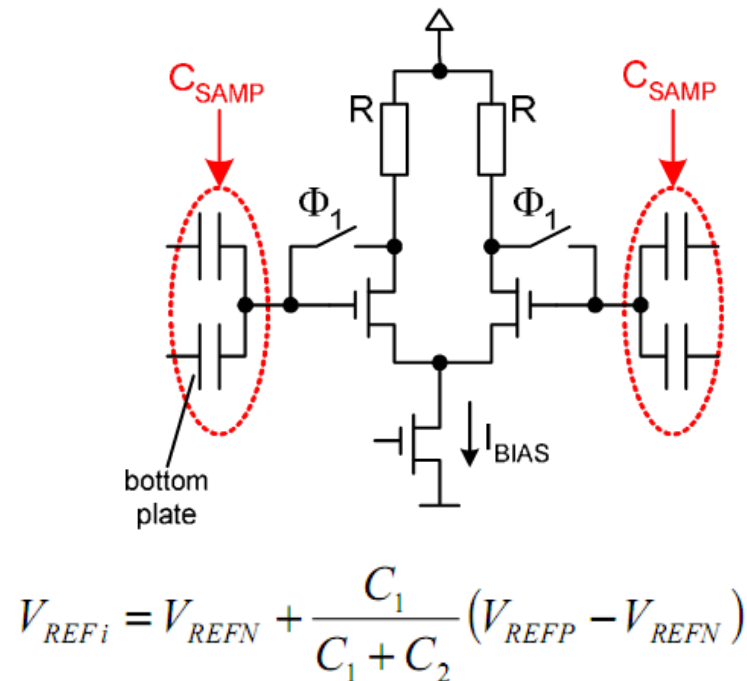
Steyaert CICC 93



Ref: Sansen: Analog
design essential. P2043-
p2048



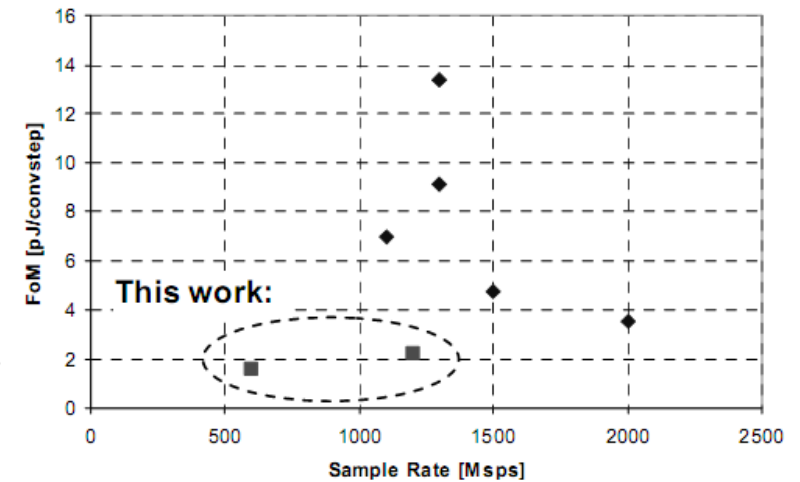
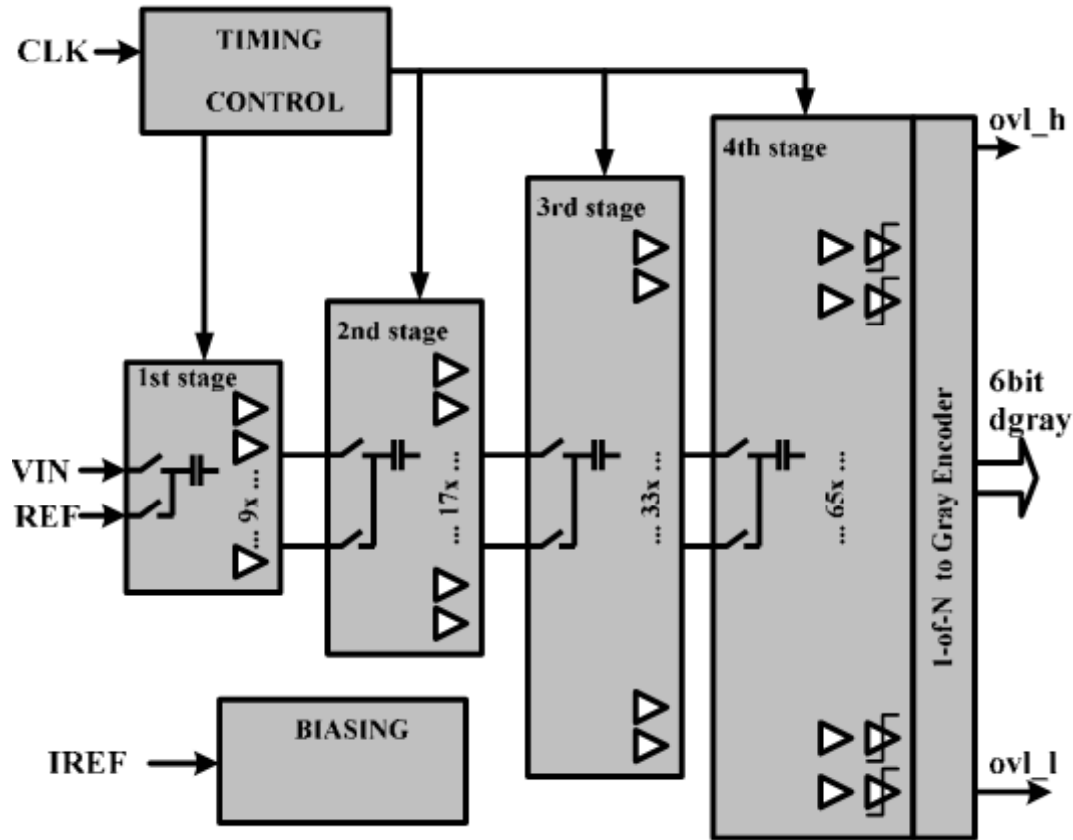
电容内插



- No over-range comparators
- No external sample-and-hold

Christoph Sandner, et. al. A 6bit, 1.2GSps Low-Power Flash-ADC in 0.13um Digital CMOS, DATE05

电容内插



Comparison to state-of-the-art 6b Flash ADCs

$$FoM = \frac{Power}{2^{ENOB,DC} \cdot 2 \cdot ERBW} [pJ/convstep]$$

Christoph Sandner, et. al. A 6bit, 1.2GSps Low-Power Flash-ADC in 0.13um Digital CMOS, DATE05

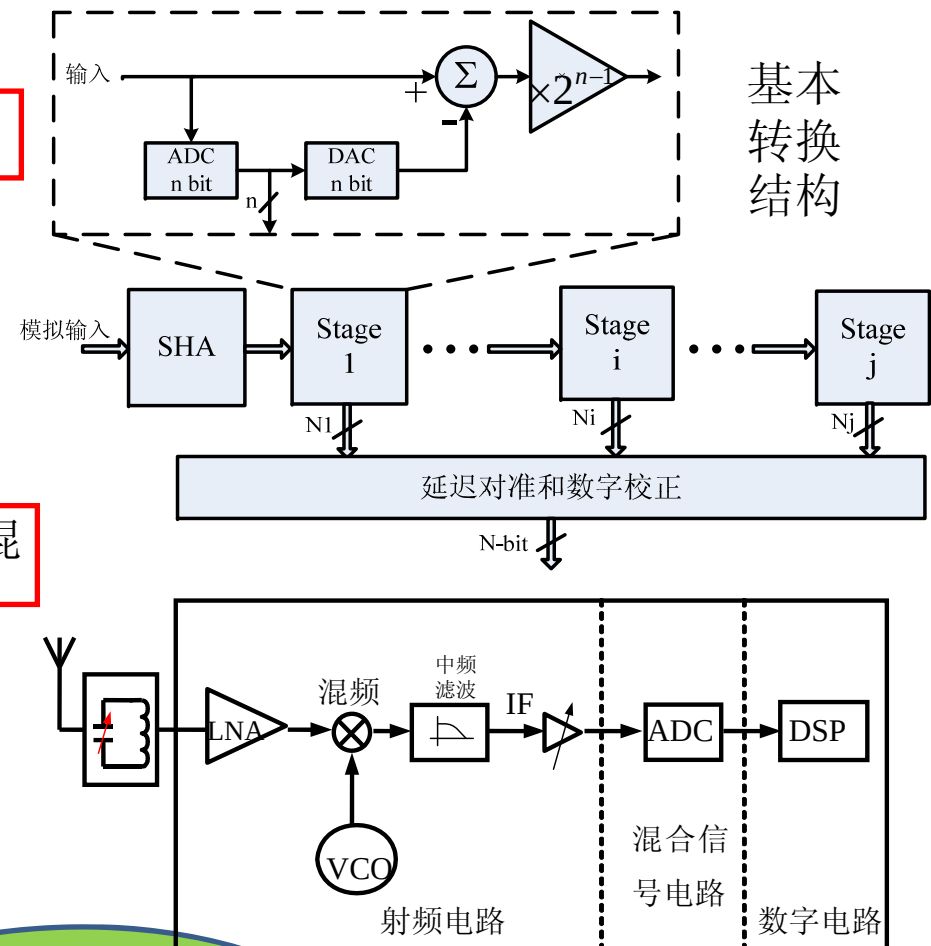
Lecture 2

ADC/DAC Structures

- Flash ADC
- **Pipeline ADC**
- SAR ADC
- Sigma-Delta Modulator
- Time-interleaved
- R2R DAC
- Current-Steering DAC

Pipeline ADC

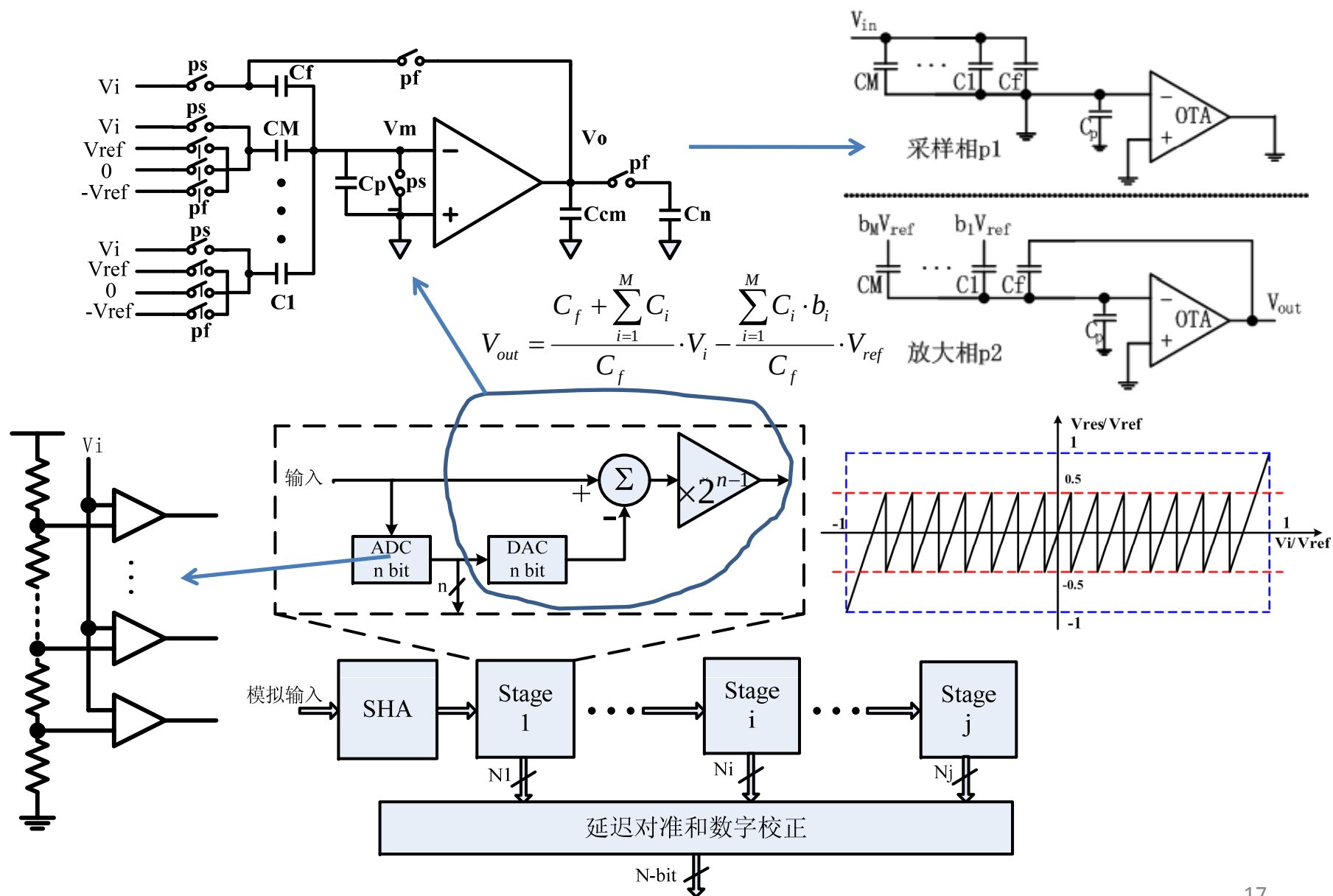
- 对同一采样串行转换，对不同采样 pipeline 并行操作
- 高精度高速结构，仍是12bit, 100MS/s 以上 ADC的首选结构
 - 宽带通信基站、雷达等高端系统
- 基于放大器的结构，在低电压工艺下设计具有挑战性
- 电路非理想因素限制精度
- 数字校准技术、低功耗设计、宽带输入、IF应用等是研究热点
- 高性能ADC本身成为一个复杂的数模混合SOC系统
 - 高性能片上时钟电路(PLL, DLL)
 - 宽带高线性输入buffer
 - 片上滤波与均衡电路
 - 高精度基准与高速参考驱动
 - 宽带低功耗开关电容电路
 - 用于校准的DSP
 - 高速接口(LVDS, SERDES)
 - 通讯控制接口(SPI, I2C)
 - 片上自测试电路



可实现性能
8~16 bit
xM ~ xGS/s

典型商用芯片:
ADI: AD9230
TI: ADS4149

Pipeline ADC Architecture



电路设计问题

- 结构级设计与优化

- SHA less?
- Stage resolution?
- Capacitor size?
- Opamp spec?
- Calibration or not?

孔径误差

带宽功耗

匹配、噪声

线性、非线性误差

Behavior Model
(MATLAB)

Monte-Carlo analysis

- 电路级设计与优化

- High performance building blocks
- Clock, Reference, I/O
- 可编程可配置

典型开关电容电路

Transistor level Simulation
(Spice, Spectre)

PVT Corner Trans.分析
很重要

- 版图设计

- 匹配设计
- 寄生优化
- 电源规划、干扰的抑制与隔离
- 面积优化

单元整数比、共心、加 dummy、差分规范、关键点保护、衬底隔离、寄生分析。。。

反复寄生提取与优化

Pipeline ADC: 向更高性能进军

- 应用需求驱动

- 新一代移动通信 (3G、LTE)
- 雷达系统
- 视频监控

TD-SCDMA BaseStation:

12bit, 80MSps,
SFDR=100dBc@200MHz

TD LTE BaseStation:

12bit, 250MSps, SFDR=82dBc@70MHz

- 高性能数据转换技术发展趋势

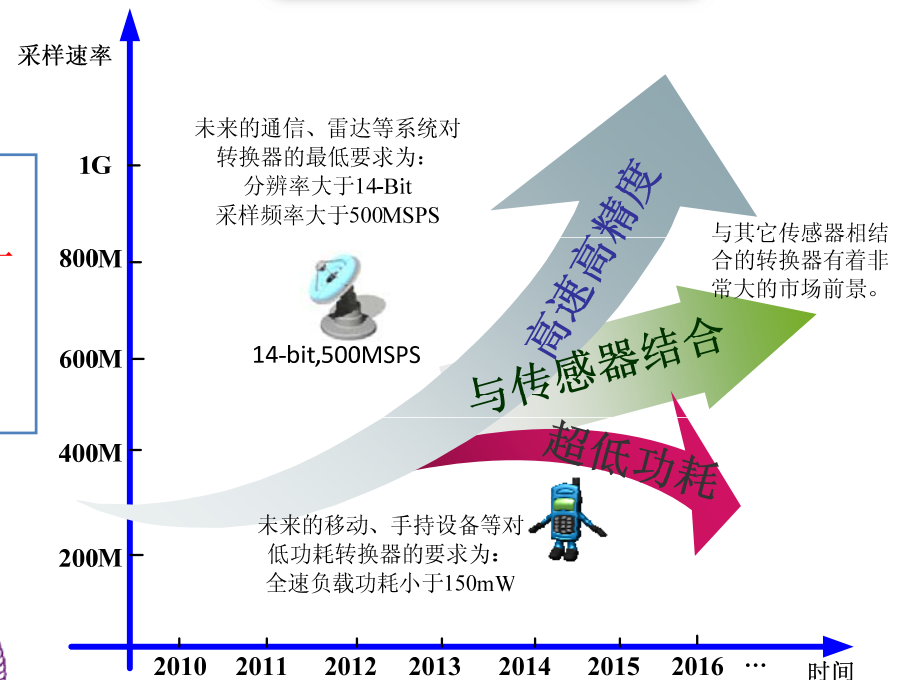
- 高速、高精度、低功耗实现
- 采用数字辅助技术
- 宽带化, 欠采样
- 多模块集成(AFE)

- 典型技术要求

- 分辨率: $\geq 12\text{bit}$;
- 速度: $\geq 80\text{MSps}$
- 输入带宽: $> 100\text{MHz}$
- 支持欠采样

ISSCC'09:
数字辅助设计
ISSCC'10:
高性能转换

中频数字化



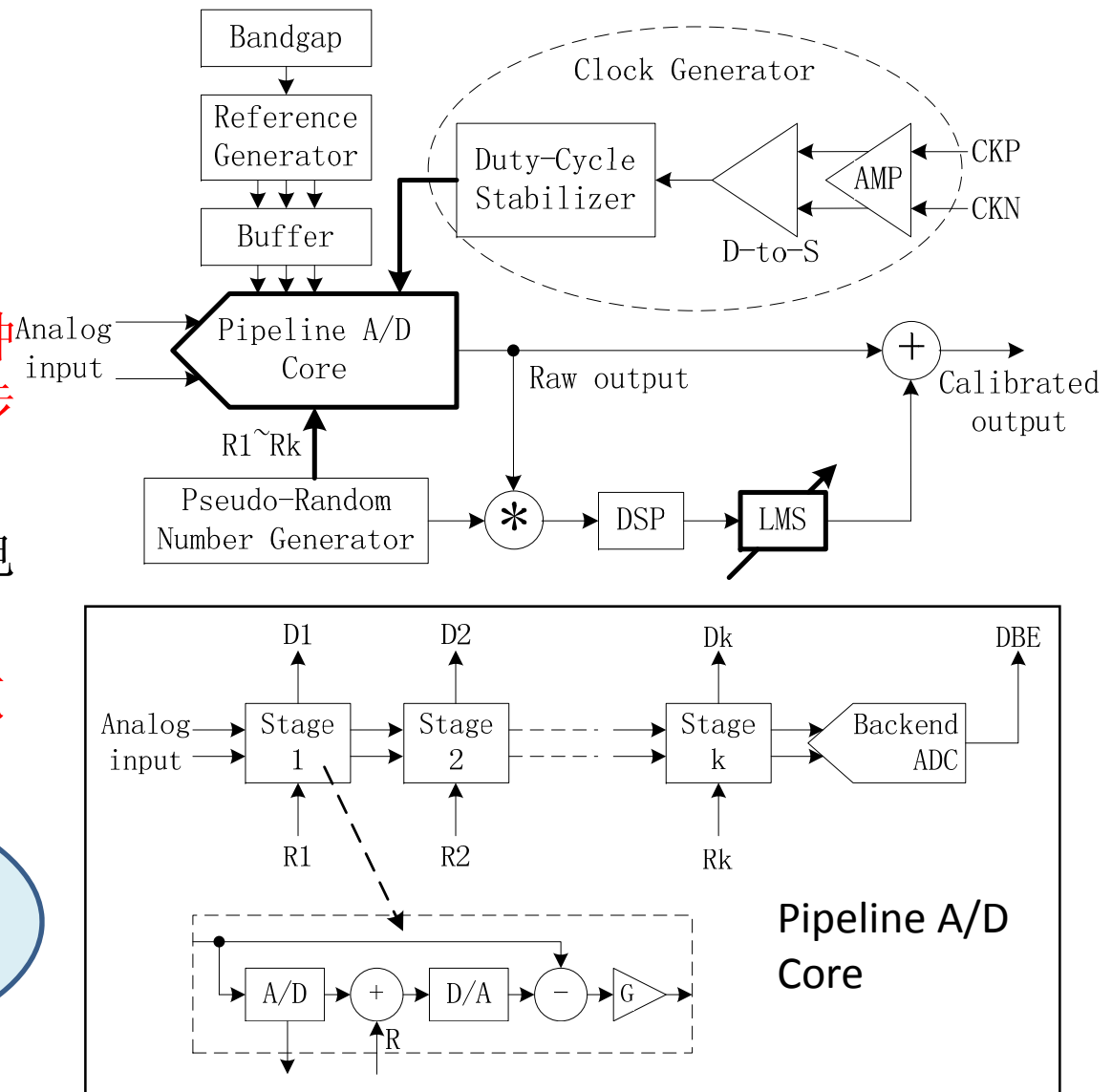
高性能转换的挑战性

难点、关键点	解决方案	研究点
低抖动时钟	低噪声放大整形、占空比稳定、 DLL 去偏移、片上 PLL 倍频	高性能低功耗 DLL, PLL
宽带、高精度采样	RCR滤波、定制I/O、自举开关、输入Buffer(BiCMOS?)	开关优化；输入电容优化；非线性效应抑制；
高精度片上参考及驱动	高稳定、低1/f Noise的Bandgap，高速高效Buffer电路	低功耗高速Buffer；降低buffer要求的结构
高速、高精度转换	Pipeline结构、Current-Steering结构、数字校准	数字校准计算量和收敛速度优化；新方法
低功耗设计	SHA-less, Scaling down技术；opa-sharing；算法优化	孔径误差抑制技术；算法优化
高性能转换器测试	标准高速接口，集成片上测试电路	片上自测试
抗工艺涨落效应	合理的版图布局设计裕度	版图布图技术
数模混合集成多模块集成	合理的电源规划与版图隔离，Deep Nwell隔离	

例：高性能ADC实施方案

- Pipeline结构
- SHA-less和scaling down
- 数字校准
- 低抖动采样时钟、时钟占空比稳定、时钟同步电路
- 片上基准、参考电压电路和高速参考buffer
- 采用0.18um CMOS工艺

0.18um, 1.8V仍是
速度、动态范围、
成本的最佳结合
点



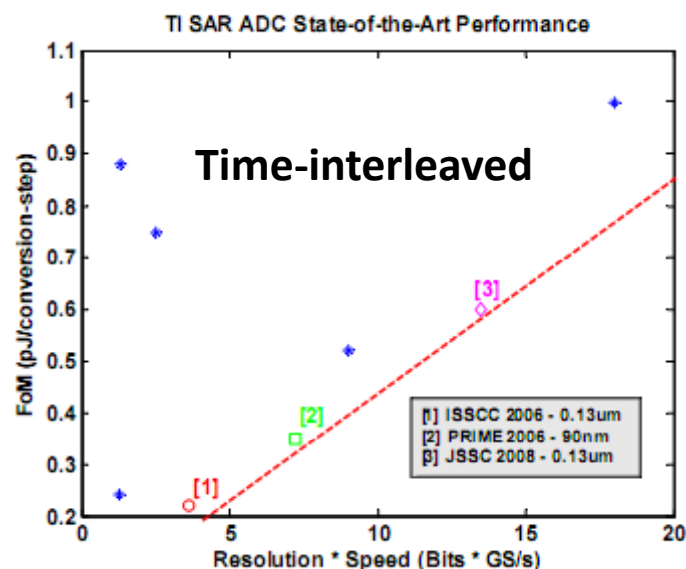
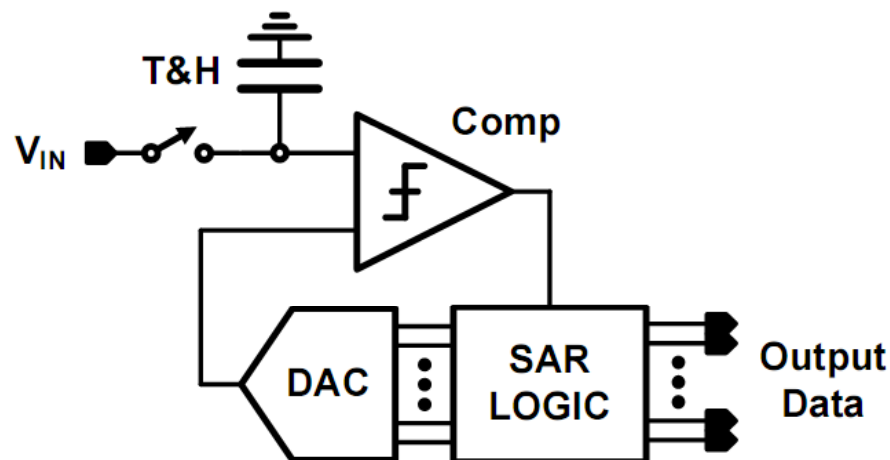
Lecture 2

ADC/DAC Structures

- Flash ADC
- Pipeline ADC
- **SAR ADC**
- Sigma-Delta Modulator
- Time-interleaved
- R2R DAC
- Current-Steering DAC

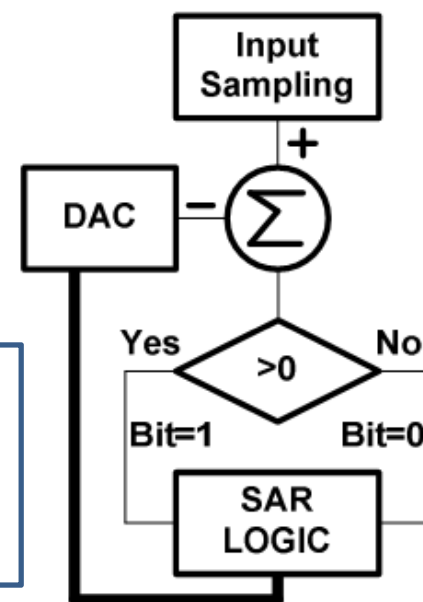
SAR ADC Principle

- 以D/A来实现A/D, 逐次逼近
- 需要N次D/A和比较实现1次N位A/D转换
- 精度主要由DAC决定
- 无运放, 低电压、低功耗
- 深亚微米CMOS工艺下很有发展潜力的结构
- 超低功耗, 高速转换是研究热点
 - 异步时序控制

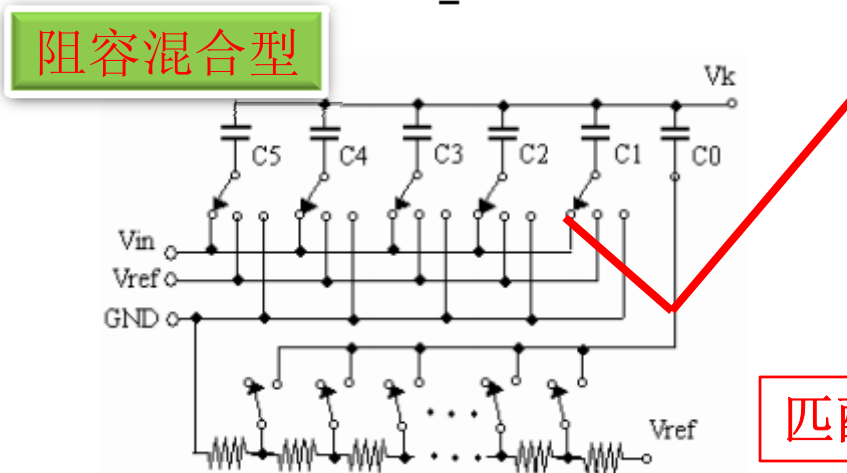
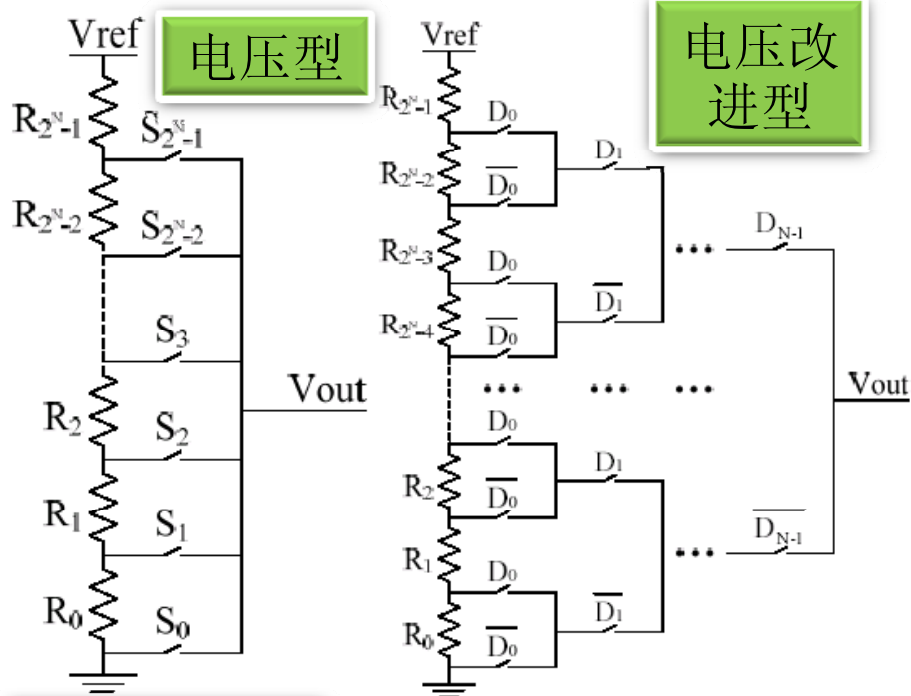


可实现性能
8~16 bit
 $x_k \sim x_{00M} \text{ S/s}$

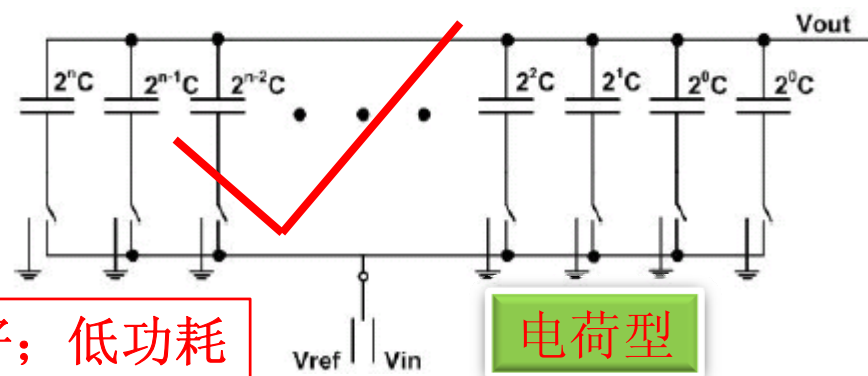
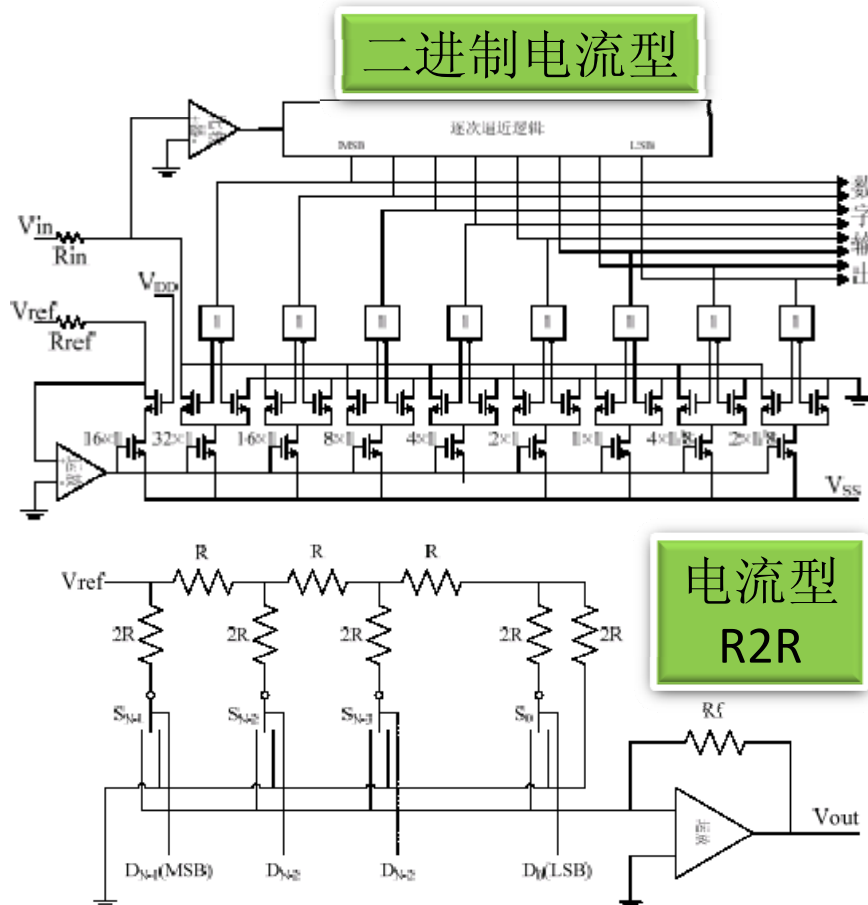
9-b, 50MS/s, 65fJ/conv.
9-b, 40MS/s, 54fJ/conv.
10-b, 10MS/s, 11fJ/conv.
No Time-interleaved!



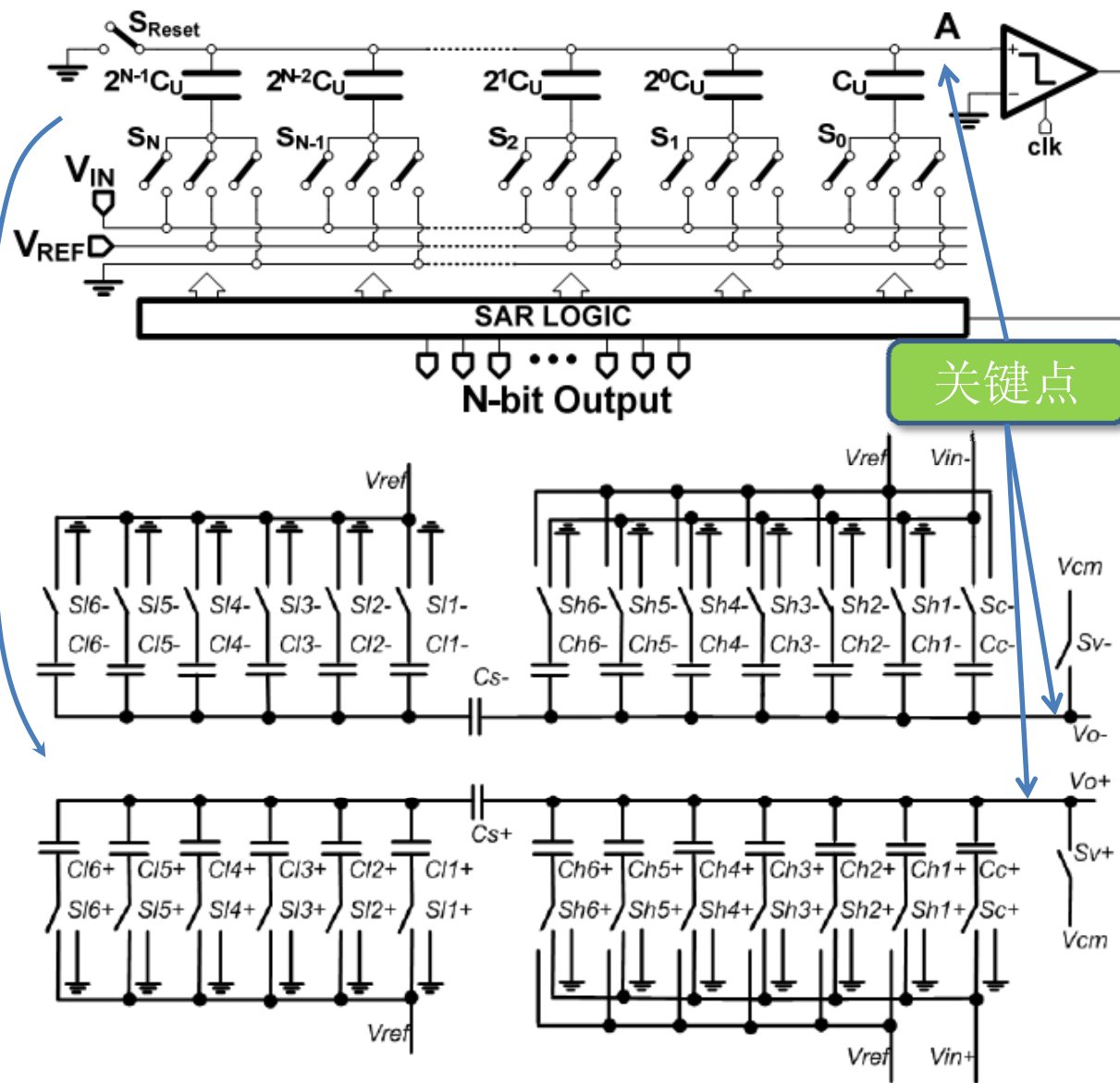
DAC topologies



匹配好；低功耗



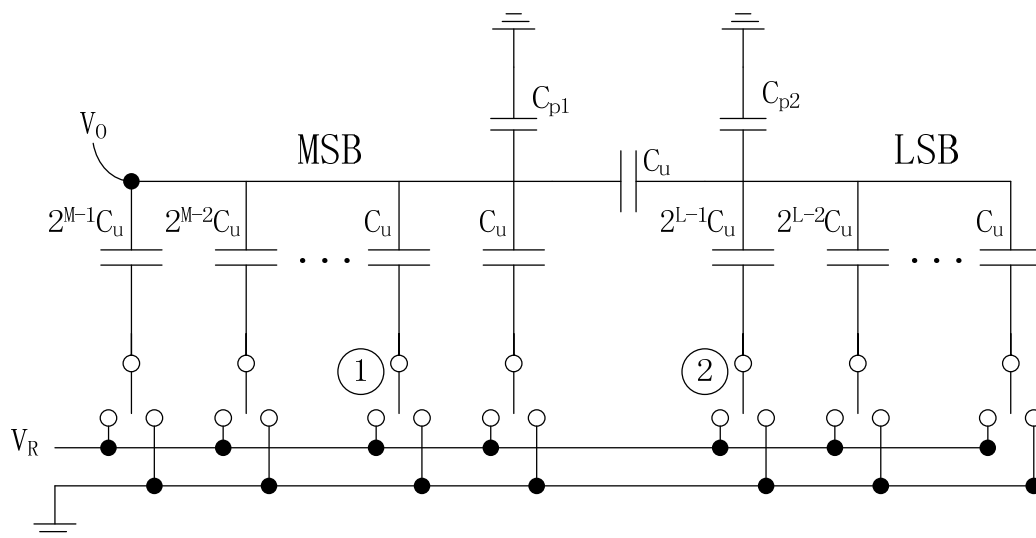
电荷型DAC



关键点

- 特点:
- 集成T/H电路
- 与输入相连的开关较多
- 输入电容较大
- 采用分段结构可减少电容数目
- 电容大小是精度与面积功耗的权衡，可通过 **mento-carlo** 仿真确定
- 高位电容可采用 **DEM** 技术进一步提高精度
- 对高精度转换，输入开关 **Ron** 线性须保证
- bootstrap**
- 高位电容可采用单元温度码控制，以减小输入端毛刺，避免电荷泄漏；以及确保单调性
- 版图关键点:
 - **DAC output**

寄生电容问题



- 考虑版图设计，跨接电容可取单位电容值，这只带来1LSB的增益偏差
- 分段结构与原二进制结构的INL是一样的
- C_{p1}不会带来非线性问题；
- C_{p2}可能导致LSB段的增益误差，带来非线性问题
- LSB段位数L小，则寄生电容影响小
- 可考虑优化电容顶板寄生电容
 - MOM电容画法

在①和②点分别输入幅度V_R的阶跃，其对DAC输出V_O的改变量分别为：

$$\Delta V_{O1} = \frac{2^L + \frac{C_{p2}}{C_u}}{2^M \cdot 2^L - 1 + (2^M + 1) \cdot \frac{C_{p2}}{C_u}} \cdot V_R$$

$$\Delta V_{O2} = \frac{2^{L-1}}{2^M \cdot 2^L - 1 + (2^M + 1) \cdot \frac{C_{p2}}{C_u}} \cdot V_R$$

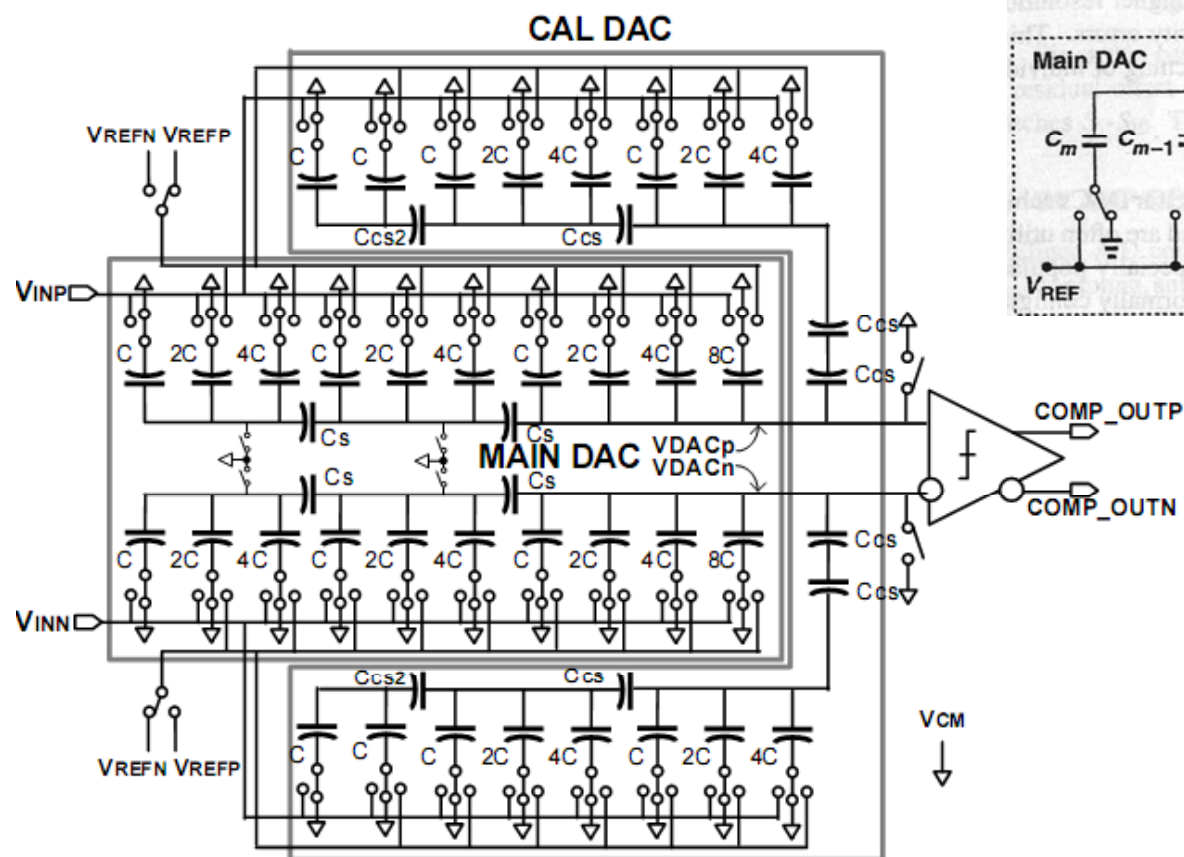
$$g_e = \frac{\Delta V_{O1}}{2\Delta V_{O2}} - 1 = \frac{C_{p2}}{C_u} \cdot \frac{1}{2^L}$$

$$C_{p1} = \alpha C_u + \beta 2^M C_u$$

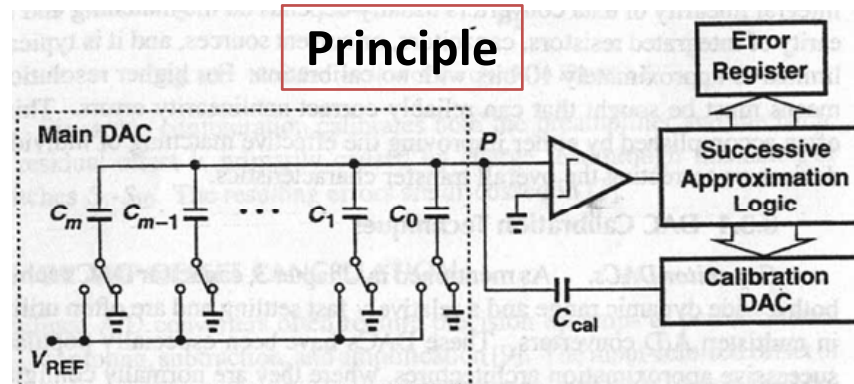
$$C_{p2} = \beta C_u + \beta(2^L - 1)C_u = \beta 2^L C_u$$

$$g_e = \beta$$

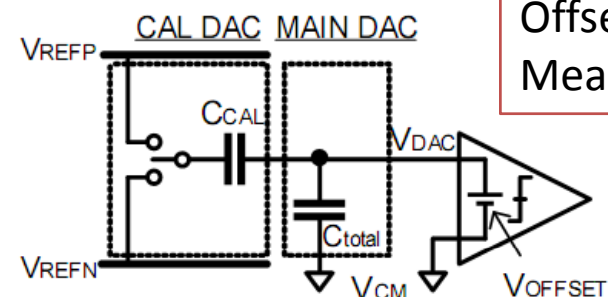
电容失配校准



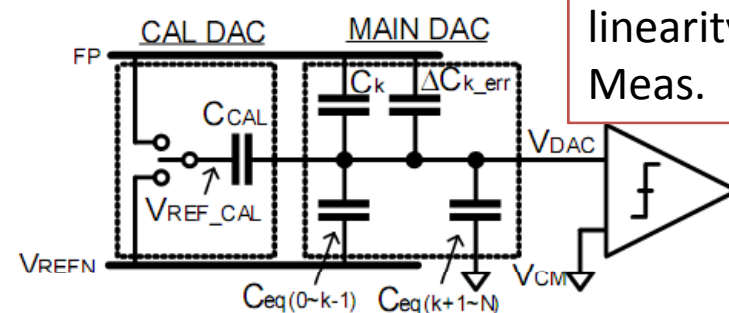
Principle



Offset Meas.

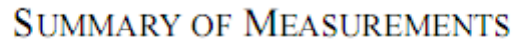


linearity Meas.

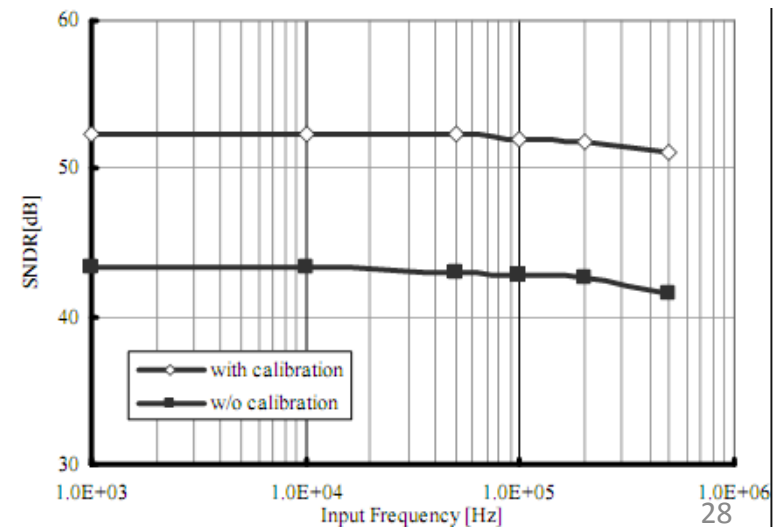


Ref: Y. Kuramochi, et, al. A 0.05-mm² 110-uW 10-b Self-Calibrating Successive Approximation ADC Core in 0.18-um CMOS, ASSC 2007

校准带来电容缩小

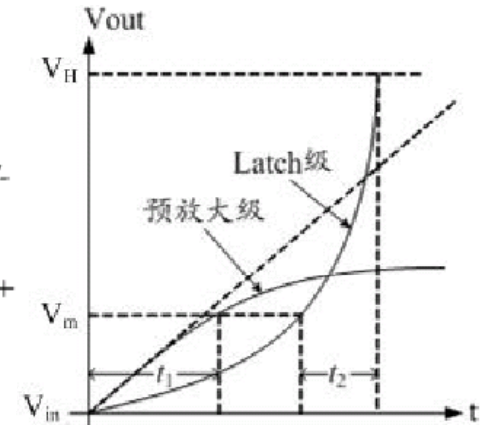
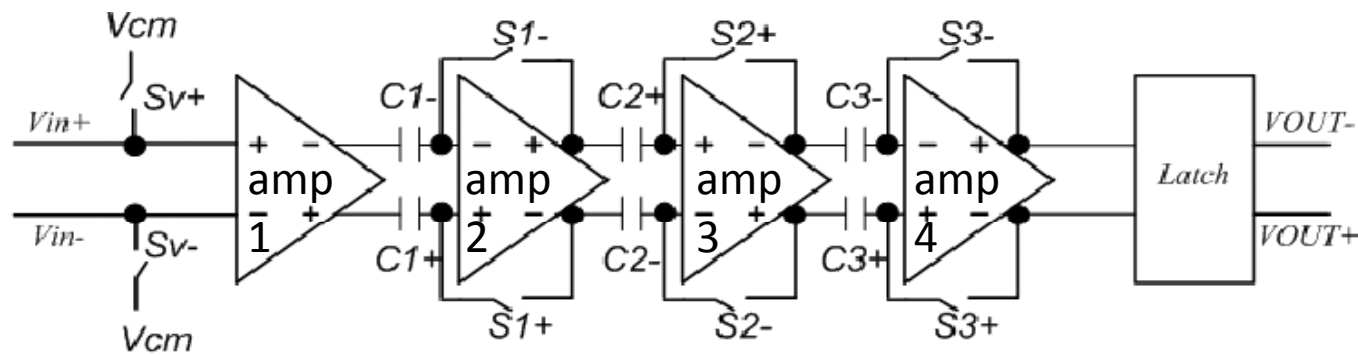


Process	0.18 μ m, 1 poly, 6metal CMOS
Resolution	10bits
Active Area	95 μ m x 550 μ m
Sampling Rate	1MSps
Analog Power	110 μ W @1.8V
SNDR @nyquist	51.1dB
SFDR @nyquist	69.8dB



Ref: Y. Kuramochi, ASSC 2007

Requirements On Comparator?

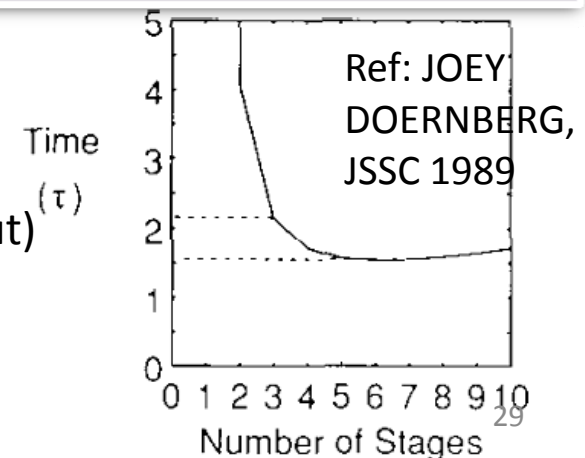


$$V_{os} \approx \frac{V_{os1}}{A_0 A_1} + \frac{V_{os2}}{A_0 A_1 A_2} + \frac{V_{os3}}{A_0 A_1 A_2 A_3} + \frac{V_{osLatch}}{A_0 A_1 A_2 A_3} + \frac{\Delta q_1}{A_0 C_0} + \frac{\Delta q_2}{A_0 A_1 C_0} + \frac{\Delta q_3}{A_0 A_1 A_2 C_0}$$

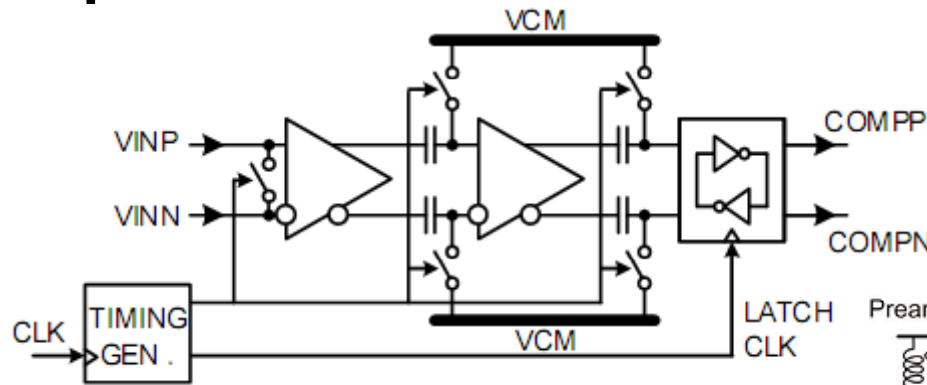
- High Speed
 - Latch尺寸小，但导致offset分布扩大
 - 前置放大器的带宽
- Low offset
 - 前置放大器的增益
 - Offset Cancellation
- Low Noise
 - 主要是第一级放大器amp1(1/f, thermal noise)
- Low kickback noise
 - Latch设计要合理(for example: Cascode, cap at input)
 - 时序设计控制(especially at latch on-to-off)
- Low Power
 - 比较器是构成SAR ADC功耗的主要单元

Offset不影响总体线性度，但考虑到转换精度，以及latch迟滞效应、噪声和分辨力，其前置放大器增益需足够

放大器级数要根据延时、增益、功耗来权衡选择



Comparator offset calibration



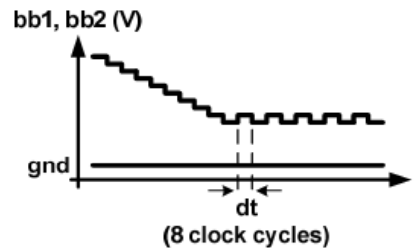
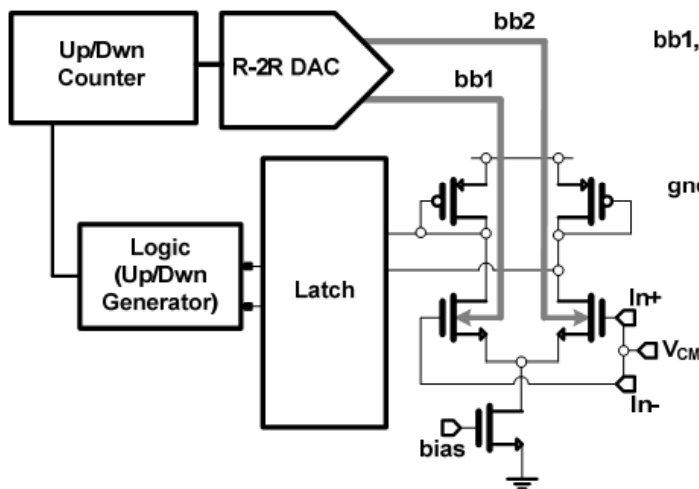
利用电容存储和抵消offset，是最常用的办法

Ref: Y. Kuramochi, ASSC 2007

在preamp输出加入电流可以矫正offset，但同时引入寄生，会降低速度；

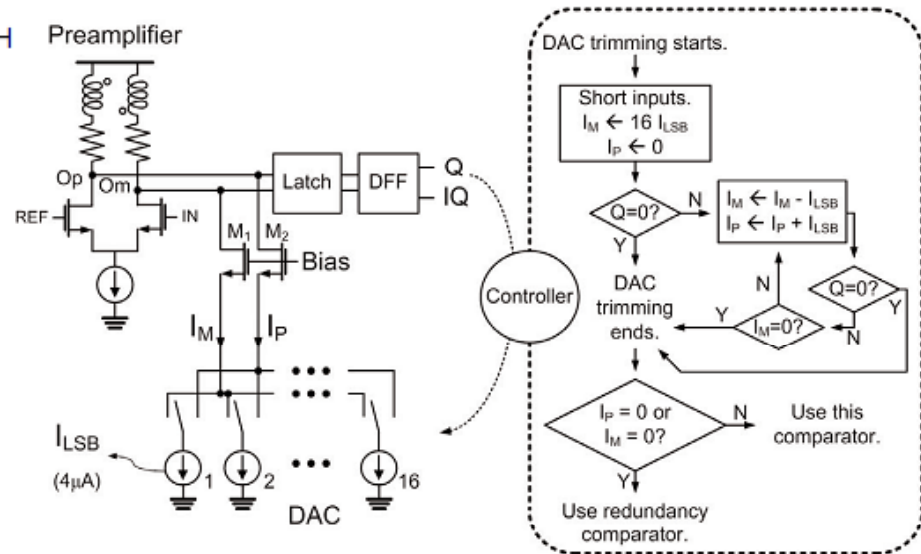
冗余设计也是提高yield的办法

Ref: S. Park, et, al. A 4GS/s 4b flash ADC in 0.18um CMOS. ISSCC 2006



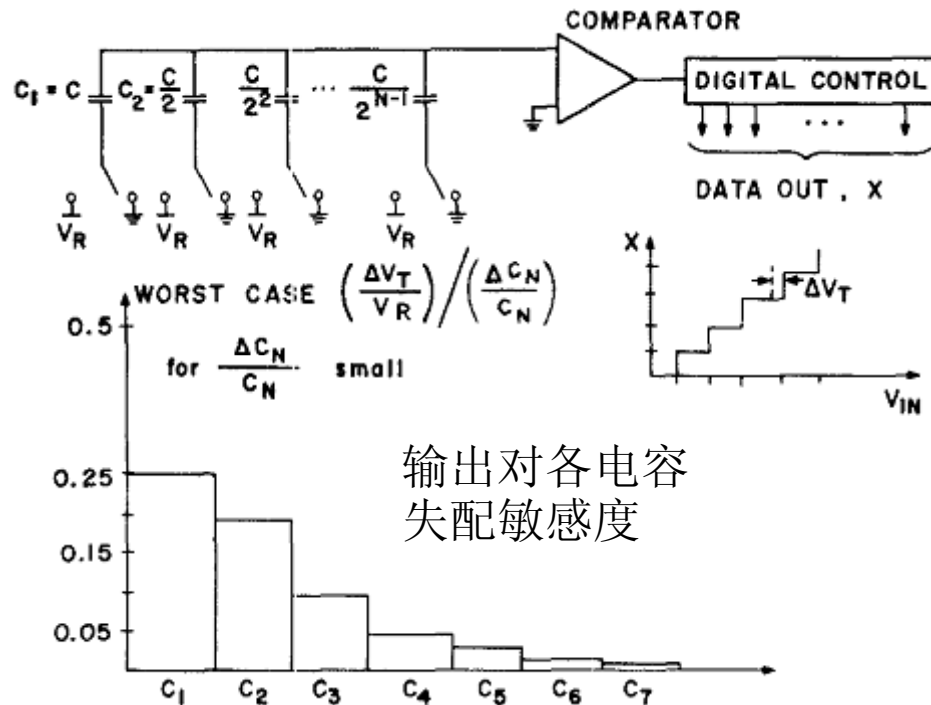
bb1: body bias 1
bb2: body bias 2

Ref:Alpman, Erkan. A 7-BIT 2.5GS/sec TIME-INTERLEAVED C-2C SAR ADC FOR 60GHz MULTI-BAND OFDM-BASED RECEIVERS. PHD Thesis



通过衬偏效应来调整输入管VT，以此来校准offset

电容失配



Ref: JAMES L. McCREARY, et, al. ALL-MOS Charge Redistribution Analog-to-Digital Conversion Techniques I. JSSC 1975

- 显然，MSB段可采用温度码控制来降低DNL
- 对于INL，MSB段采用温度码与否都一样
- 对于匹配与INL Yield的关系实际上与Current-steering DAC的分析是一样的

$$\Delta C = \sqrt{2^N} \sigma(C_u)$$

$$\frac{\Delta C}{C} = \frac{\sqrt{2^N} \sigma(C_u)}{2^N C_u} = \frac{1}{\sqrt{2^N}} \cdot \frac{\sigma(C_u)}{C_u}$$

对于分段结构：

$$\frac{\Delta C_M}{C_M} = \frac{1}{\sqrt{2^M}} \cdot \frac{\sigma(C_u)}{C_u}$$

$$\Delta V_T = 0.25 \cdot V_R \cdot \frac{\Delta C_M}{C_M} = \frac{2^{N-2}}{\sqrt{2^{M-1}}} \cdot \frac{\sigma(C_u)}{C_u} \text{ LSB}$$

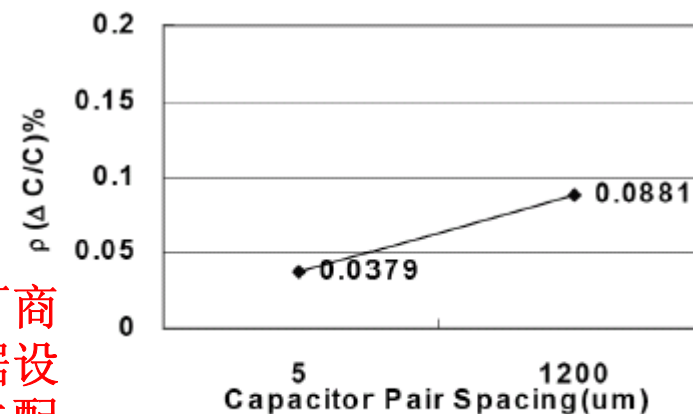
代表DNL

$$\frac{\sigma(C_u)}{C_u} = \frac{1}{2\alpha\sqrt{2^N}}$$

例如： $\alpha \sim 6.2e-4$
for
INL_yield=90%

电容取值

电容面积 (um ²)	电容失配 $\rho (\Delta C/C) \%$
10×10	0.0933
15×15	0.0646
20×20	0.0658
25×25	0.0402
30×30	0.0379
35×35	0.0178
100×100	0.0055



根据工艺厂商提供的数据设定容值和失配

10u*10u: 0.11%

调整容值

系统级模型
Mento-Carlo分析

N

满足要求?

若由噪声决定:

$$V_n^2 = \frac{kT}{2^{N/2} C_U}$$

$$C_U > \frac{12kT}{V_R^2} 2^{2N-N/2}$$

10u*10u的分析结果

满足12bit要求

DNL		INL		SNDR	
$\leq 0.5\text{LSB}$	$\leq 1\text{LSB}$	$\leq 0.4\text{LSB}$	$\leq 1\text{LSB}$	$< 70\text{dB}$	$\geq 70\text{dB}$
86.21%	99.54%	85.4%	100%	4.5%	95.5%

Asynchronous SAR ADC

- 同步采样，异步转换
- 控制电路需要的脉冲由电路自动产生
- 好处：假设10bit 100MSPS，则对于

同步SAR时钟频率至少1G

比较器时间： $T_{\text{cmp}} = K \cdot \ln \frac{V_{\text{FS}}}{V_{\text{res}}}$

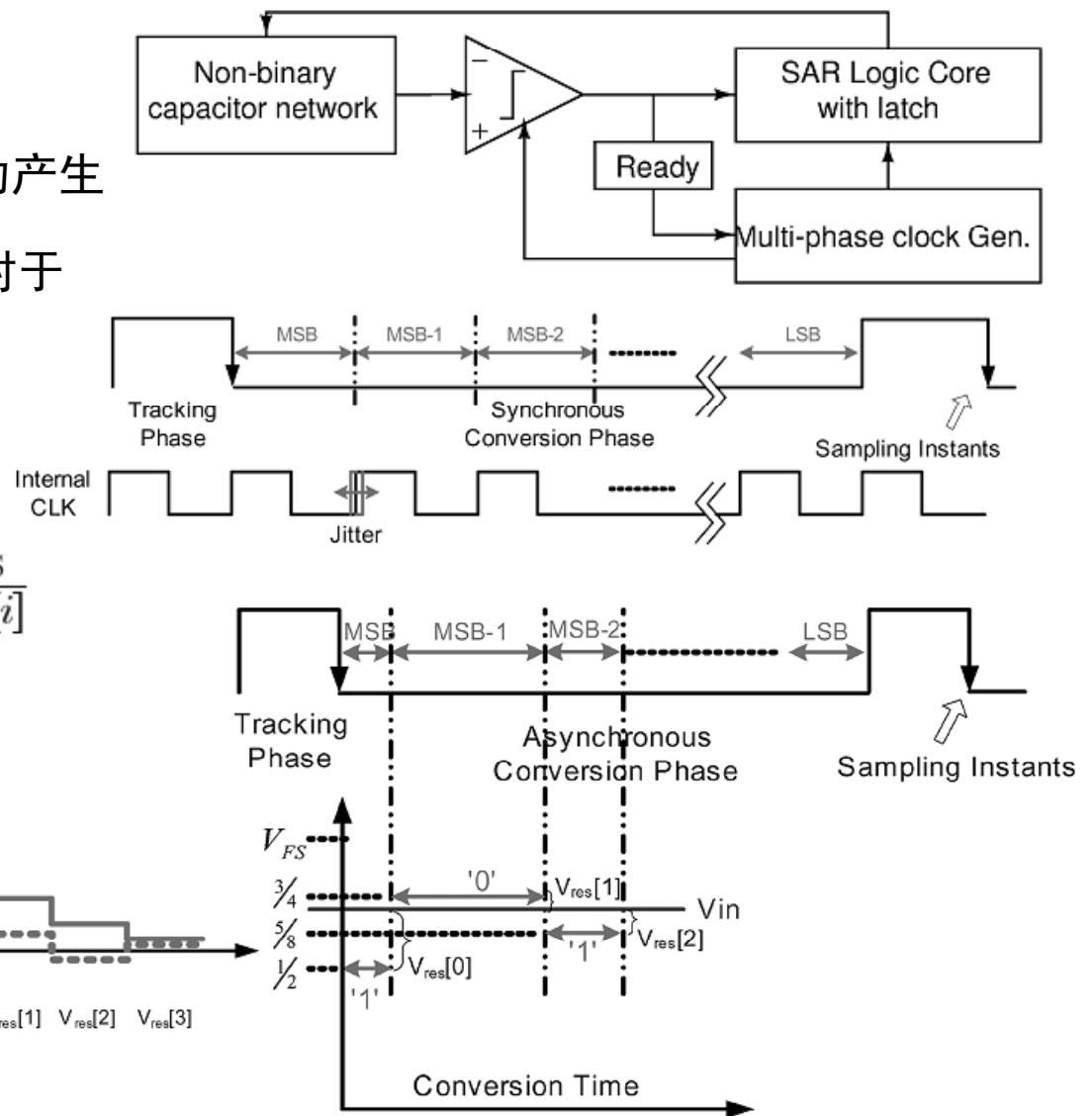
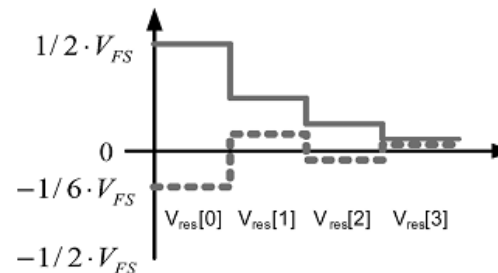
$$T_{\text{async}} = \sum_{i=0}^{N-1} K \cdot \ln \frac{V_{\text{FS}}}{V_{\text{res}}[i]}$$

$$T_{\text{sync}} = N \cdot K \cdot \ln \frac{V_{\text{FS}}}{V_{\text{min}}}$$

当N比较大时，有：

$$\left(\frac{T_{\text{async}}}{T_{\text{sync}}} \right)_{\text{max}} \approx \frac{1}{2}$$

$$\left(\frac{T_{\text{async}}}{T_{\text{sync}}} \right)_{\text{min}} \approx \frac{1}{2}$$



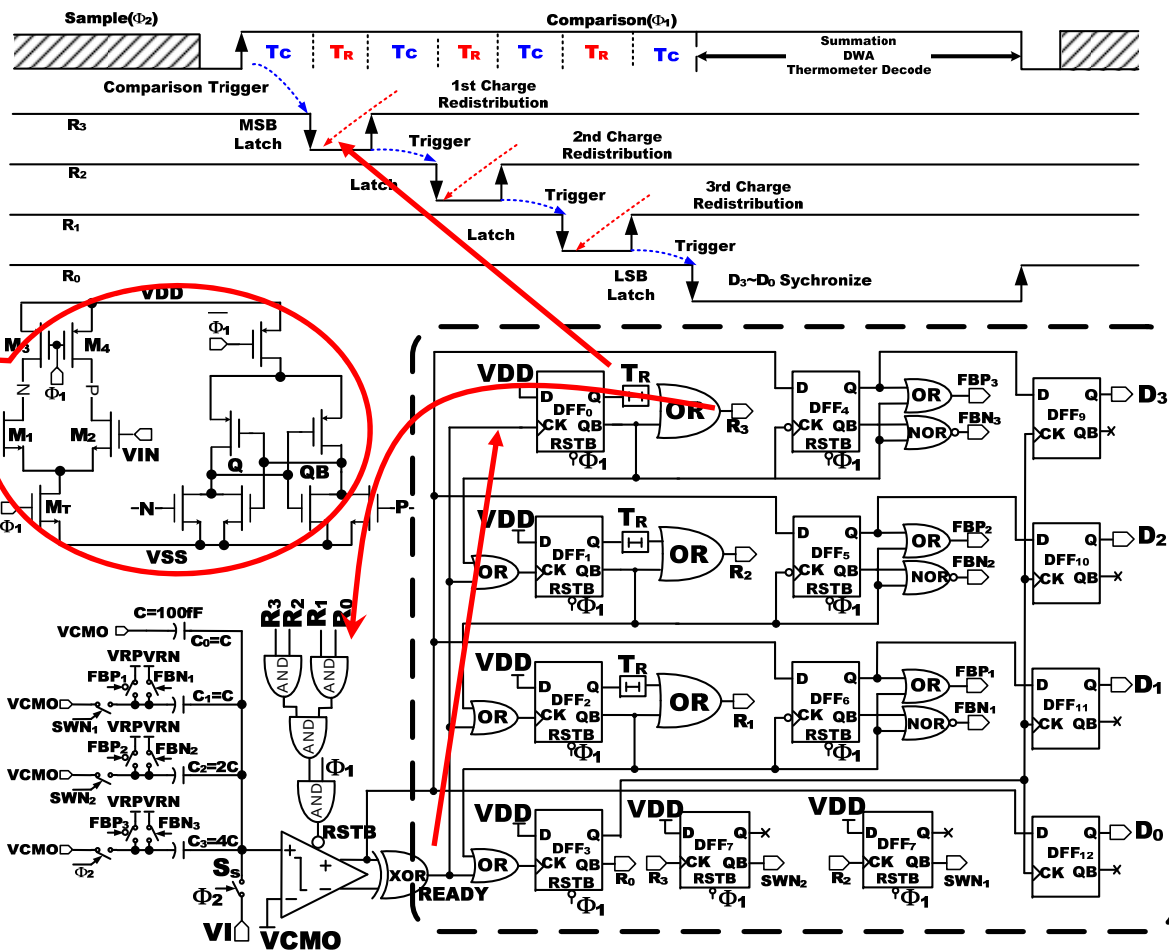
Ref: Shuo-Wei Michael Chen, et, al. A 6-bit 600-MS/s 5.3-mW Asynchronous ADC in 0.13-um CMOS. JSSC 2006

异步 SAR

- 异步时序
 - 比较器比较结束后能自动触发电荷重分配
 - 电荷重分配结束后自动触发下一次比较
- 下面给出一个4bit的设计例子
 - 异步逻辑由一个时钟沿触发
 - 异步控制逻辑自动产生后续的所有脉冲

4-bit设计例子

- SAR结构
 - 电荷重分布型DAC
- 1个动态比较器
- 异步控制逻辑



- 异步逻辑由一个时钟沿触发
- 异步控制逻辑自动产生后续的所有脉冲

Lecture 2

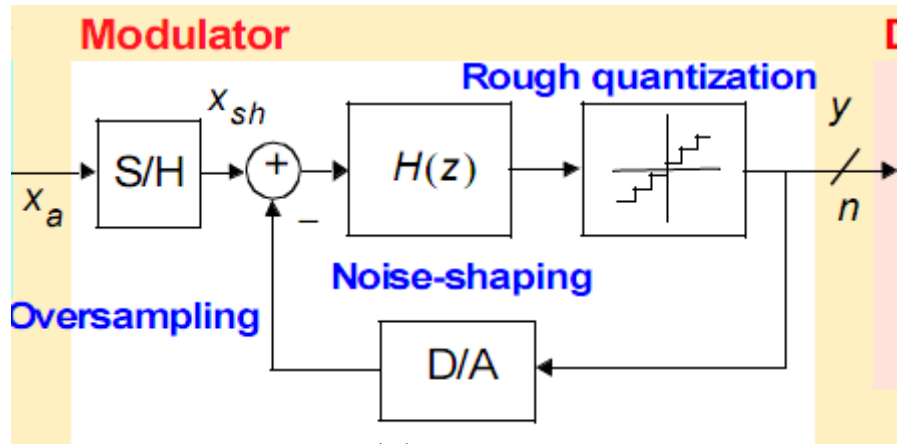
ADC/DAC Structures

- Flash ADC
- Pipeline ADC
- SAR ADC
- **Sigma-Delta Modulator**
- Time-interleaved
- R2R DAC
- Current-Steering DAC

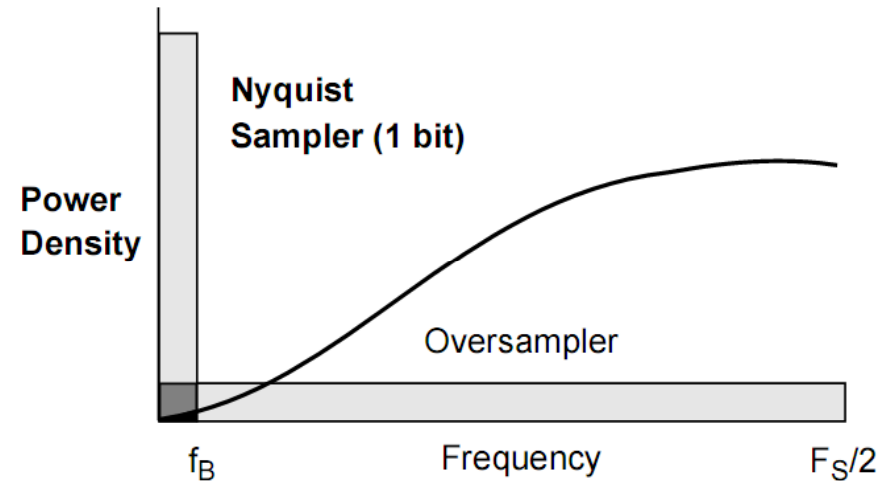
Sigma-Delta Modulator

- Principle
- Architecture
- Circuit non-idealities
- Sigma-Delta Converter
 - Example: Audio DAC Design

Principle



$$Y(z) = \frac{H(z)}{1 + H(z)} X(z) + \frac{1}{1 + H(z)} Q(z)$$



$$S_{NR_{\max}} = 10 \log_{10} \left[\frac{3\pi}{2} \cdot (2L + 1) \cdot \left(\frac{M}{\pi} \right)^{2L+1} \right]$$

- Oversampling
- Noise-shaping
- Low pass filtering

三部曲☺

For High SNR

- High order
 - Single loop
 - MASH
- Multi bit

Sigma-Delta Modulator

- Principle
- Architecture
- Circuit non-idealities
- Sigma-Delta Converter
 - Example: Audio DAC Design

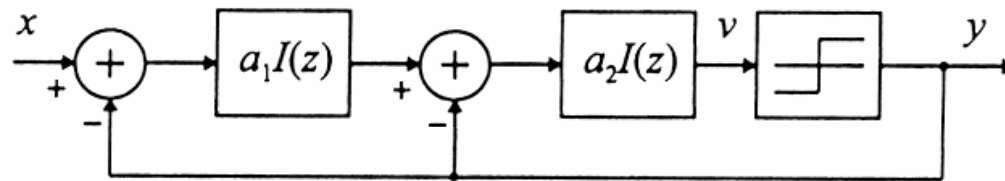
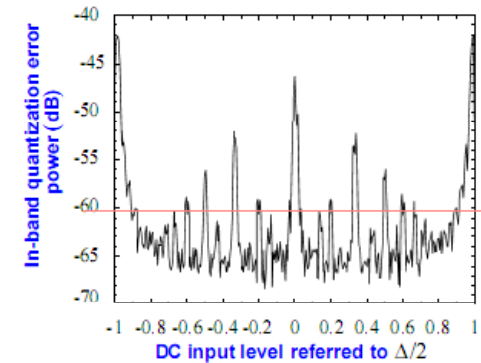
Architectures for high SNR

- Low order, single loop, single bit
 - 高过采样率，稳定，简单
- High order, single loop, single bit
 - 稳定性问题，输入信号幅度受限
- MASH
 - 稳定，对匹配要求高
- Multi bit
 - D/A线性问题，可校准或DEM

Single-loop, single bit

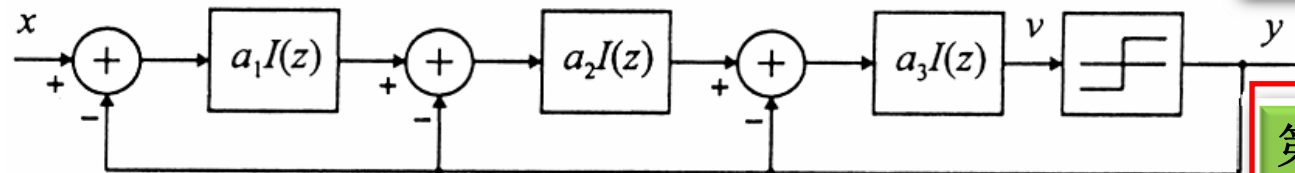
1阶调制器, patterns in spectrum, 不用在第一级

$$Y(z) = z^{-1}X(z) + (1 - z^{-1})E(z)$$



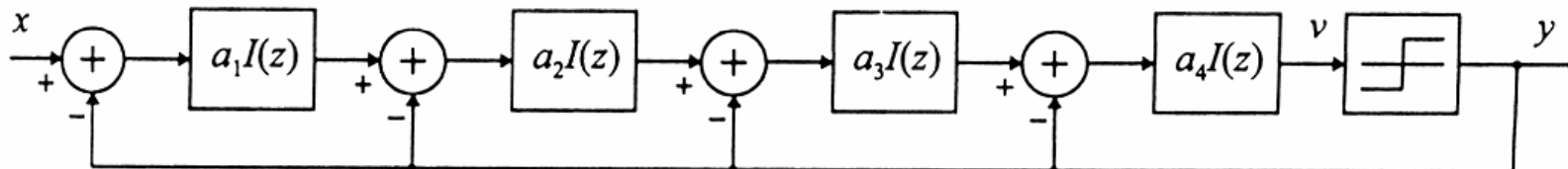
2阶调制器 $Y(z) = z^{-2}X(z) + (1 - z^{-1})^2 E(z)$

2阶、3阶用得比较多, 4阶对稳定性要仔细考虑, 建议用MASH结构来实现



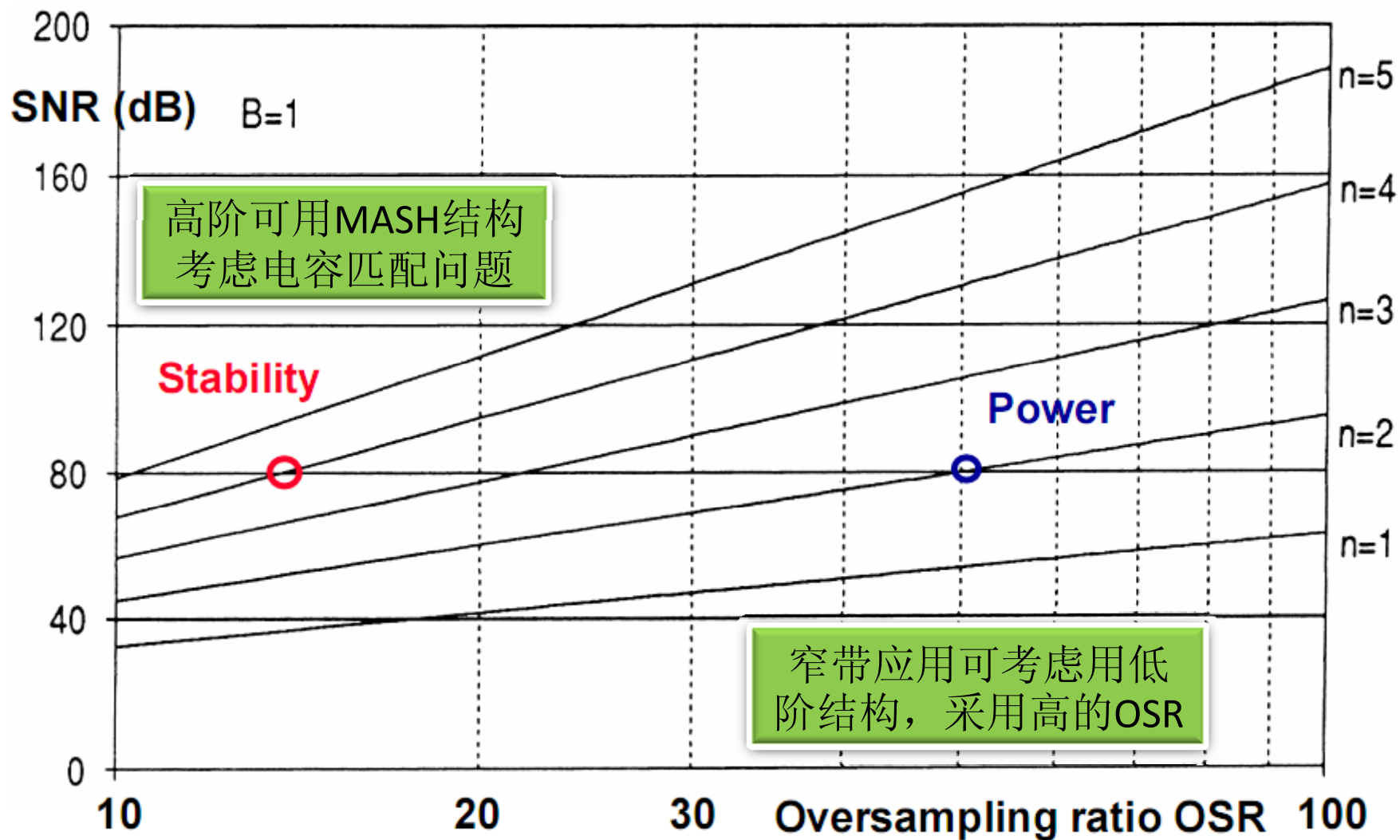
3阶调制器 $Y(z) = z^{-3}X(z) + (1 - z^{-1})^3 E(z)$

第一个积分器是最重要的, 后续积分器误差可被noise shaping



4阶调制器 $Y(z) = z^{-4}X(z) + (1 - z^{-1})^4 E(z)$

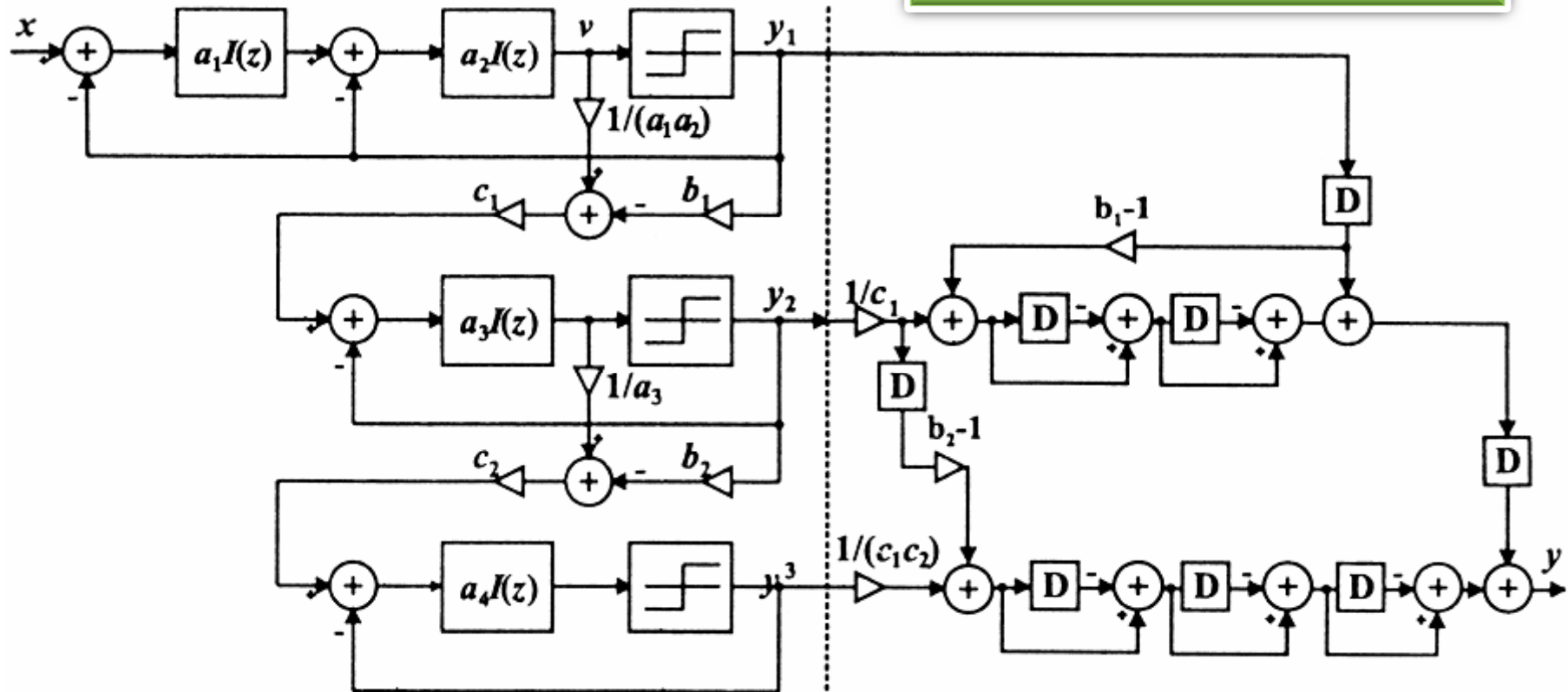
SNR vs. OSR



Ref: Sansen. Analog design essential, chapter 21

MASH (2-1-1)

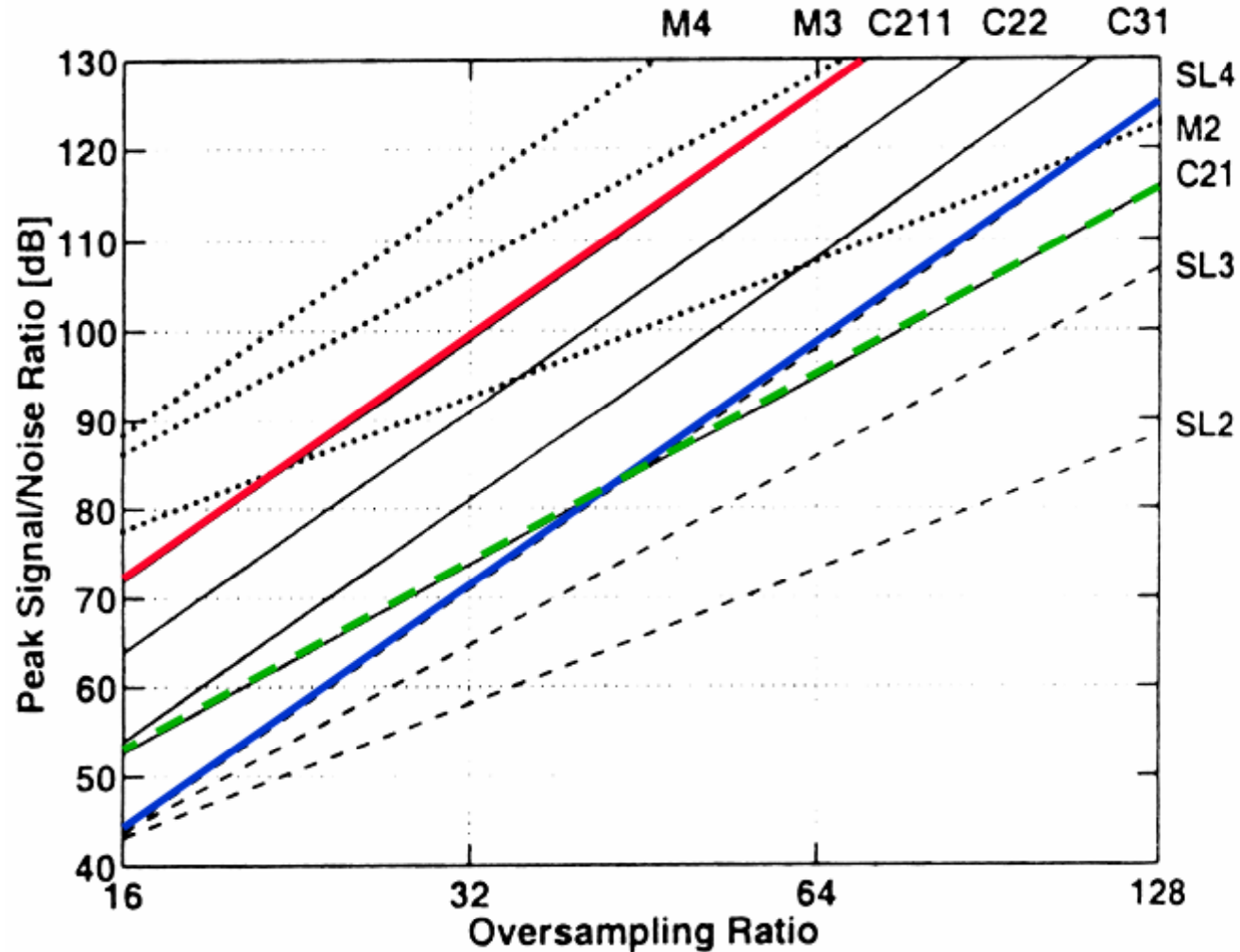
第一积分器是最重要的，后续积分器误差可被noise shaping



$$Y(z) = z^{-4} \cdot X(z) + \frac{(1 - z^{-1})^4 \cdot E_3(z)}{c_1 \cdot c_2}$$

请推导传输函数，并分析电容失配效应？

Multi-bit vs. single-bit



Cascaded 2-1-1

Single loop 4th

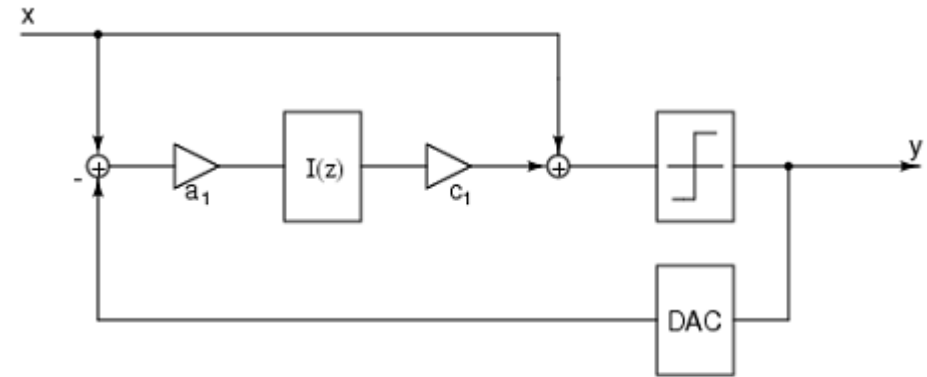
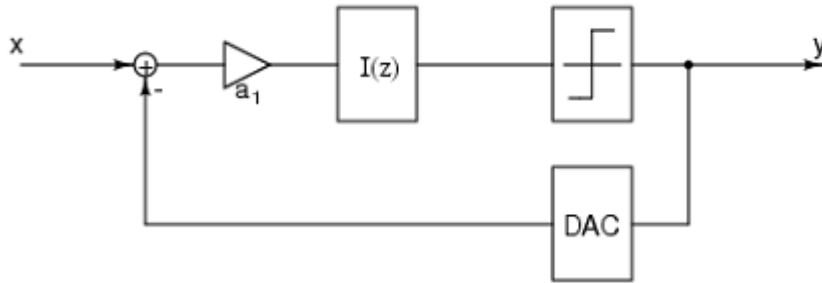
Cascaded 2-1

Multibit (4 bit)

Ref. Marquez
CAS Sept 98,
1232-1241

Ref: Sansen. Analog design essential, chapter 21

Full-feedforward architecture



Convent. Sigma-Delta topology

Full feedforward topology

$$H_x(z) = \frac{a_1 I}{1 + a_1 I}$$

$$H_e(z) = \frac{1}{1 + a_1 I}$$

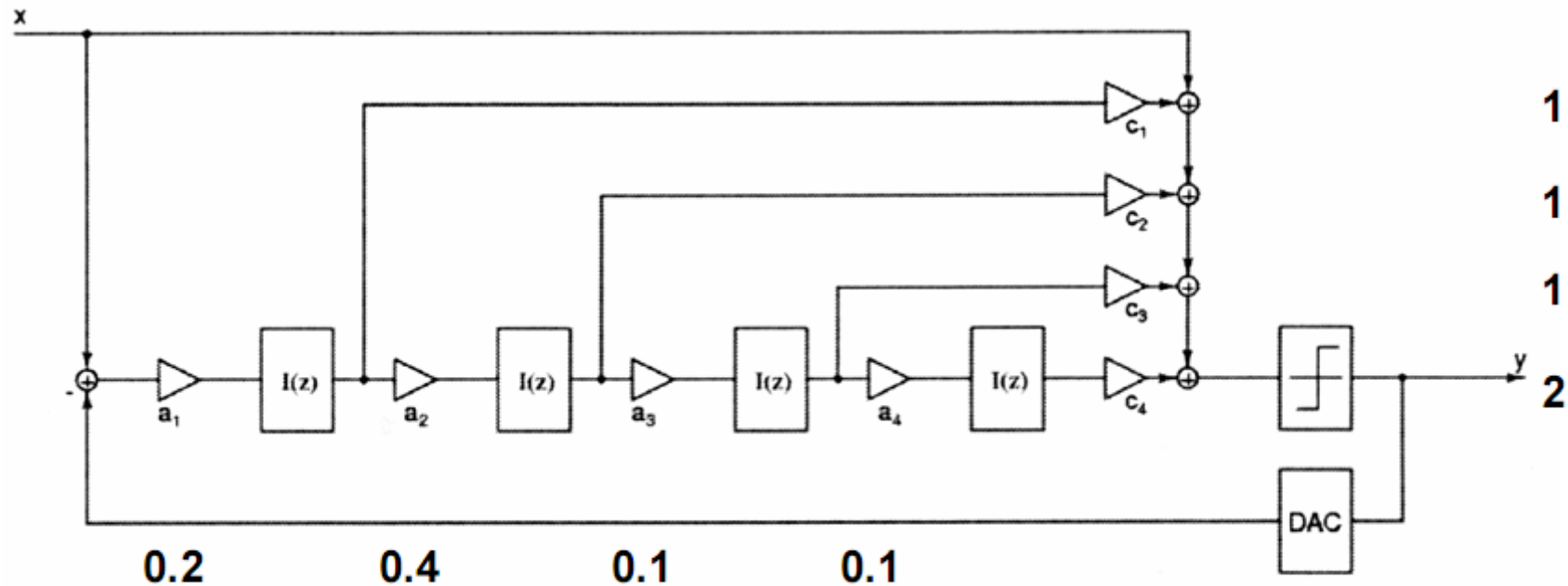
$$H_x(z) = 1$$

$$H_e(z) = \frac{1}{1 + a_1 c_1 I}$$

Silva, Electronic Letters, June 01, 737-738

Ref: Sansen. Analog design essential, chapter 21

4th-Order single-bit 1 MS/s sigma-delta Modulator



Single feedback loop : processes quantization noise only

Full feedforward : unity-gain transfer

4th order - single bit

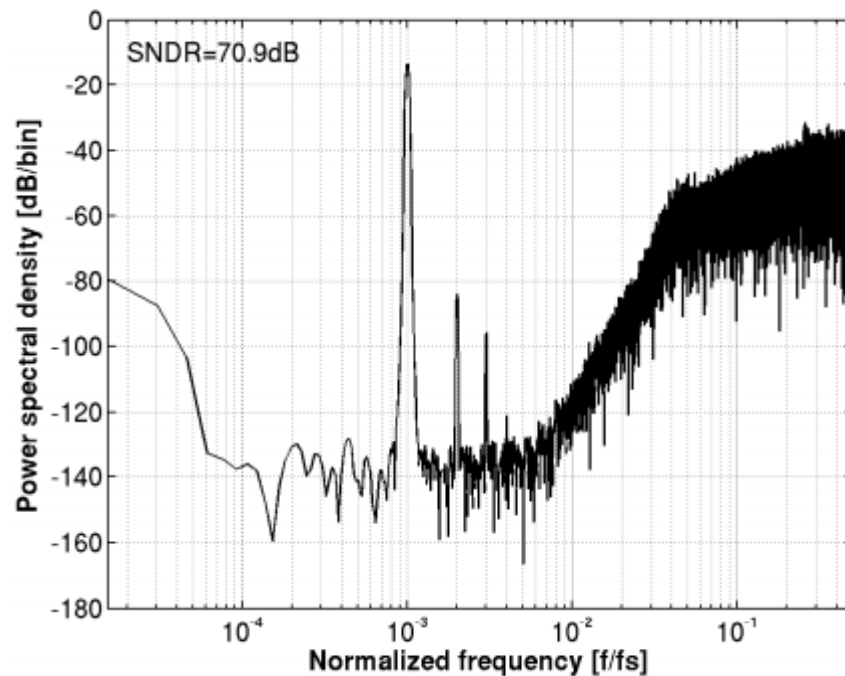
Optimization coefficients or equal swing

Ref: Sansen. Analog design essential, chapter 21

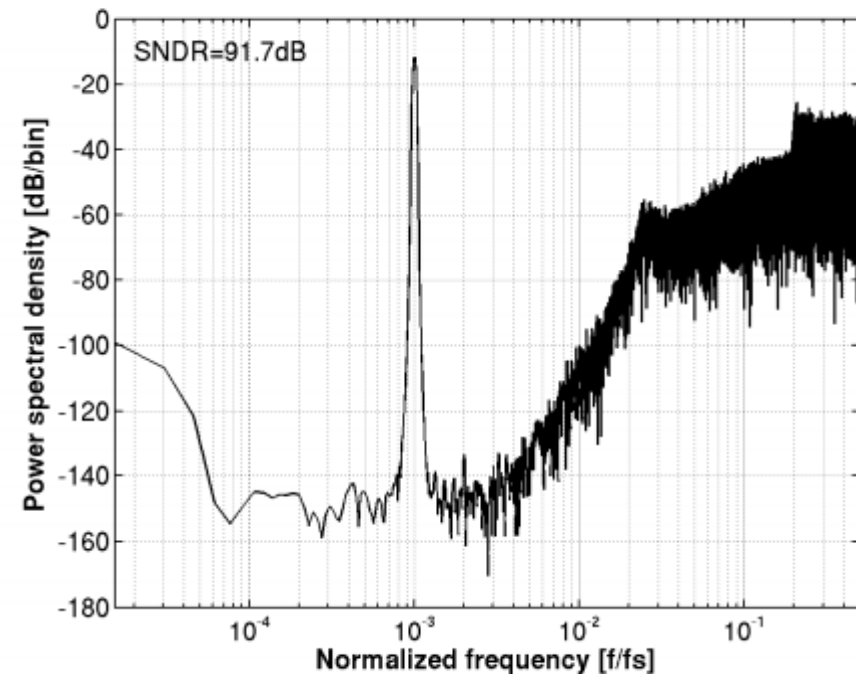
Yao, .., VLSI Circuits '05
Yao. etal. "Low-Power Low-Voltage $\Sigma\Delta$ modulators in Nanometer CMOS", Springer '06

Performance Comparison

- ◆ 4th-order conventional topology
- ◆ Behavioral simulation with:
 $a_1=-0.1$, $a_2=-0.1$, $A_0=40$ dB

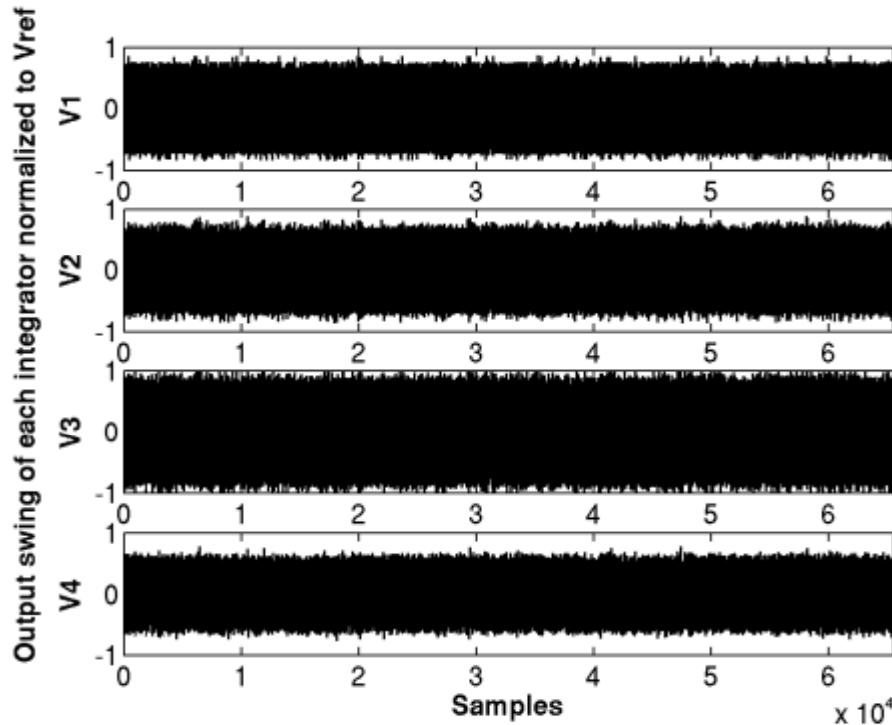


- ◆ 4th-order full feedforward topology
- ◆ Behavioral simulation with:
 $a_1=-0.1$, $a_2=-0.1$, $A_0=40$ dB

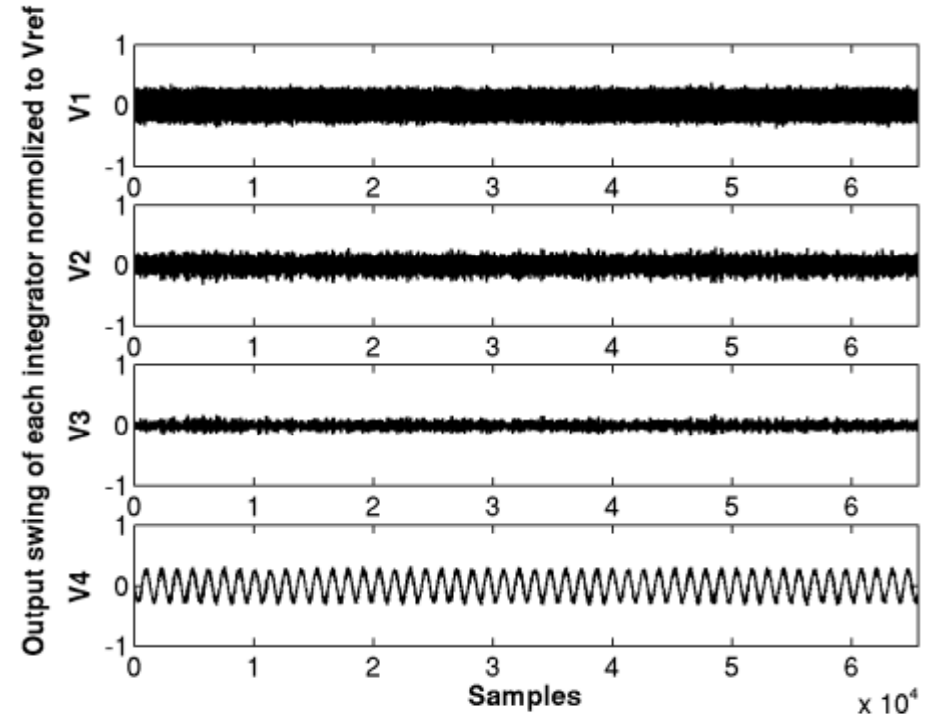


Ref: Sansen. Analog design essential, chapter 21

Output Swing of Integrators



Output swings of each integrator in the conventional topology



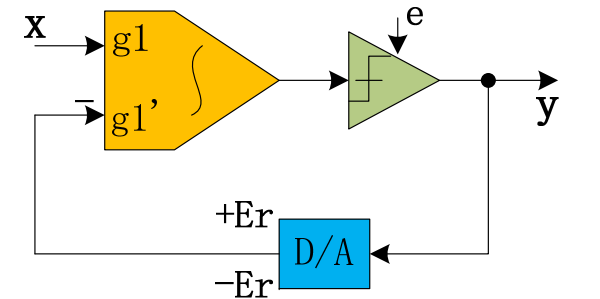
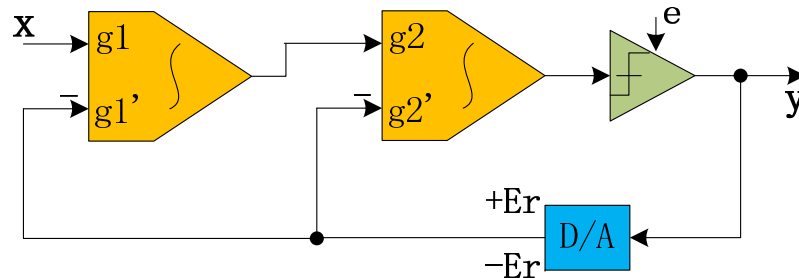
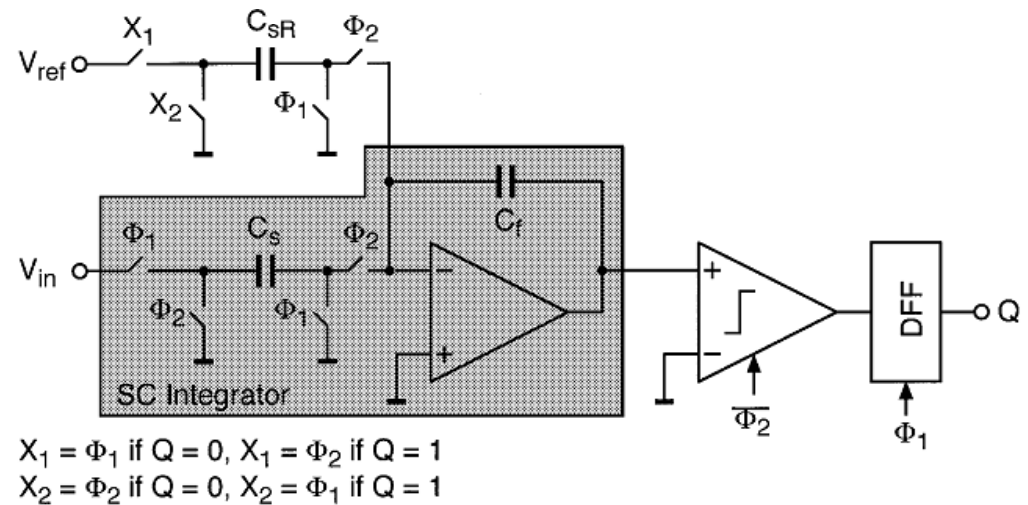
Output swings of each integrator in the full-feedforward topology

Sigma-Delta Modulator

- Principle
- Architecture
- Circuit non-idealities
- Sigma-Delta Converter
 - Example: Audio DAC Design

Circuit non-idealities

- Sampling clock jitter
- Switch thermal noise
- Nonlinear switch R_{on}
- Opamp noise
- Opamp finite gain
- Opamp nonlinearity
- Opamp BW & SR
- Comparator offset & hysteresis



$$g1 = C_s/C_f$$

$$g1' = C_{sR}/C_f$$

第一积分器是最重要的，后续积分器误差可被noise shaping

Opamp finite gain

➤理想传输函数

$$H(z) = \frac{z^{-1}}{1 - z^{-1}} \quad (\text{理想积分器})$$

$$Y(z) = z^{-2} X(z) + (1 - z^{-1})^2 E(z)$$

➤实际传输函数

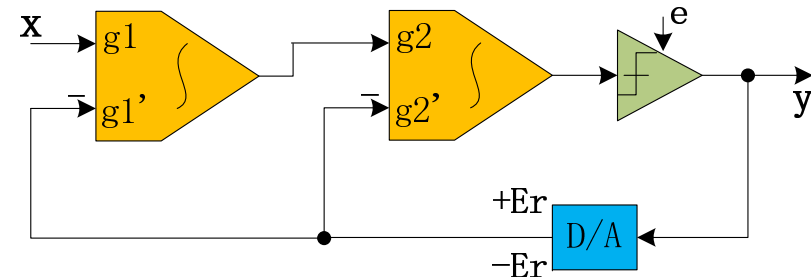
$$H(z) = \frac{g_1 z^{-1}}{1 - (1 - \mu) z^{-1}} \quad \mu \approx \frac{1}{A_v} \quad (\text{有损积分器})$$

$$Y(z) \approx z^{-2} X(z) + (1 - z^{-1} + \mu z^{-2})^2 E(z)$$

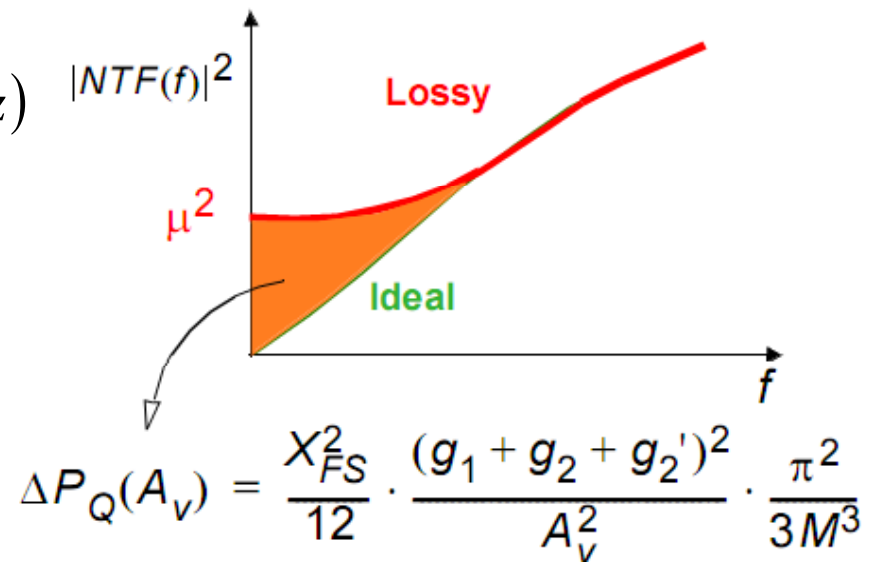
➤结论：对有限增益效应不敏感

➤不完全建立也会导致电荷不完全转移

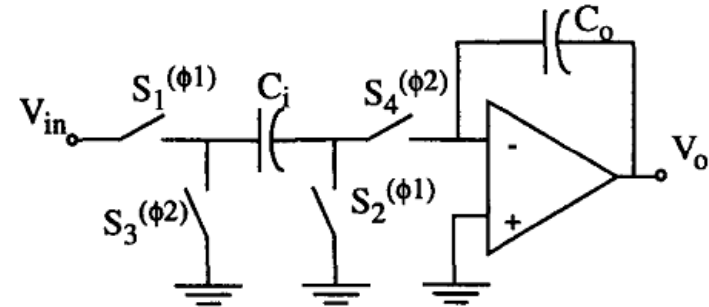
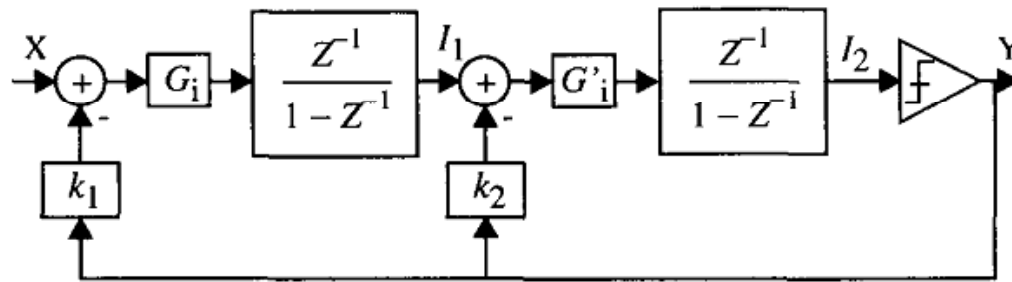
- 线性增益误差，不敏感
- 非线性建立，带来谐波失真
- Solution: 提高SR



Incomplete Charge Transfer



Opamp settling



➤ No SR limited: $v_o(t) = v_o(nT - T) + \frac{C_i}{C} V_s (1 - e^{-t/\tau}) ; nT - T < t < nT$

➤ With SR limited: $v_o(nT) = v_o(nT - T) + G_i V_s - \text{sgn}(V_s) SR\tau e^{-\left(\frac{T}{2\tau} - \frac{G_i |V_s|}{SR\tau} + 1\right)}$

$$g_i(V_s) = \begin{cases} G_i(1 - \beta e); & |V_s| \leq V_L \\ G_i(1 - \beta e^{|V_s|/V_L}); & |V_s| > V_L \end{cases}$$

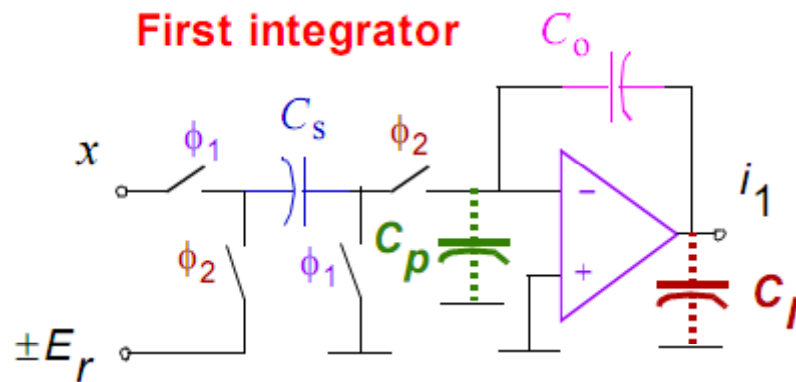
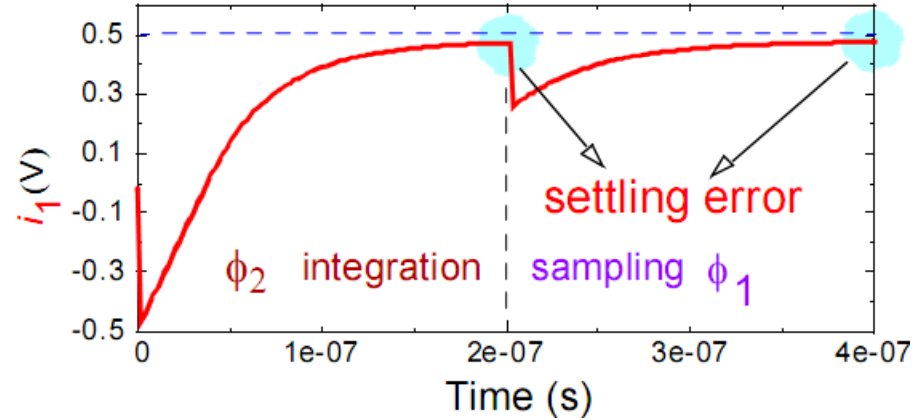
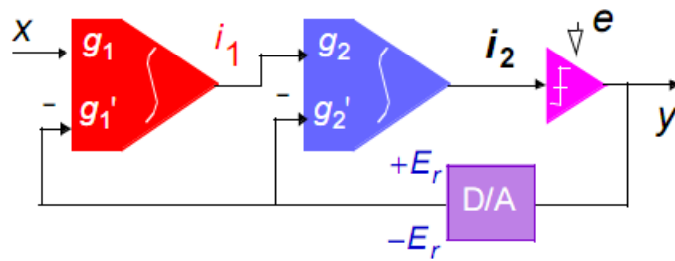
$$\beta = e^{-(T/(2\tau) + 1)}; V_L = SR\tau/G_i$$

$$g_i(v) = G_i \cdot (\alpha_1 + \alpha_3 v^2 + \alpha_5 v^4)$$

➤ 线性建立, no SR limited, 建立误差可假设为白噪声
➤ 若SR limited, 则可能会引入与输入相关的增益误差, 表现为失真, 主要是HD3, HD5

F. Medeiro, et, al. Modeling Opamp –Induced Harmonic Distortion for Switched-Capacitor Σ - Δ Modulator Design. ISCAS 94.

Opamp settling

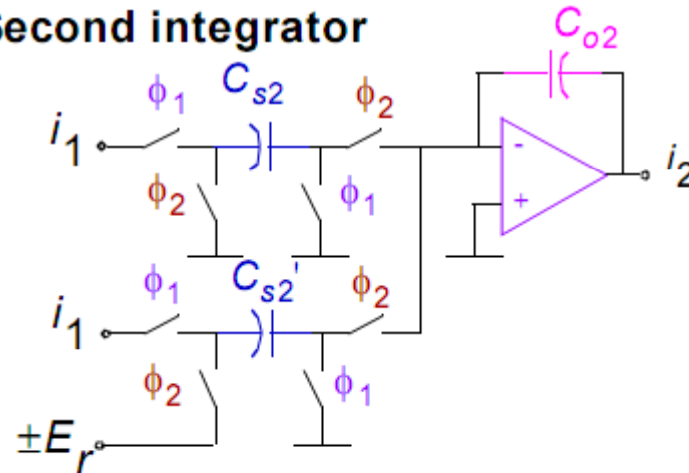


➤ 线性建立, no SR limited,
建立误差可假设为白噪声

➤ 足够的SR

➤ 可采用Class AB放大器

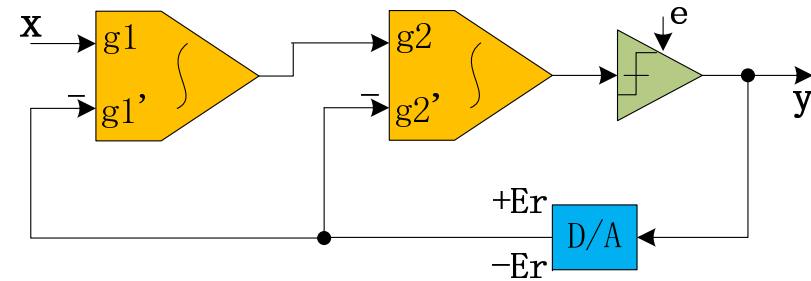
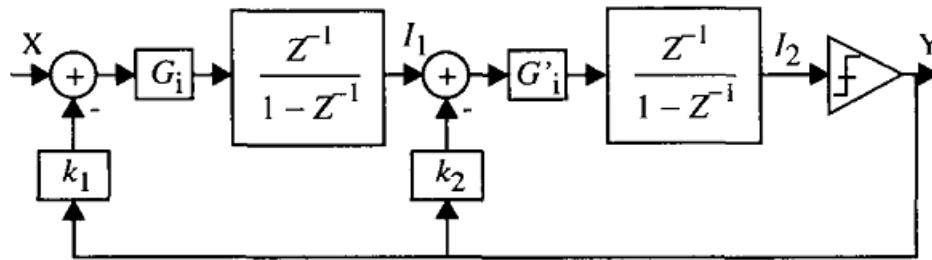
Second integrator



$$P_{s\text{amp}} = \frac{X_{FS}^2}{9M} \left(1 + \frac{C_p}{C_o}\right)^2 \left(\frac{C_{I1}}{C_{eq,1}}\right)^2 \exp\left(-\frac{g_m}{C_{eq,1}} \frac{1}{f_S}\right)$$

$$P_{int} = \frac{X_{FS}^2}{9M} \left(1 + \frac{C_p}{C_s}\right)^2 \left(\frac{C_s}{C_{eq,2}}\right)^2 \left(1 + \frac{C_{I2}}{C_o}\right)^2 \exp\left(-\frac{g_m}{C_{eq,2}} \frac{1}{f_S}\right)$$

Opamp nonlinearity



$$A_V = A_0 (1 + \beta_1 v_o + \beta_2 v_o^2)$$

$$A_2 = \frac{|\beta_1| (1 + k_1) k_2^2}{4A_0} \frac{1}{k_1^3} A^2$$

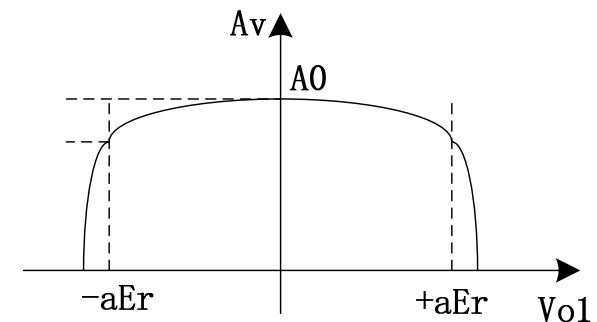
$$A_3 = \frac{|\beta_2| (1 + k_1) k_2^3}{8A_0} \frac{1}{k_1^4} A^3$$

$$A_2 = \frac{2|\beta_1|}{A_0} g_1^2 A^2$$

$$A_3 = \frac{2|\beta_2|}{A_0} g_1^3 A^3$$

- 全差分结构: $\beta_1 \ll 1$
- 例: $A_0 = 1000$, $\beta_2 = 10\%$, resolution = ?
- 运放设计: 提高增益 A_0 ; 输出 rail-to-rail

g_1, g_1'	0.5	0.25	1/3	0.25
g_2	0.5	0.5	0.6	1
g_2'	0.5	0.25	0.4	0.5
Total int. OS / E_{ref}	3.5	2	2.4	2



Other nonlinearities

- 电容电压系数
 - $C(v) \approx C_0(1 + k_v v)$ $k_v \ll 1$ for MIM, MOM, PIP
 - $HD_2 \approx (1/2)k_v A$ for single-end circuitry
 - $HD_2 \approx 0$ for fully-differential circuitry
 - MOS CAP 不适合于做高精度工作电容
- 采样开关非线性 R_{on}
 - 与输入电压相关的导通电阻，带来与输入相关的电荷注入误差
 - 即使低频输入也存在
 - Solution: bootstrapped sampling switch

The rest non-idealities

- Comparator offset & hysteresis

- Offset可被积分器DC GAIN抑制

- Hysteresis:

$$\Delta P_Q(h) = 4h^2 \cdot \frac{\pi^4}{5M^5}$$

- Sampling clock jitter

- 增加噪声基底

$$P_j = \frac{A_{inp}^2}{2} \frac{(2\pi f_B \sigma_T)^2}{M}$$

- Switch thermal noise

- 增加噪声基底

$$P_{sw} = \frac{kT}{MC_s}$$

- Opamp thermal noise

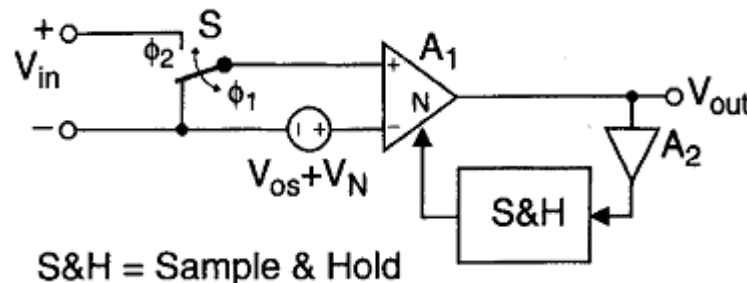
- 增加噪声基底

- V_n 为积分器输出相输出噪声

- 可采用 AZ & CDS

$$P_{op} = \frac{V_n^2}{Mg_1^2}$$

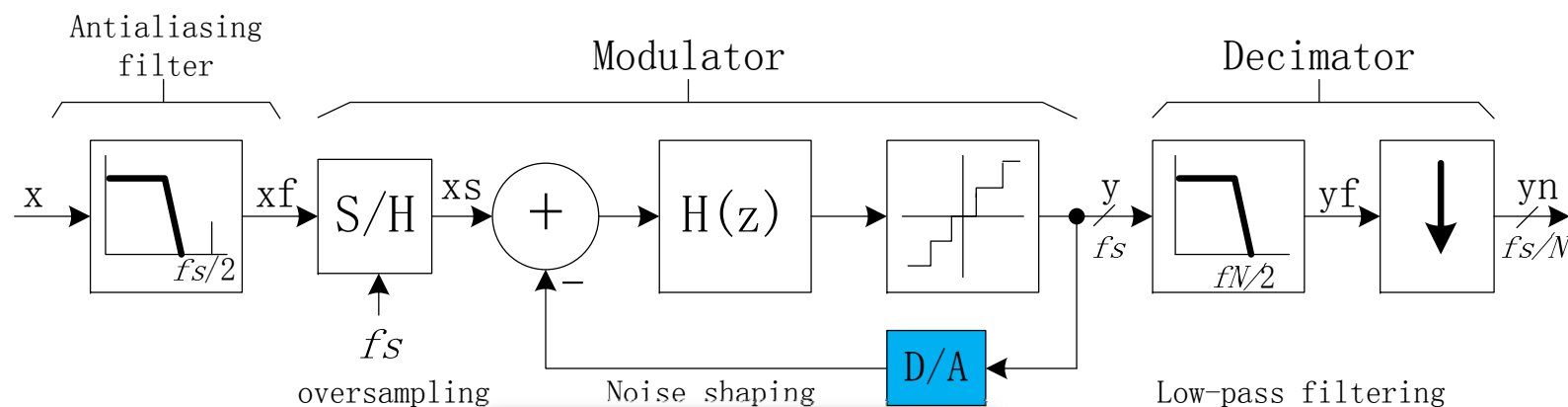
➤考虑第一个积分器；
➤都可被过采样系数抑制



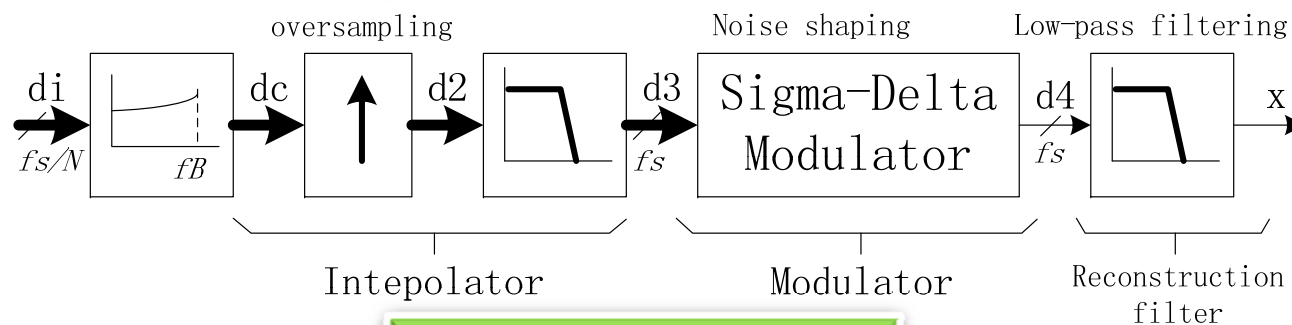
Sigma-Delta Modulator

- Principle
- Architecture
- Circuit non-idealities
- Sigma-Delta Converter
 - Example: Audio DAC Design

Sigma-Delta Converter



Sigma-Delta ADC



Sigma-Delta DAC

练习：画出各节点功率谱示意图

Example: Audio DAC

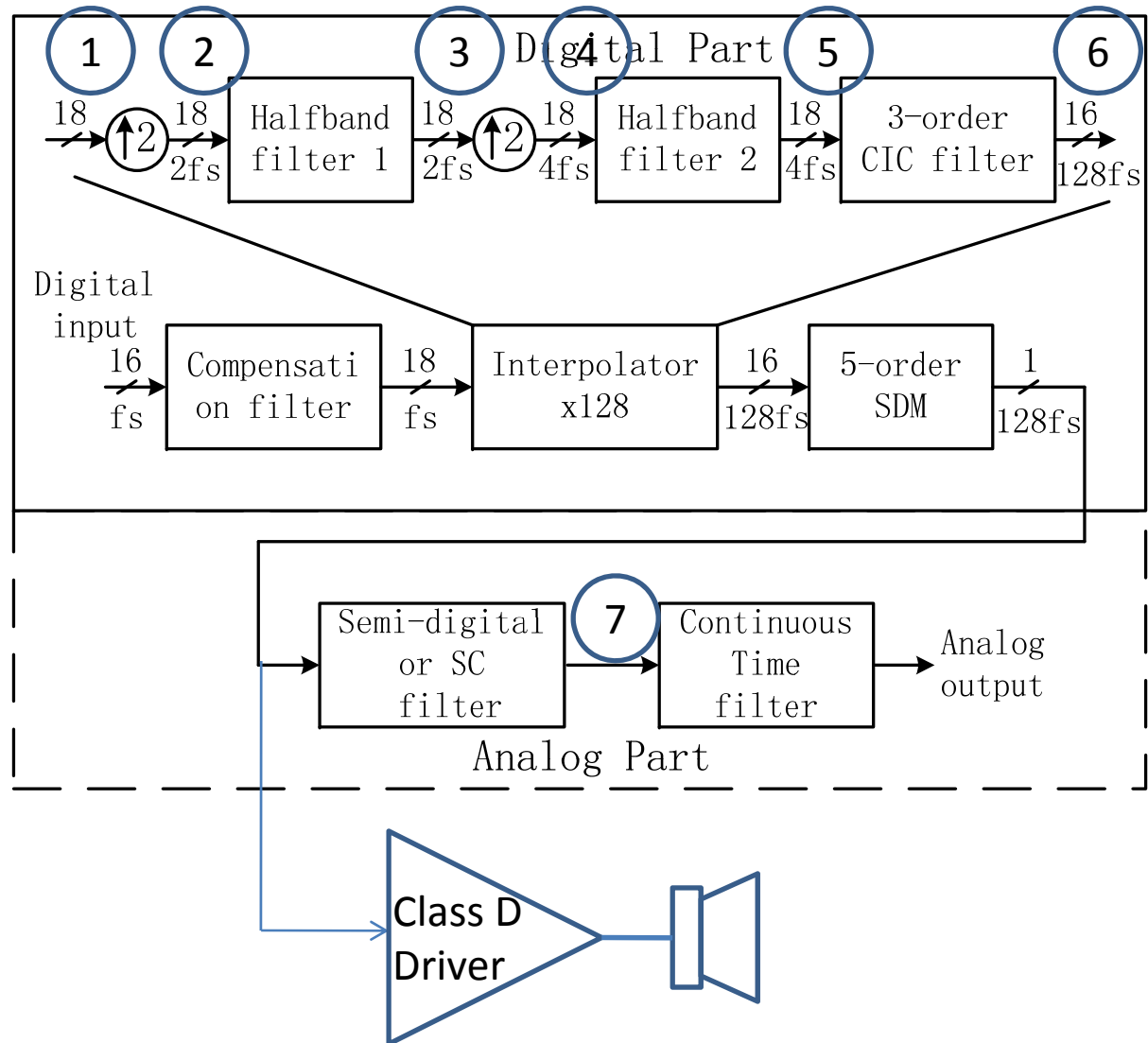
指标:

分辨率: 16-bit

带宽: 20kHz

应用: 音频领域

注意各模块传输特性与各点频谱



插值滤波器

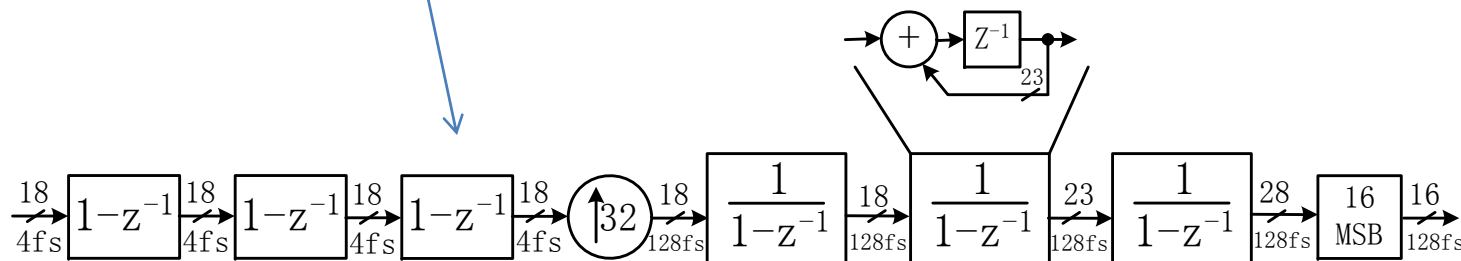
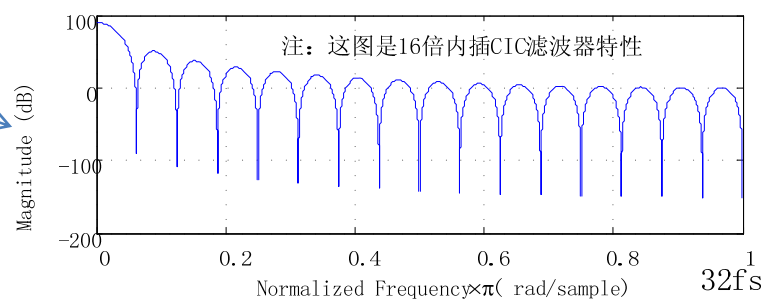
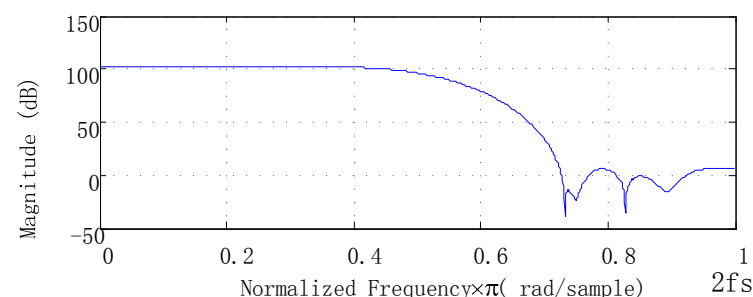
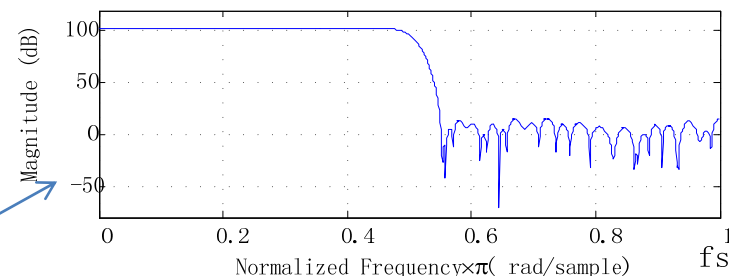
数据流特点：速率低，滤波要求高
速率高，滤波要求低

1) 第一半带滤波器：123阶FIR

2) 第二半带滤波器：27阶FIR

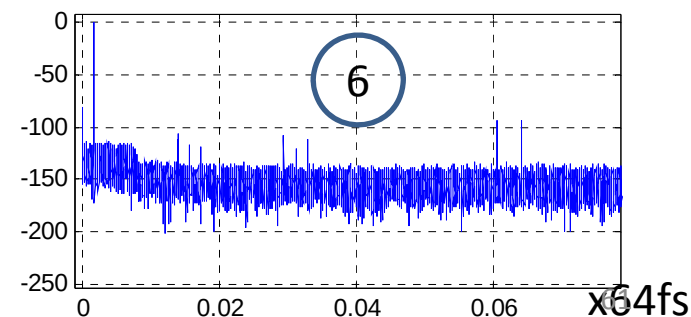
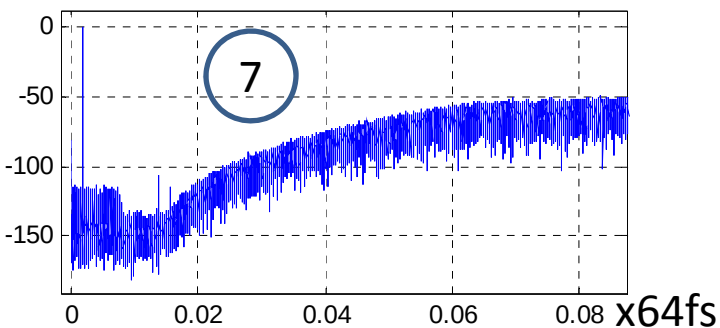
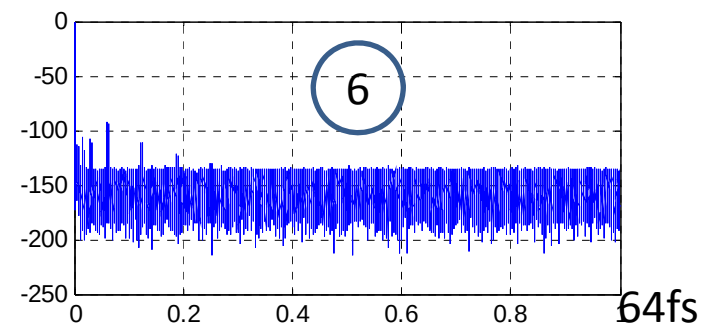
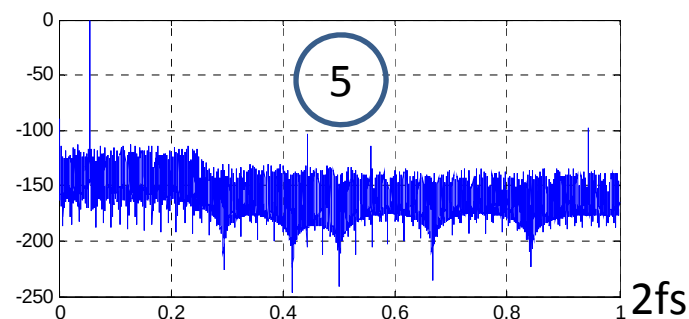
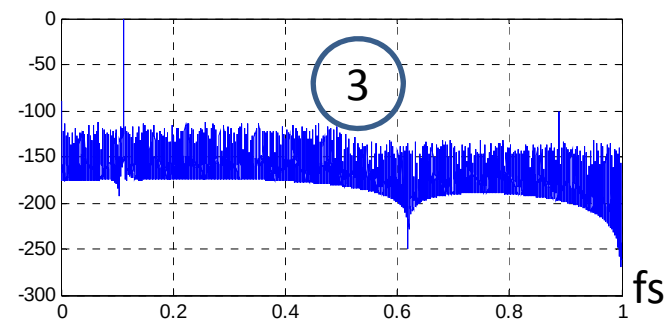
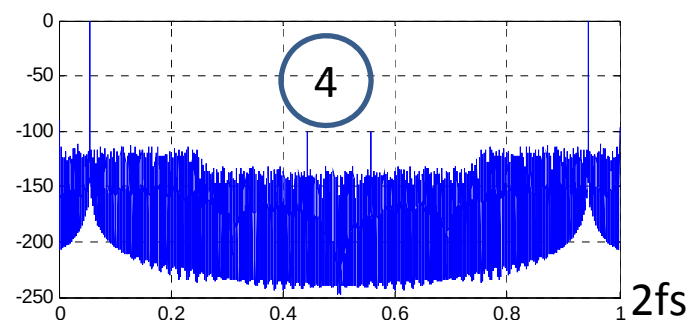
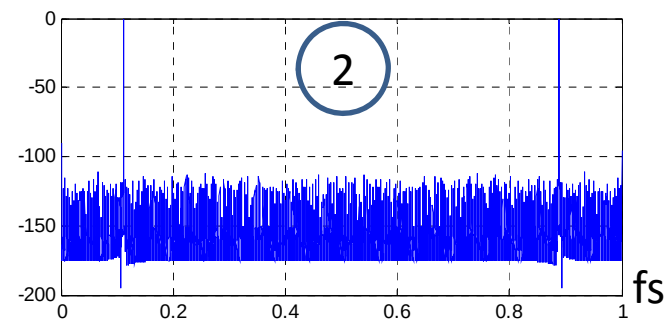
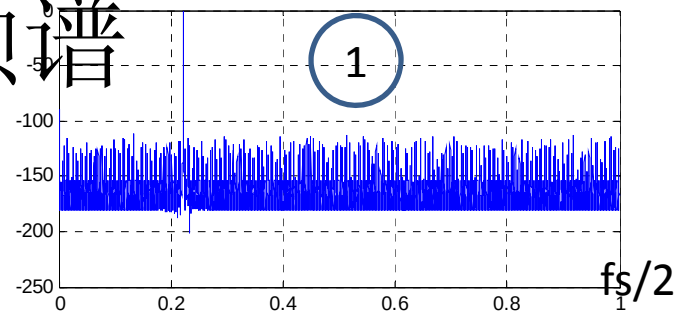
3) 3阶CIC滤波器

在 $2n \cdot \pi / D$ 处(D为内插系数)为零点



各节点频谱

输入为单
频信号时的
各节点
输出频谱



Lecture 2

ADC/DAC Structures

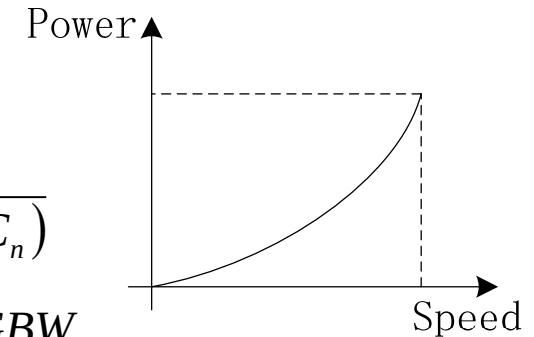
- Flash ADC
- Pipeline ADC
- SAR ADC
- Sigma-Delta Modulator
- **Time-interleaved**
- R2R DAC
- Current-Steering DAC

Time-interleaved

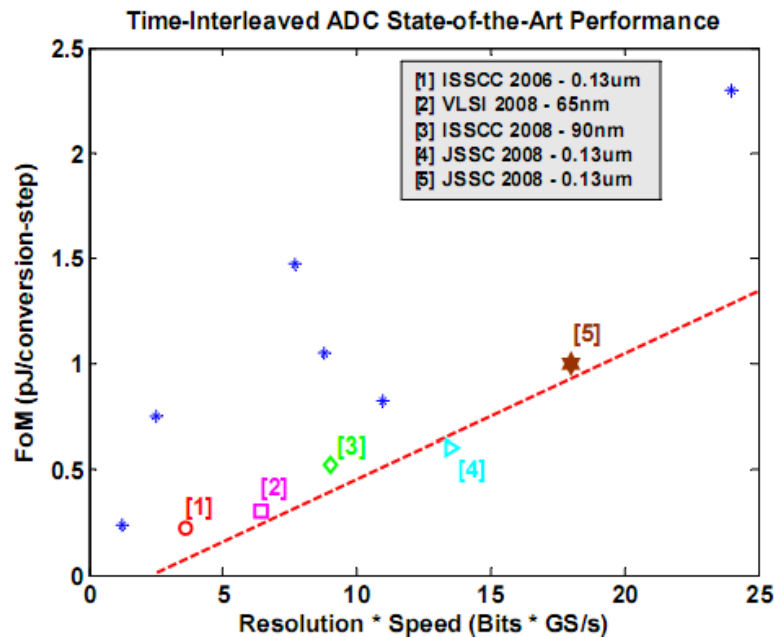
- 多路时间交织实现高速转换
- 通道ADC可采用Flash, SAR, Pipeline等
- 通道间失配导致性能下降
- Clock skew带来非均匀采样问题
 - 可采用同一输入SHA
- 需要校准

$$GBW_0 = \frac{g_m}{2\pi(C_L + C_n)}$$

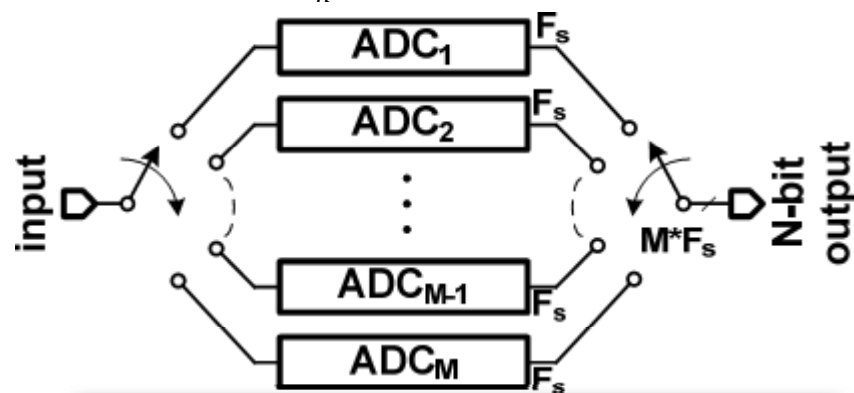
$$GBW = \frac{Mg_m}{2\pi(C_L + MC_n)} < M \cdot GBW_0$$



单通道ADC的速度
功耗关系



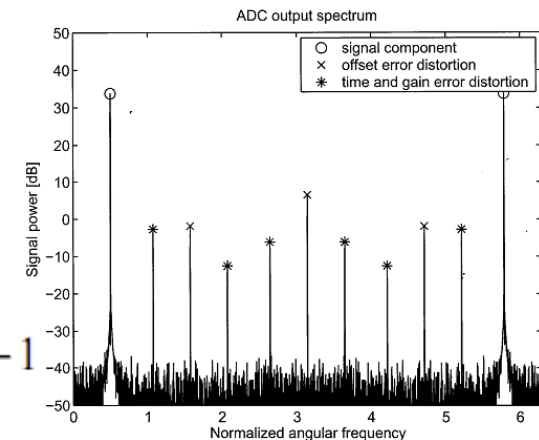
$$GBW_{max} \approx \frac{g_m}{C_n}$$



Time-Interleaved在实现高速时较单通道在功耗上有优势，且可实现单通道所达不到的速度

Non-idealities

- 1) Offset mismatch $\rightarrow \frac{k}{M} \omega_s \quad k = 1, \dots, M-1$
 - 2) Gain mismatch $\rightarrow \pm \omega + \frac{k}{M} \omega_s \quad k = 1, \dots, M-1$
 - 3) Clock skew $\rightarrow \pm \omega + \frac{k}{M} \omega_s \quad k = 1, \dots, M-1$
 - 4) Bandwidth mismatch $\rightarrow \pm \omega + \frac{k}{M} \omega_s \quad k = 1, \dots, M-1$
- Distortion at these frequencies**



Ref: J. Elbornsson, et, al. Blind Adaptive Equalization of Mismatch Errors in a Time-Interleaved A/D Converter System. TCAS I, 2004

Deterministic mismatches among ADCs

	Offset Mismatch	Gain Mismatch	Clock Skew	Bandwidth Mismatch
Input Amplitude	Independent	Linearly dependent	Linearly dependent	Nonlinearly dependent
Input Frequency	Independent	Independent	Linearly dependent	Nonlinearly dependent

ADC IP Example

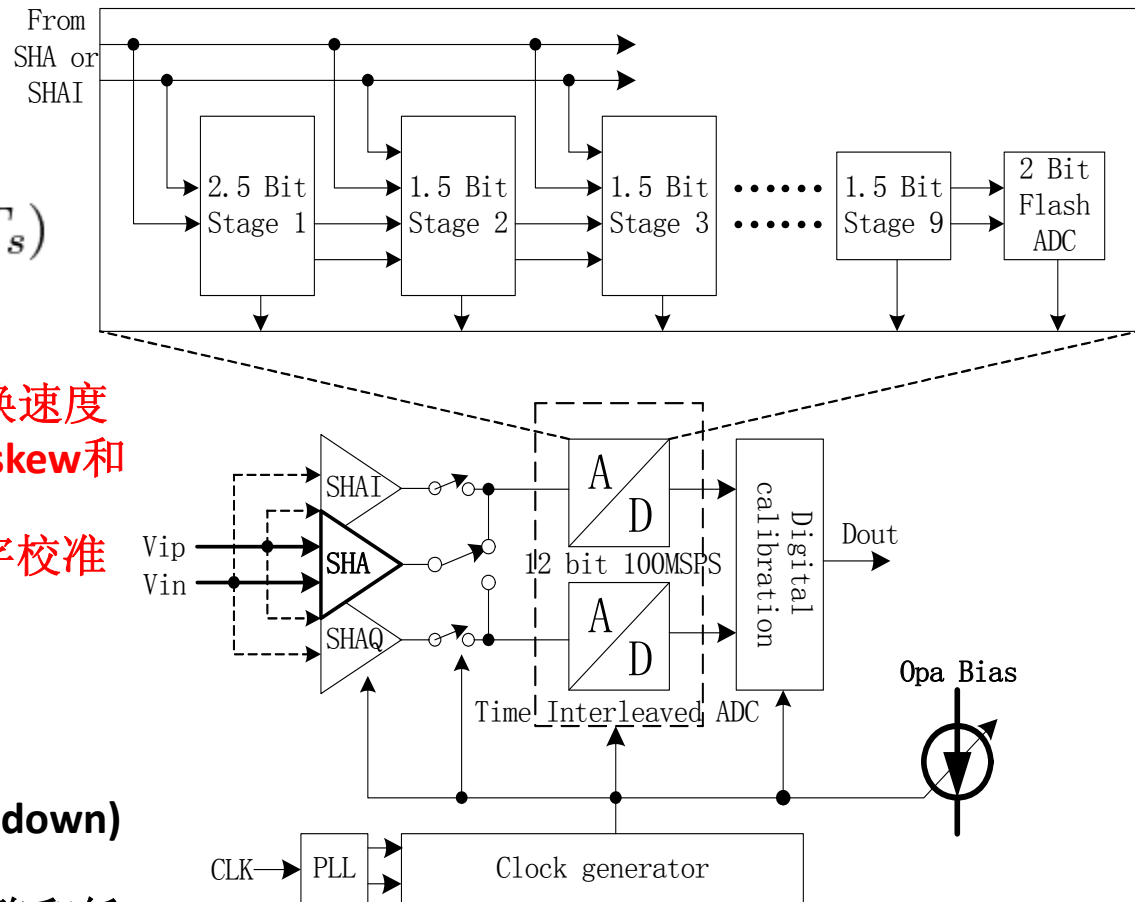
$$y_i[k] = u((kM + i)T_s) + \Delta_{o_i}^0$$

$$\bar{m}_{y_i} = \bar{m}_u + \Delta_{o_i}^0$$

$$y_i[k] = (1 + \Delta_{g_i}^0)u((kM + i)T_s)$$

$$\bar{\sigma}_{y_i}^2 = (1 + \Delta_{g_i}^0)^2 \bar{\sigma}_u^2$$

- 双路交织结构实现**200Msps**转换速度
- 用双路共用一个**SHA**解决**clock skew**和**bandwidth mismatch**
- **Offset**、**Gain mismatch**通过数字校准来解决
- 支持**I/Q**双路转换
- **A/D**转换核心采用**Pipeline**结构
- 可配置的精度/速度
- 从前级到后级采用**缩减(scaling down)**技术来优化功耗
- 片上集成基准源、参考电压电路和低抖动的时钟发生电路

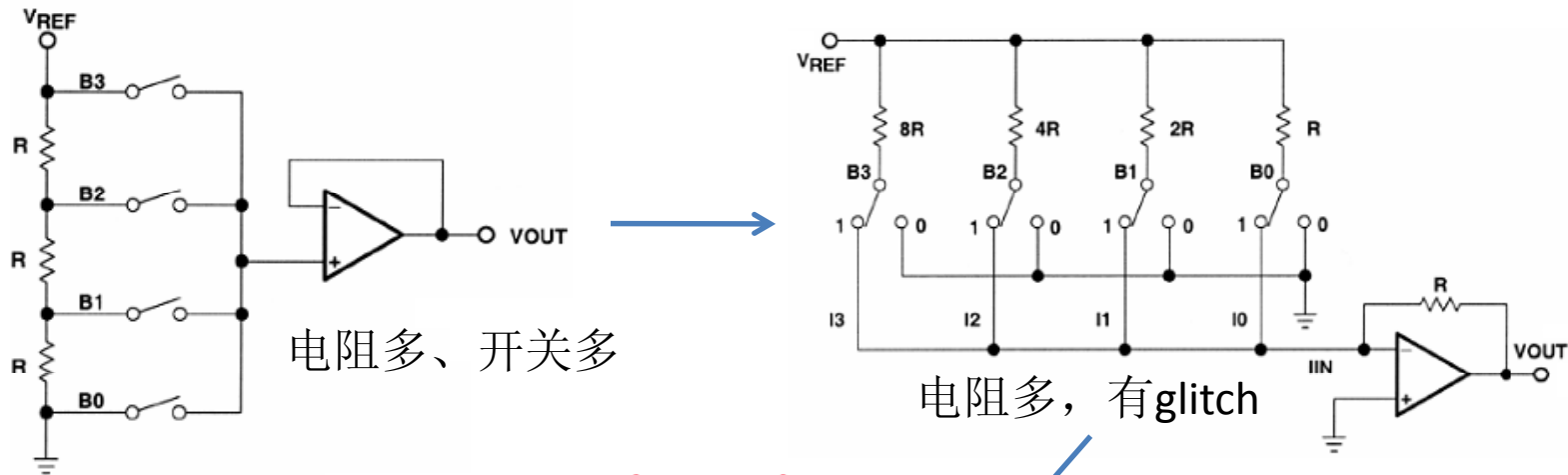


Lecture 2

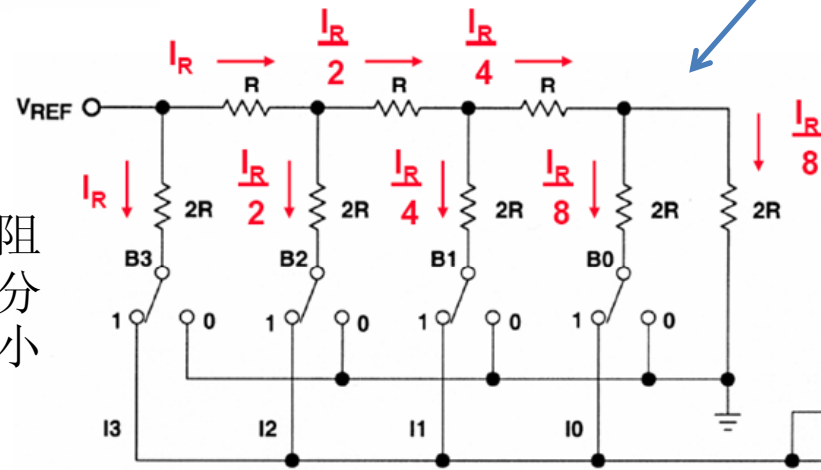
ADC/DAC Structures

- Flash ADC
- Pipeline ADC
- SAR ADC
- Sigma-Delta Modulator
- Time-interleaved
- **R2R DAC**
- Current-Steering DAC

R2R DAC



与传统二进制电阻阵列相比，阻值分散小，总体面积小



问题：实际实现时需要两个参考电压；且运放也需要驱动阻性负载

此为>0的直流电压，否则VOUT<0

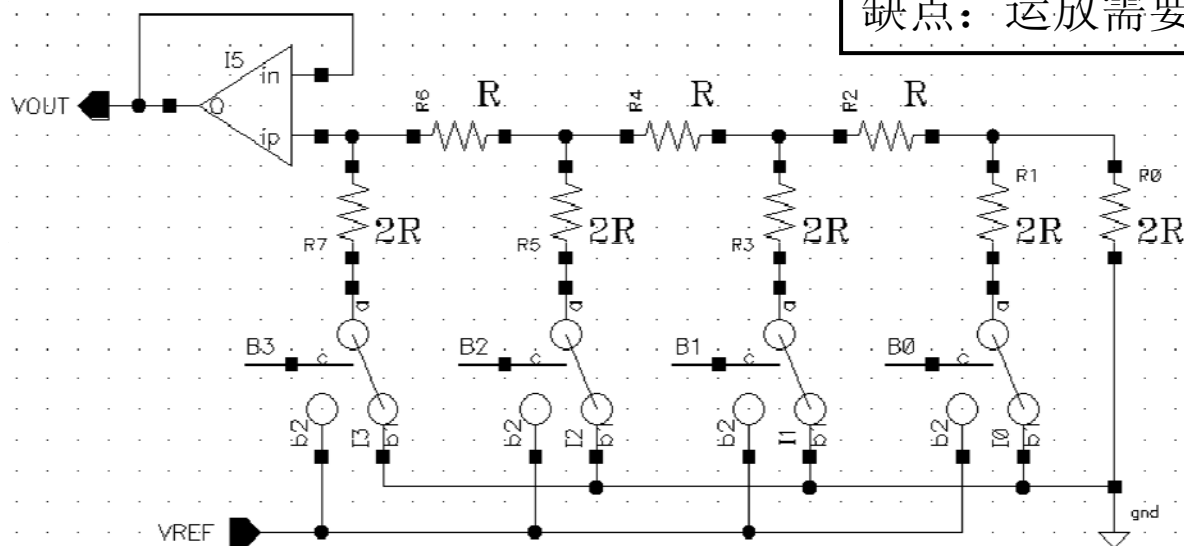
反转结构R2R DAC

优点:

只需一个参考电压;

当负载为容性负载时, 运放也只需驱动容性负载

缺点: 运放需要rail-to-rail设计



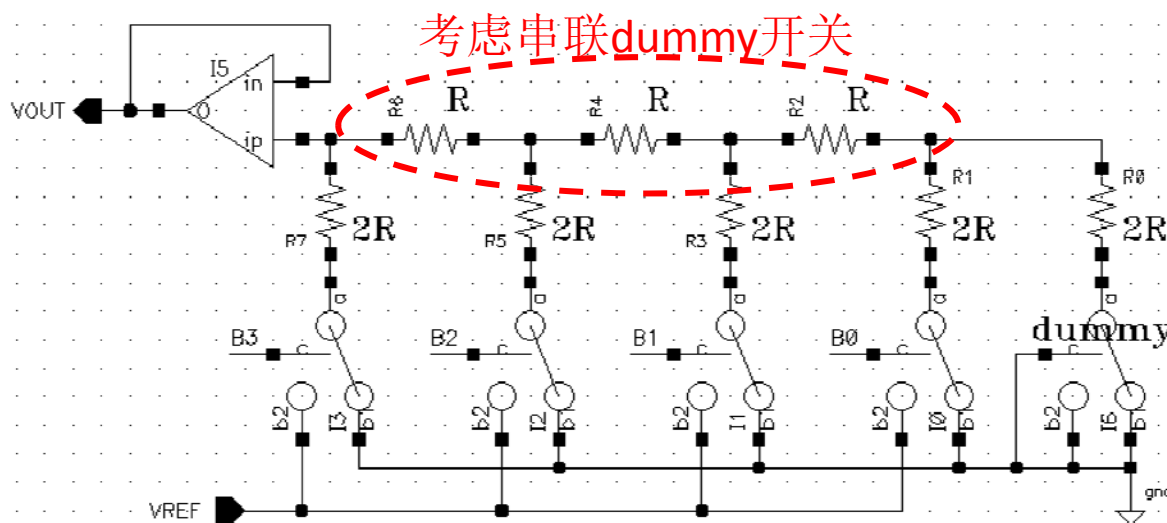
设计要点

电阻:

注意电阻之间的匹配设计;

电阻大小确定速度, 例: $R=5k$, $N=12\text{bit}$, $f_s=40\text{kHz}$

电阻增大, 开关更易设计, 但影响速度



开关:

理论上电阻从高位到低位应保持倍增关系

为减小开关尺寸, 可考虑在图中横向的 R 电阻上串联dummy开关

加上导通电阻后, 尽量保持原有电阻比例关系

尽量减小开关导通电阻与 R 的比值

Lecture 2

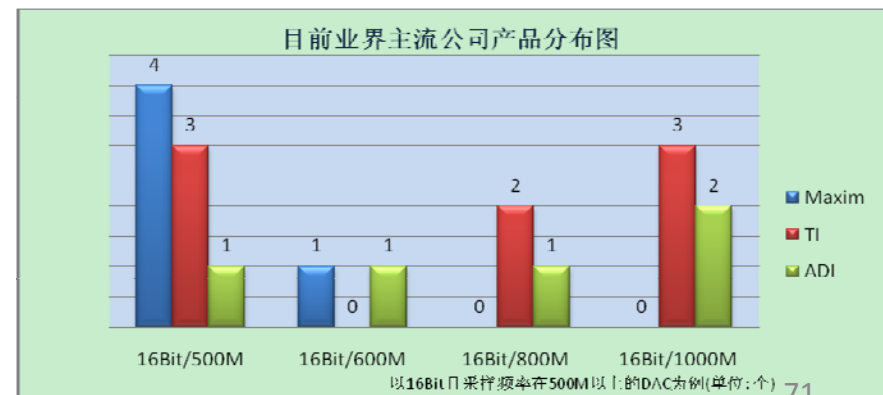
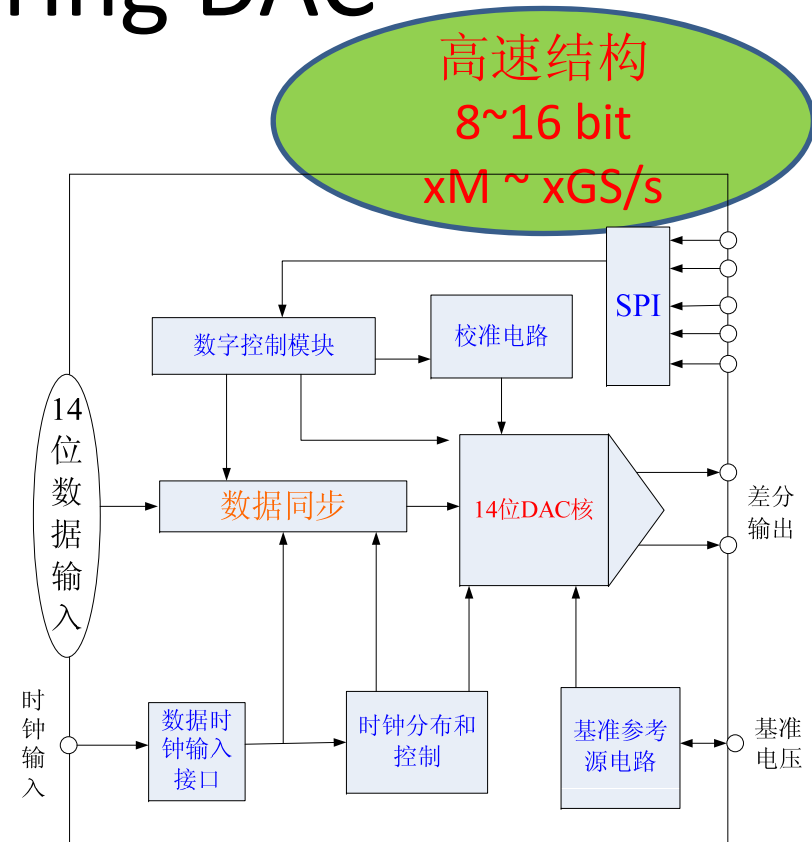
ADC/DAC Structures

- Flash ADC
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- Time-interleaved
- R2R DAC
- **Current-Steering DAC**

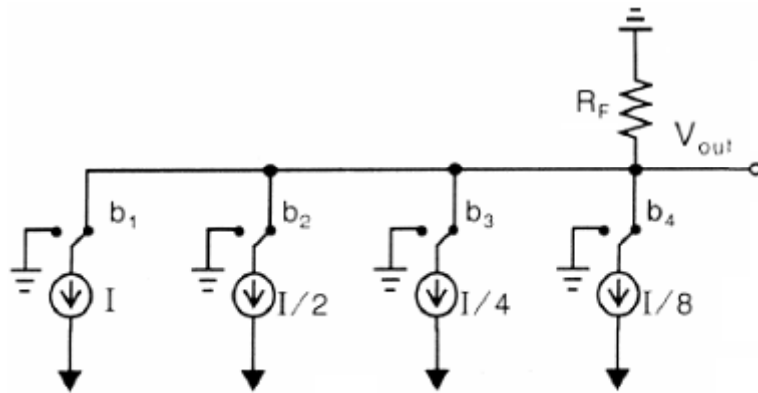
Current-Steering DAC

- 通过电流相加，输出为电流，实现高速操作
- 无需输出模拟Buffer，外部接电阻即可得到电压信号
- 版图布图技术、宽带输出、IF DDS应用、低功耗设计 等是研究热点
- 高性能DAC本身成为一个复杂的数模混合SOC系统
 - 高性能片上时钟电路(PLL)
 - 高精度基准
 - 宽带低功耗开关电容电路
 - 数字控制与校准
 - 数字信号处理（补偿、插值滤波）
 - 高速接口(LVDS, SERDES)
 - 通讯控制接口(SPI, I2C)
 - 片上自测试电路

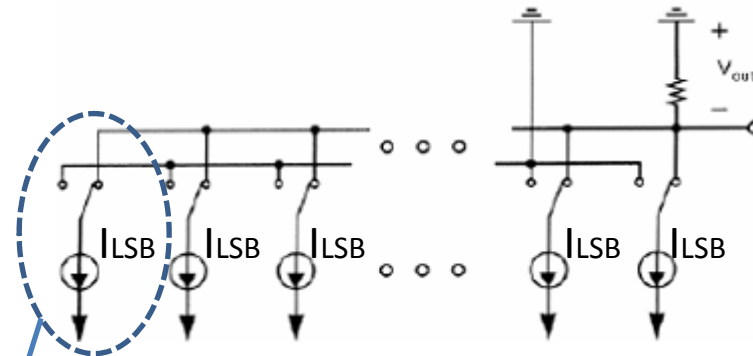
典型指标：
14bit, 2GS/s, 60dBc@500MHz
 商用芯片：AD9739



Principle

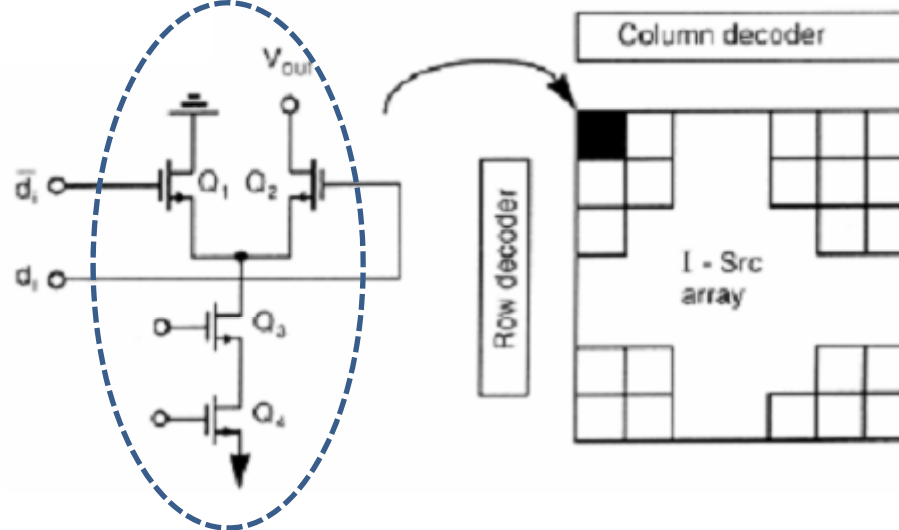


二进制电流舵DAC原理

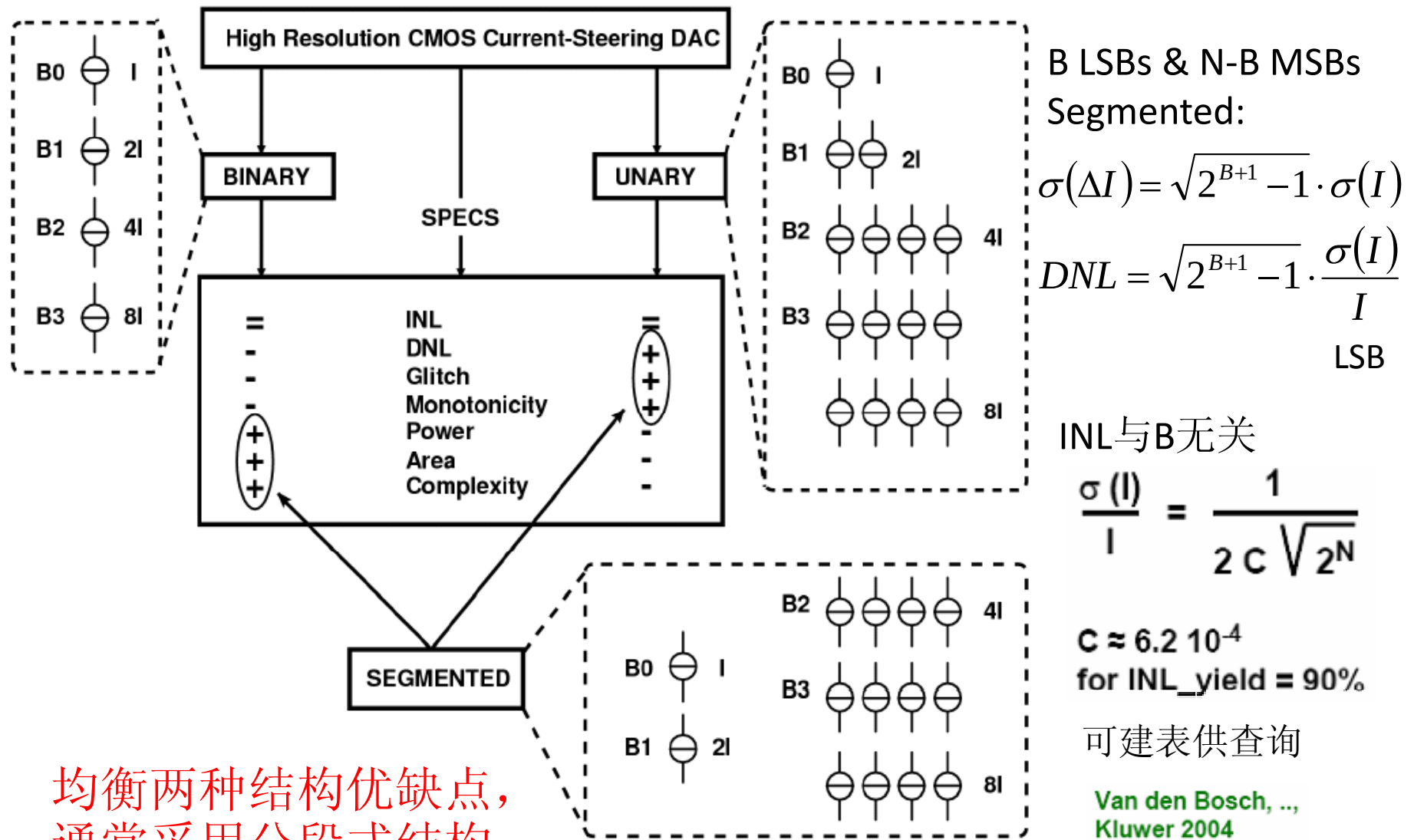


一元电流舵DAC原理

二进制：在高位切换时有较大glitch，和大的DNL，可能存在单调性问题
一元结构：DNL, glitch小，确保单调性，单元件数太多，布线复杂
解决方法：分段式



Segmented architecture



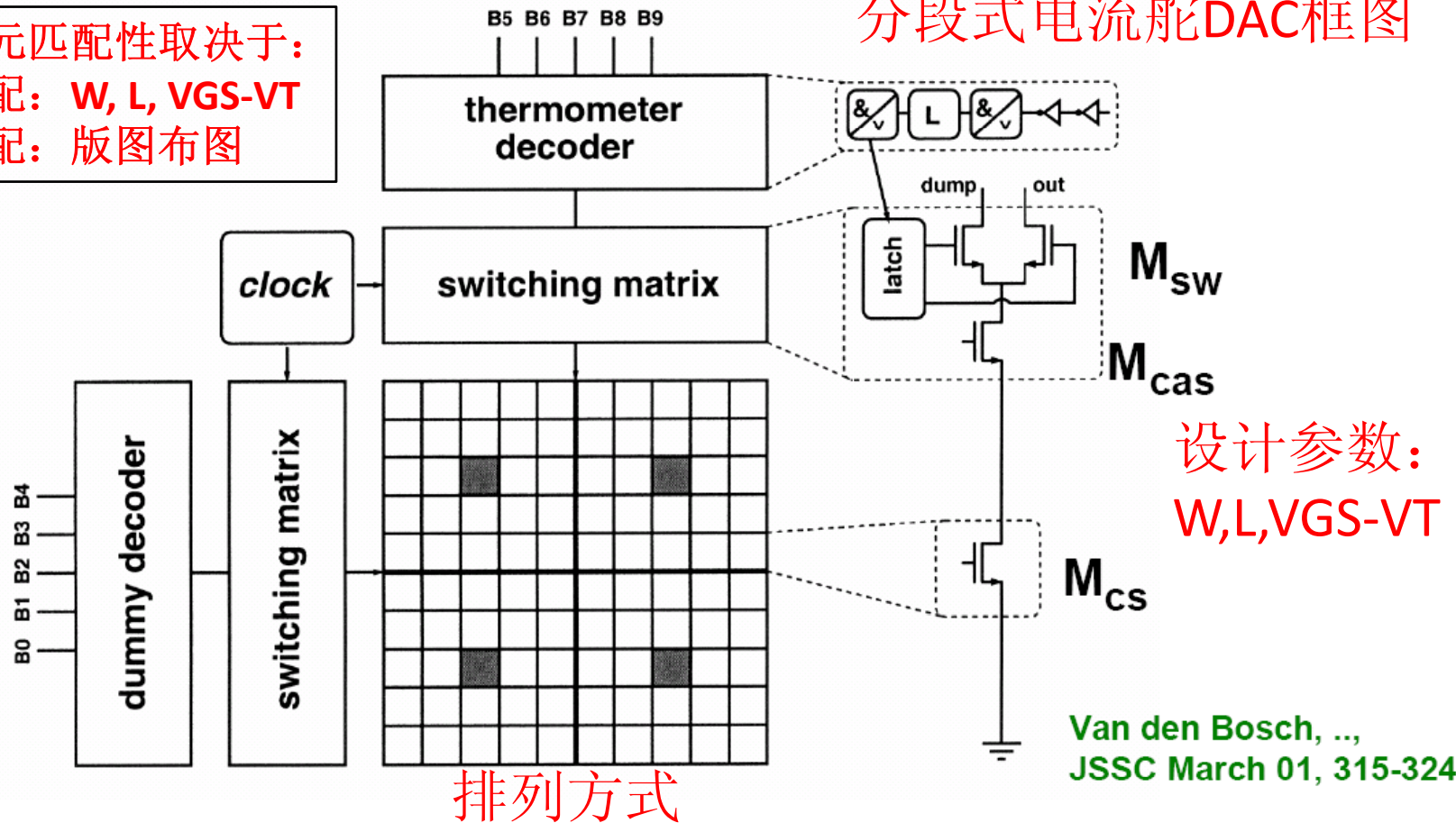
均衡两种结构优缺点，
通常采用分段式结构

Current Steering DAC

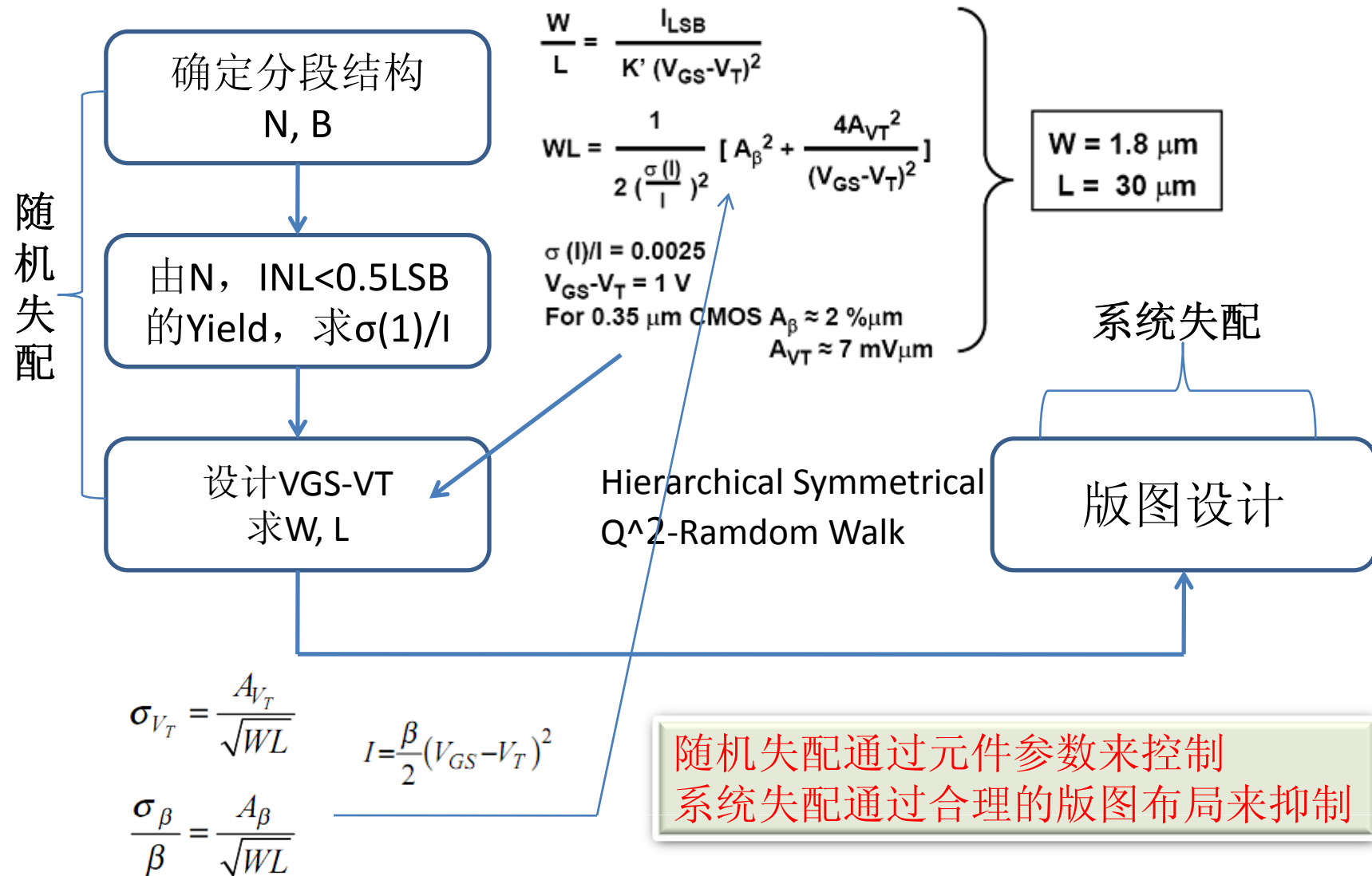
Floorplan 10-bit segmented DAC

电流单元匹配性取决于：
随机失配：W, L, VGS-VT
系统失配：版图布图

分段式电流舵DAC框图



设计过程



END